

Syllabus

PN Junction diode:

Working of PN junction diode,

V-I characteristics of

PN Junction diode,

Zener diode.

avalanche breakdown.

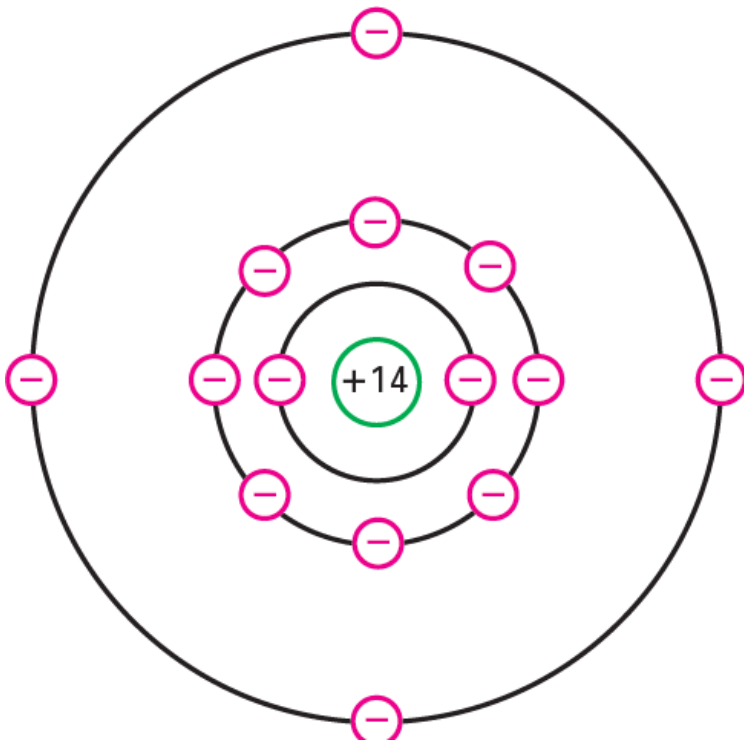
Basics of Zener voltage regulator.

Raw material

- Germanium
- Silicon
- Purity : **99.9999999999% (Eleven nine purity)**

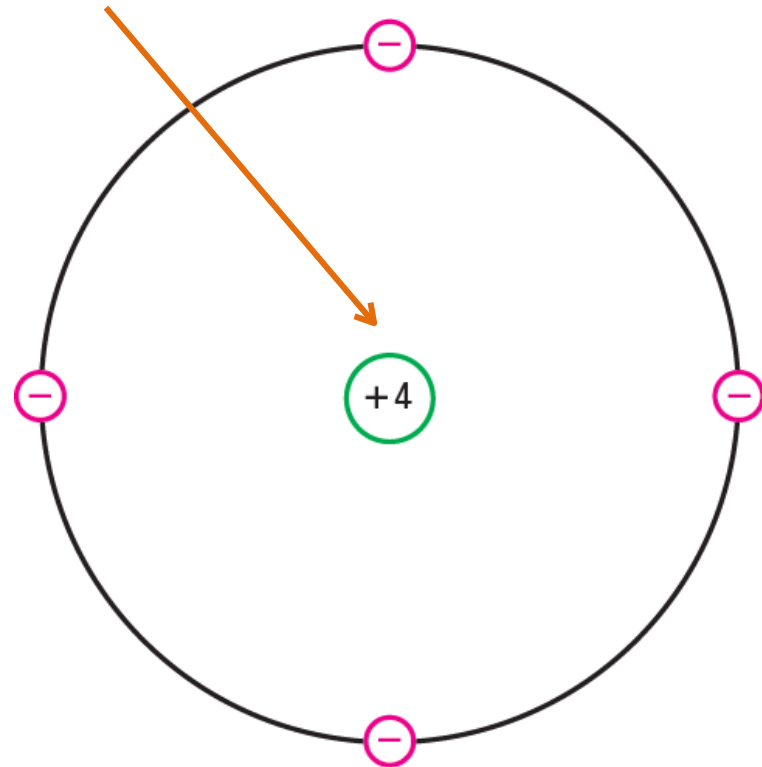
Core diagram of silicon

**Atomic number -
14**



Silicon atom

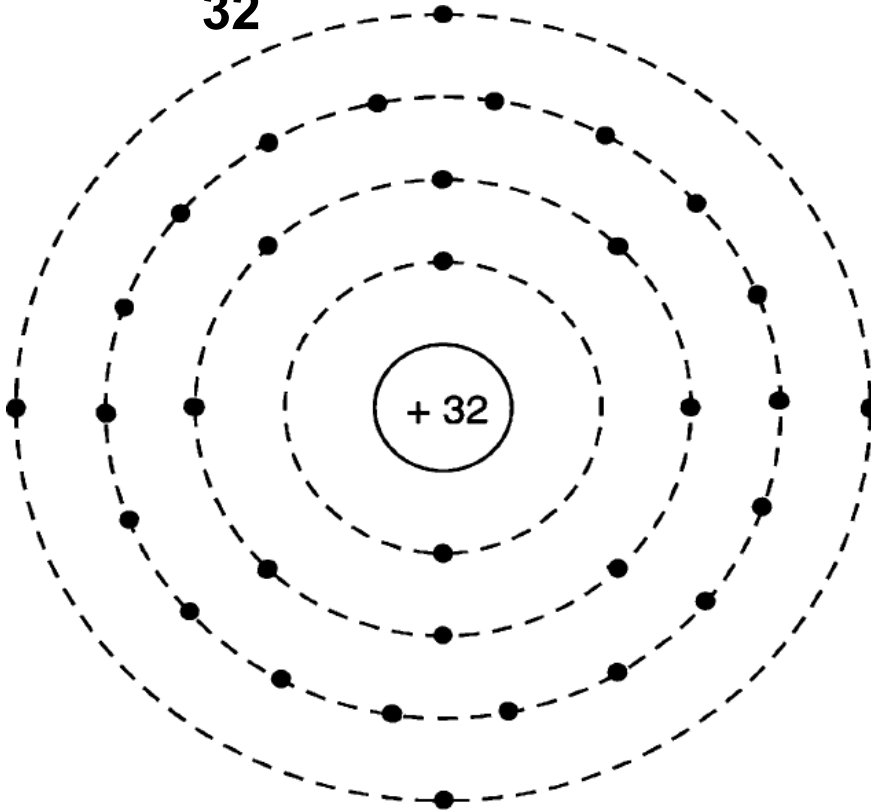
14 protons in the nucleus
10 electrons in the first two orbits.
net charge of +4



core diagram.

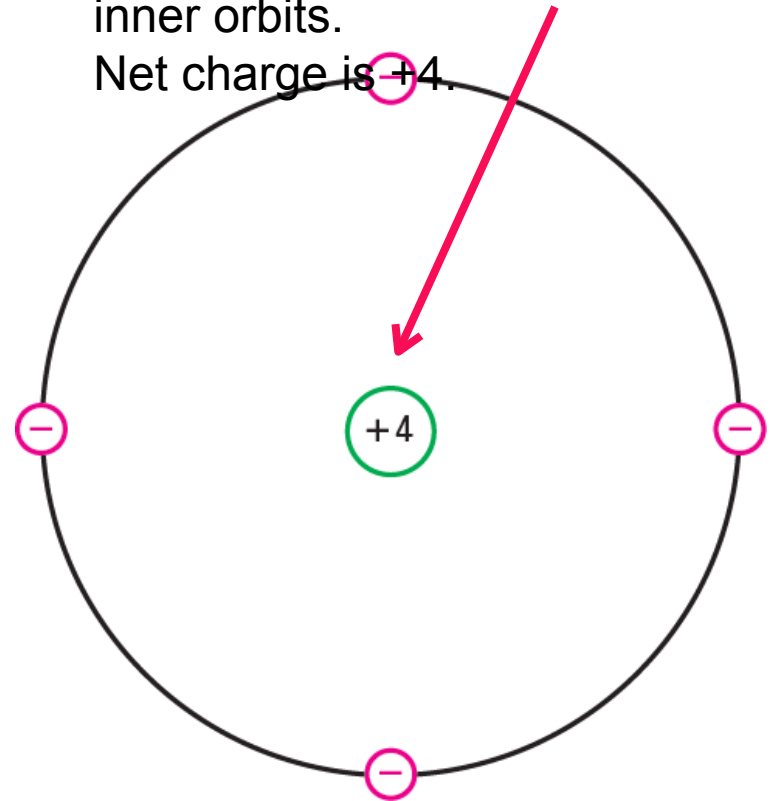
Core diagram of germanium

**Atomic number -
32**



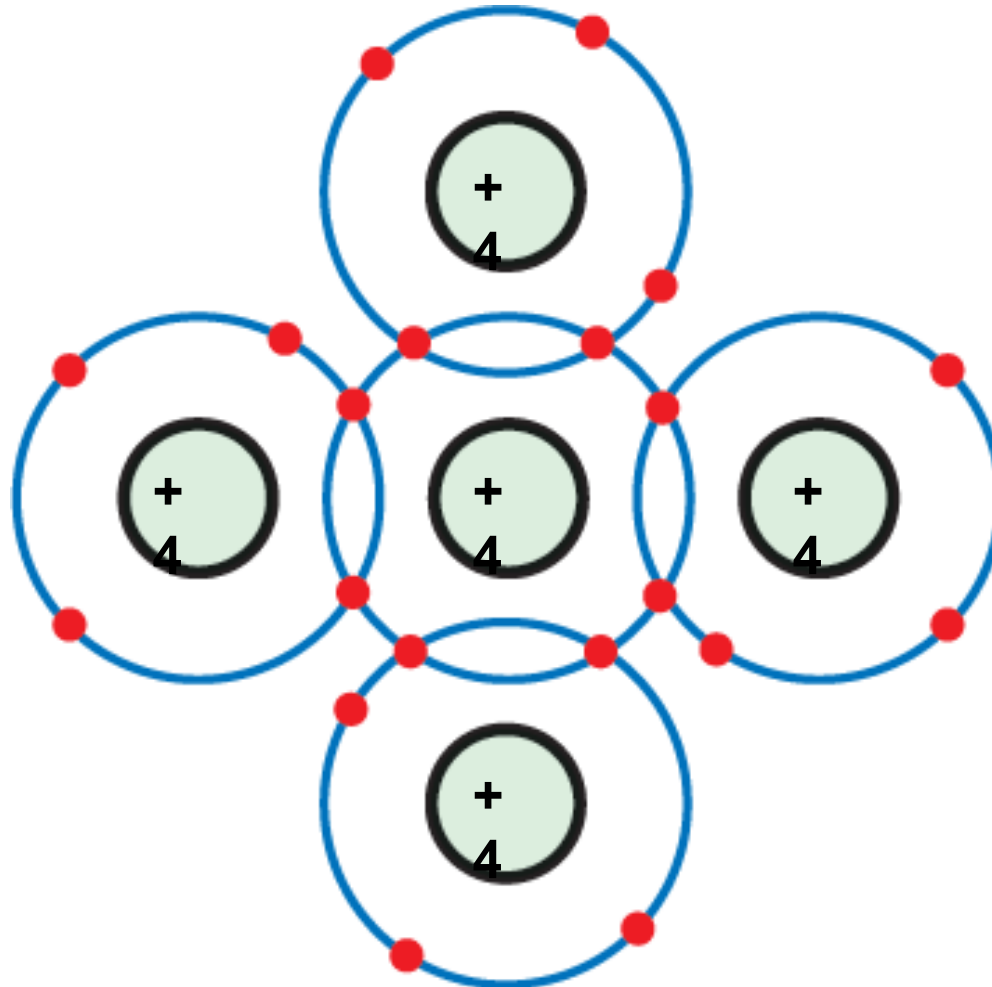
germanium atom

32 protons in the nucleus
and 28 electrons in the 3
inner orbits.
Net charge is +4.

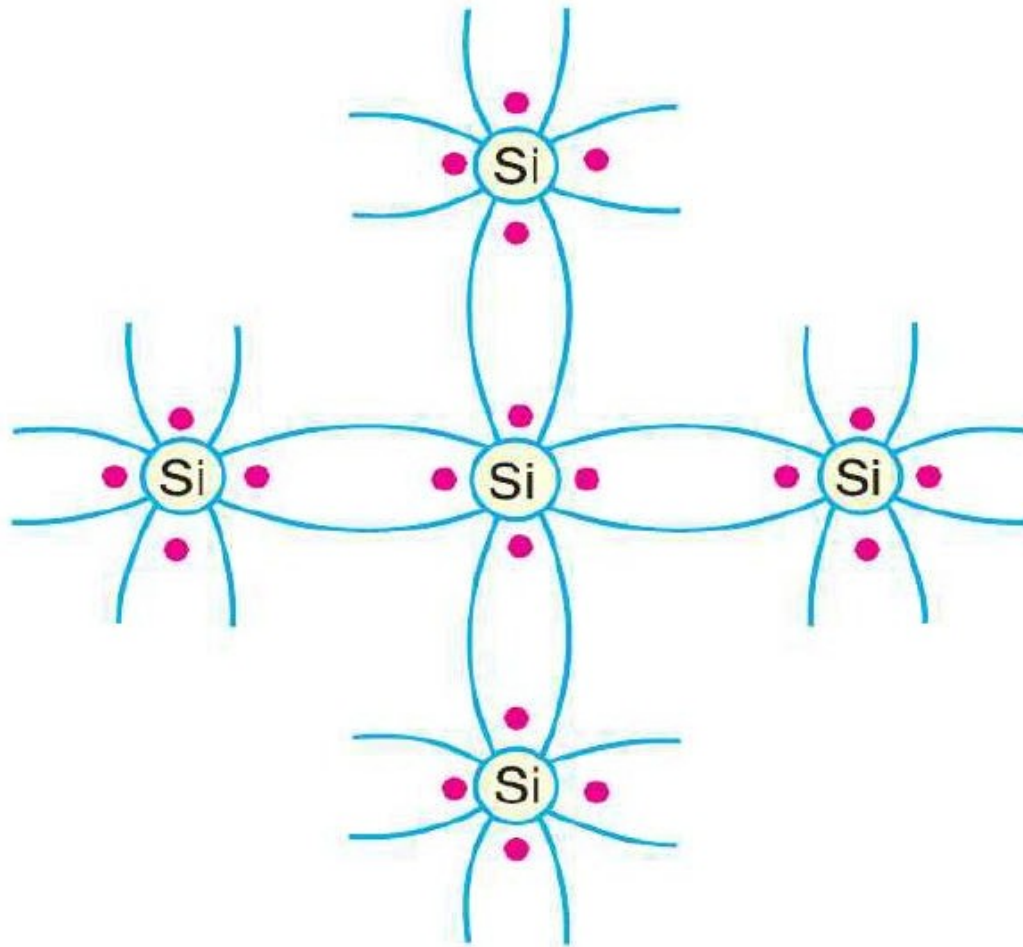


core diagram.

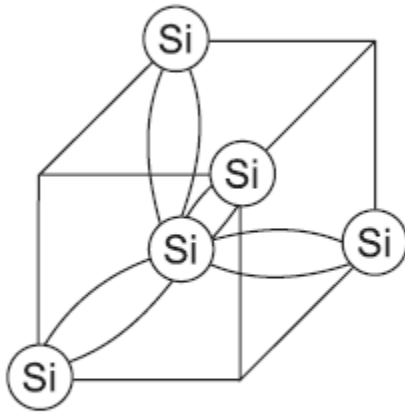
Germanium/Silicon crystals



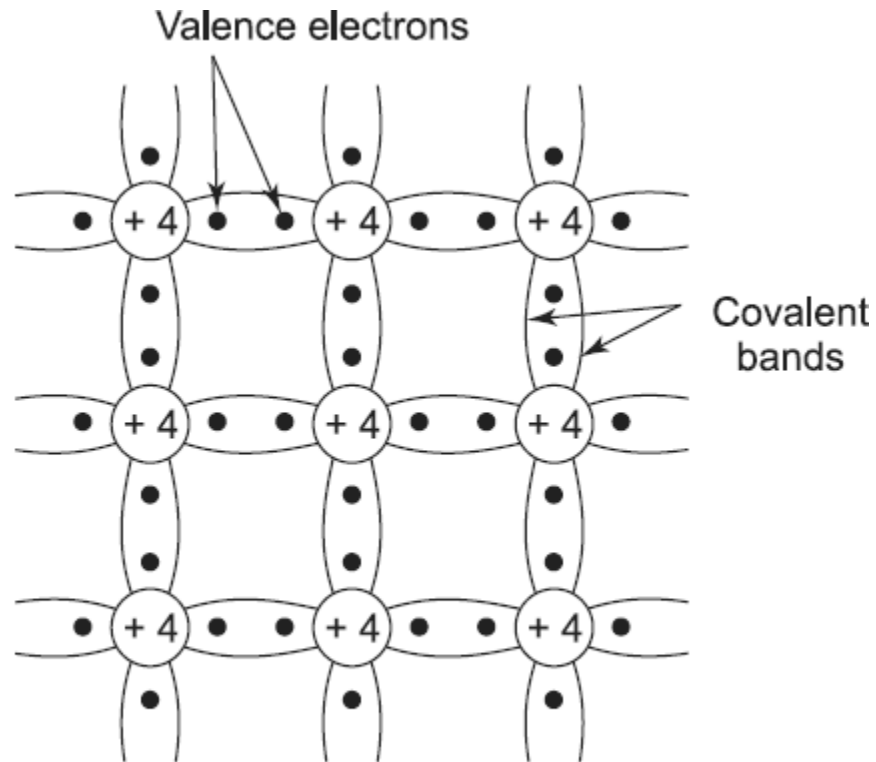
Silicon crystals



Silicon crystals

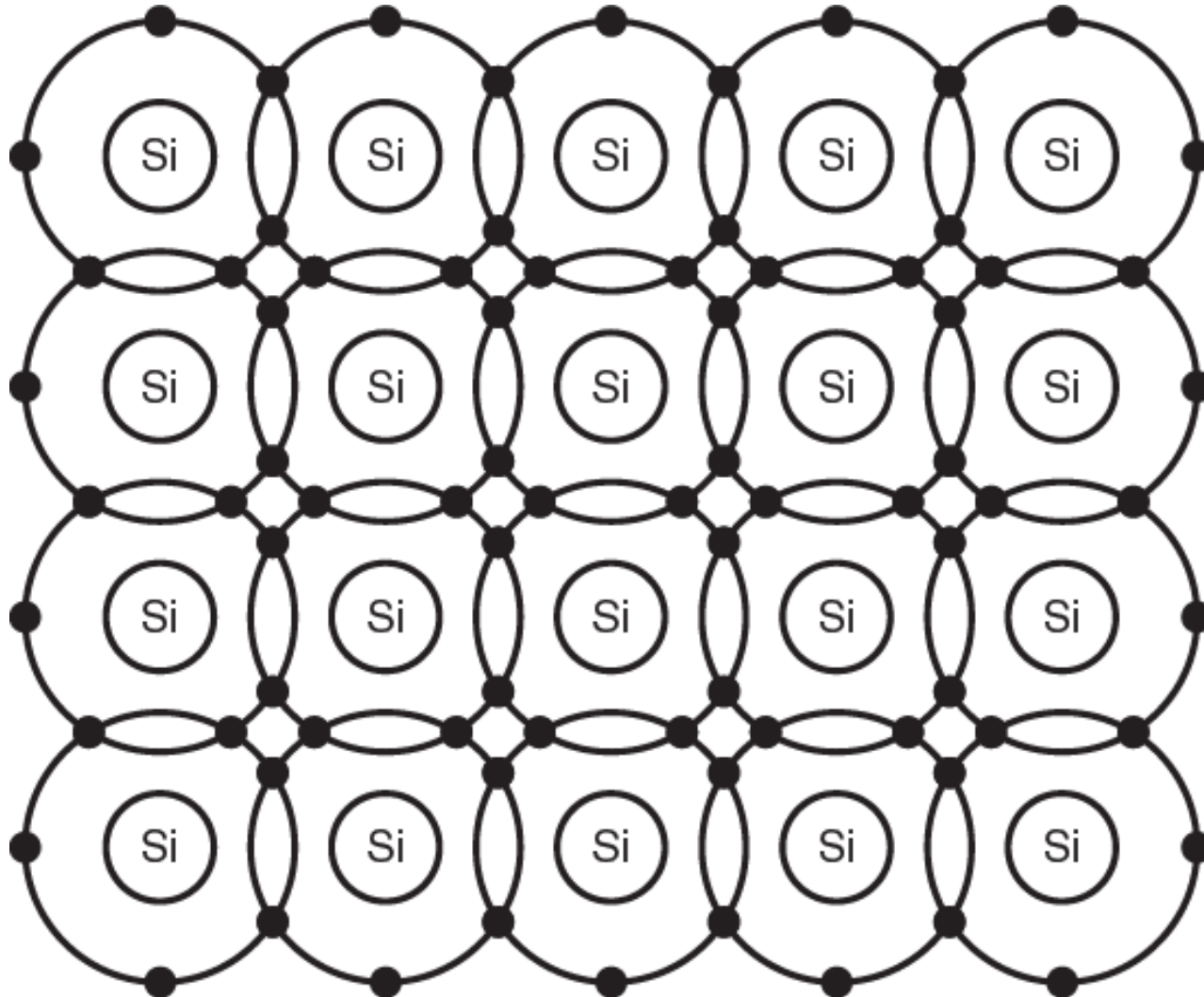


(a) Tetrahedral pattern

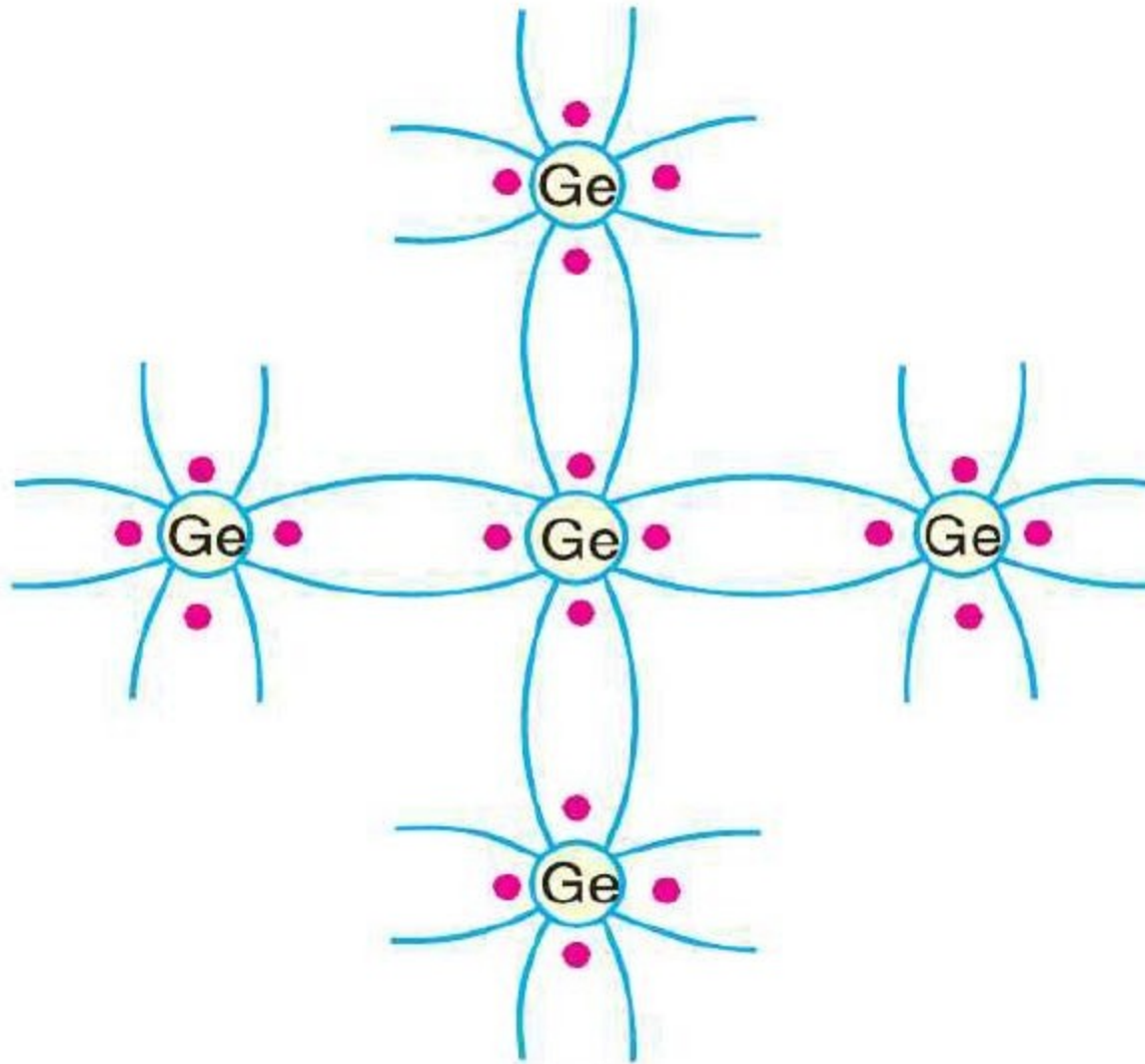


(b) Two-dimensional representation

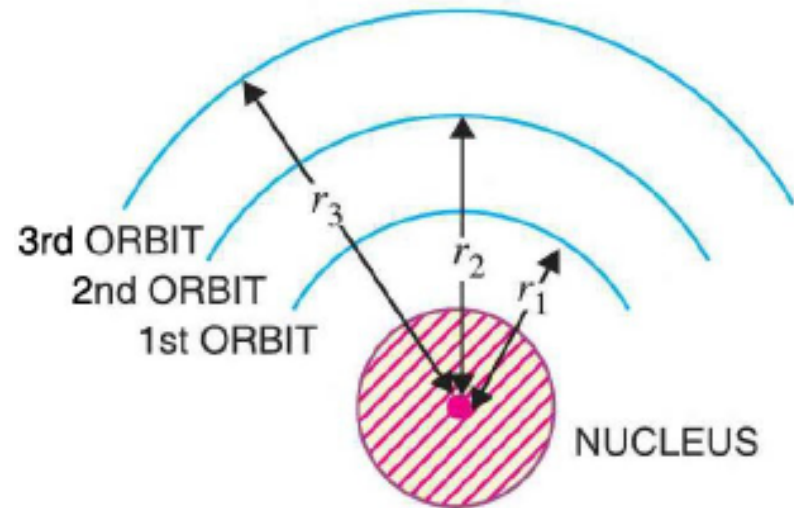
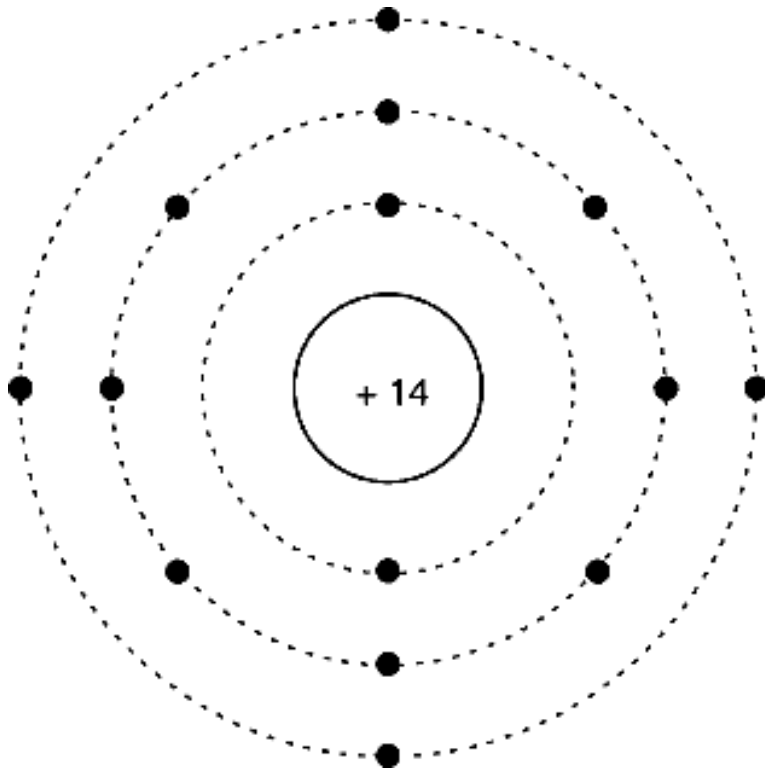
Silicon crystals



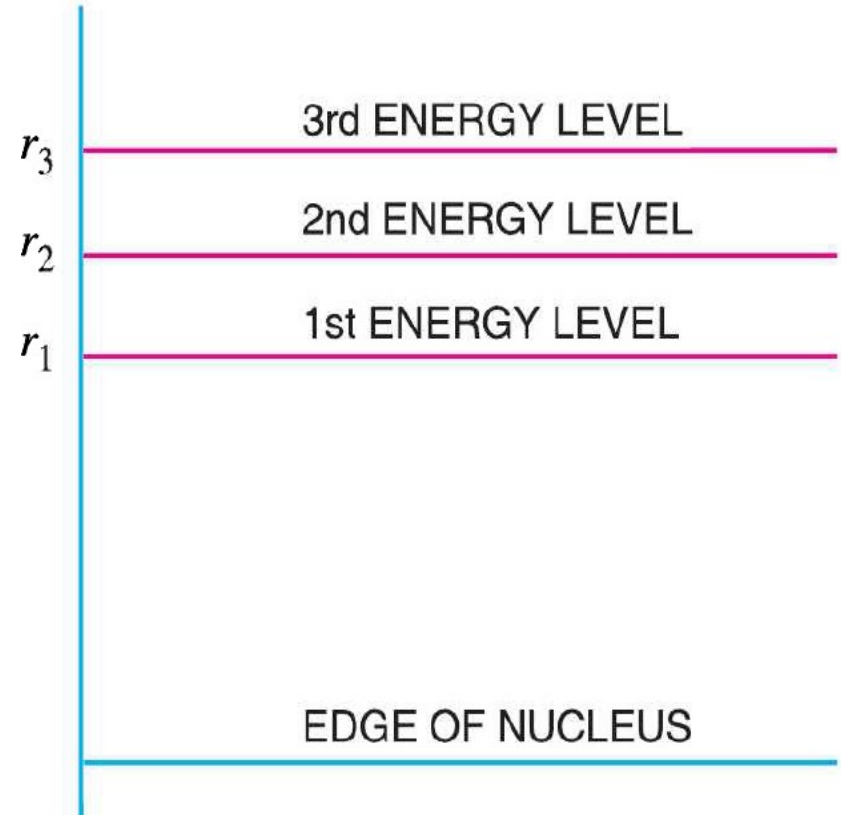
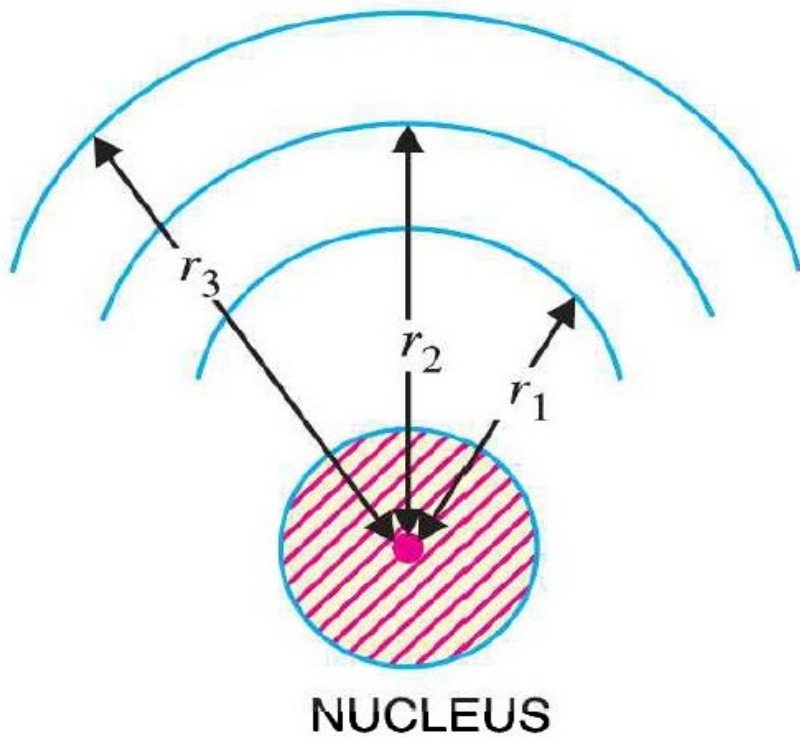
Germanium crystals



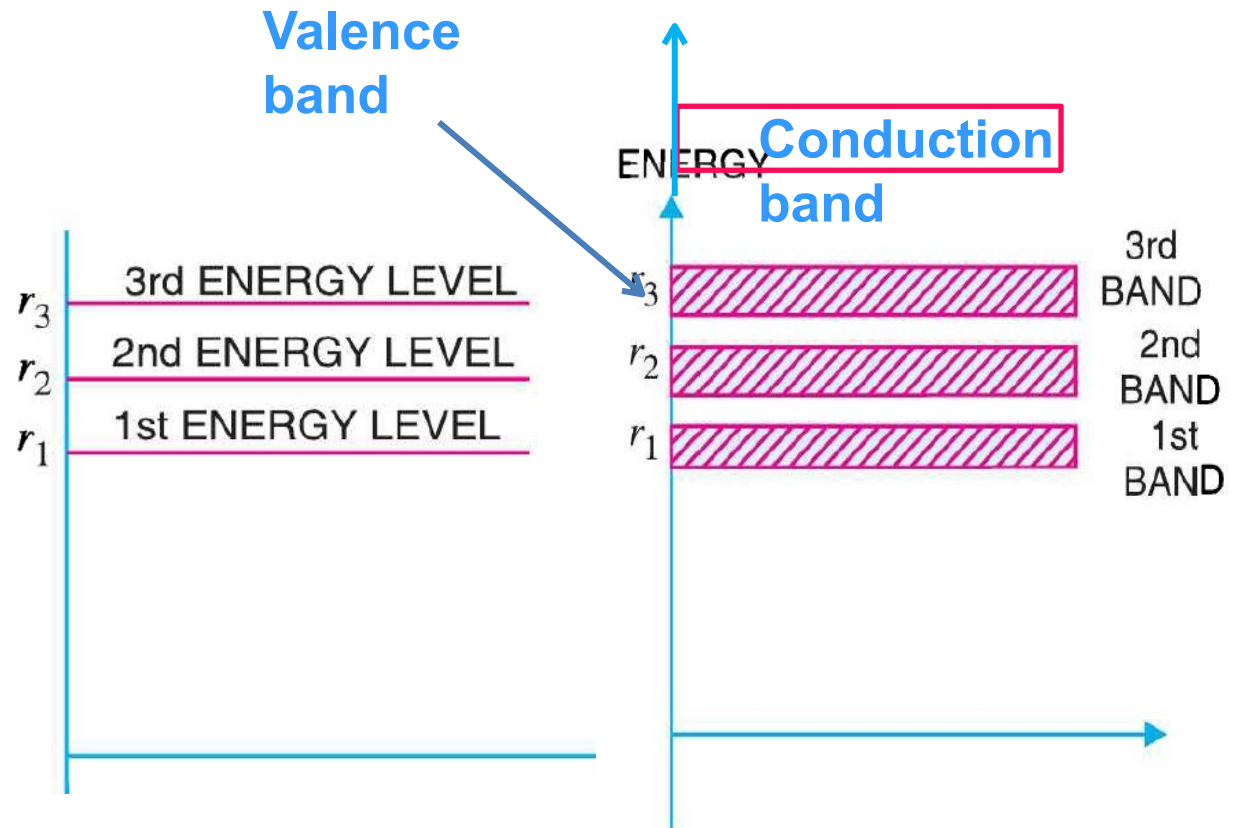
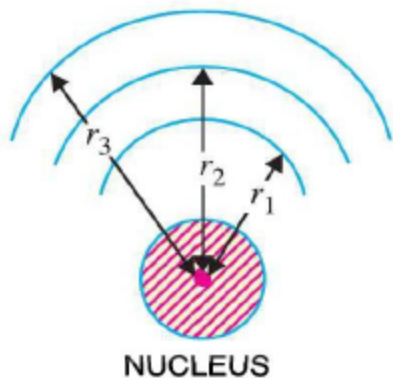
Energy level diagram



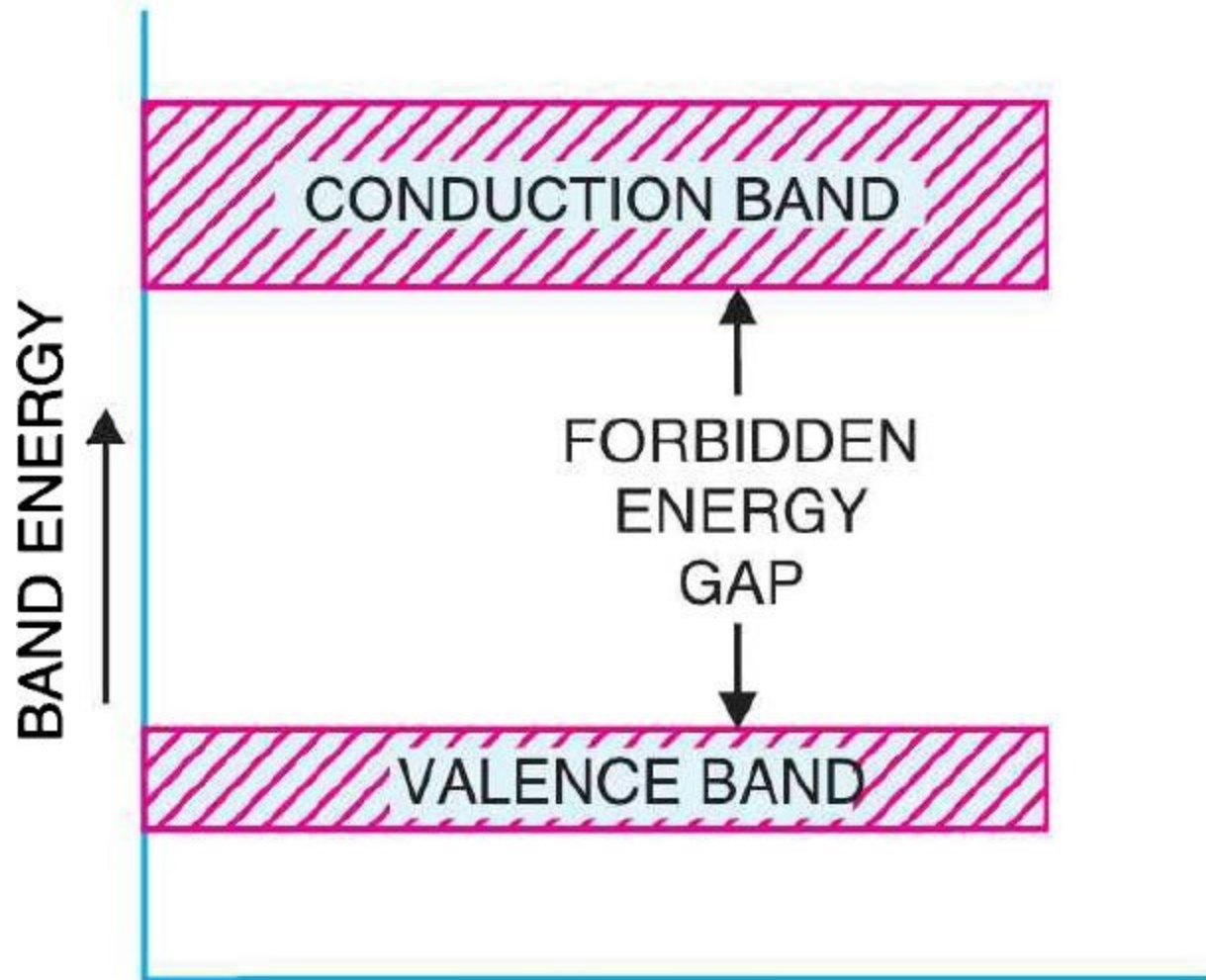
Energy level diagram



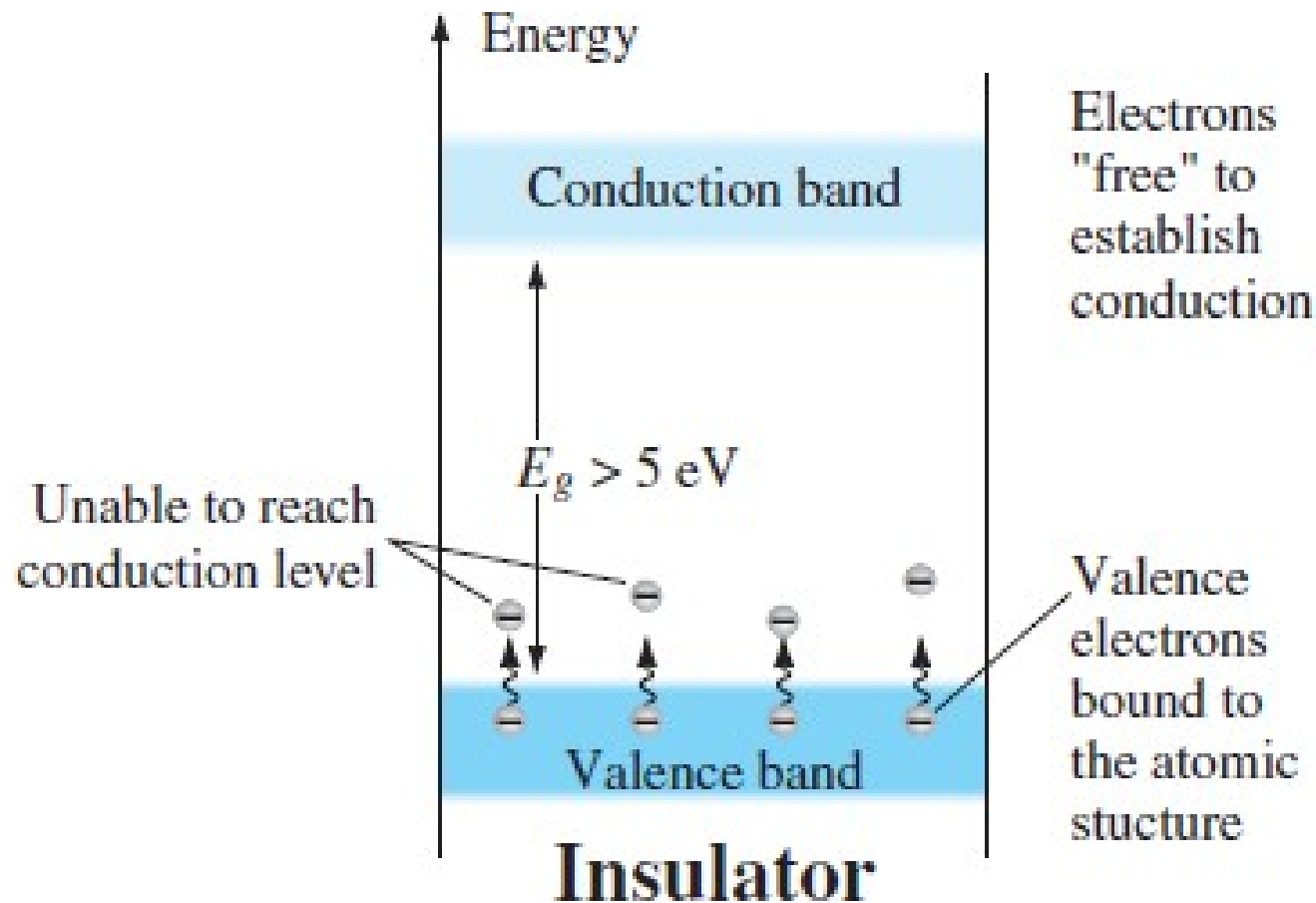
Energy band diagram



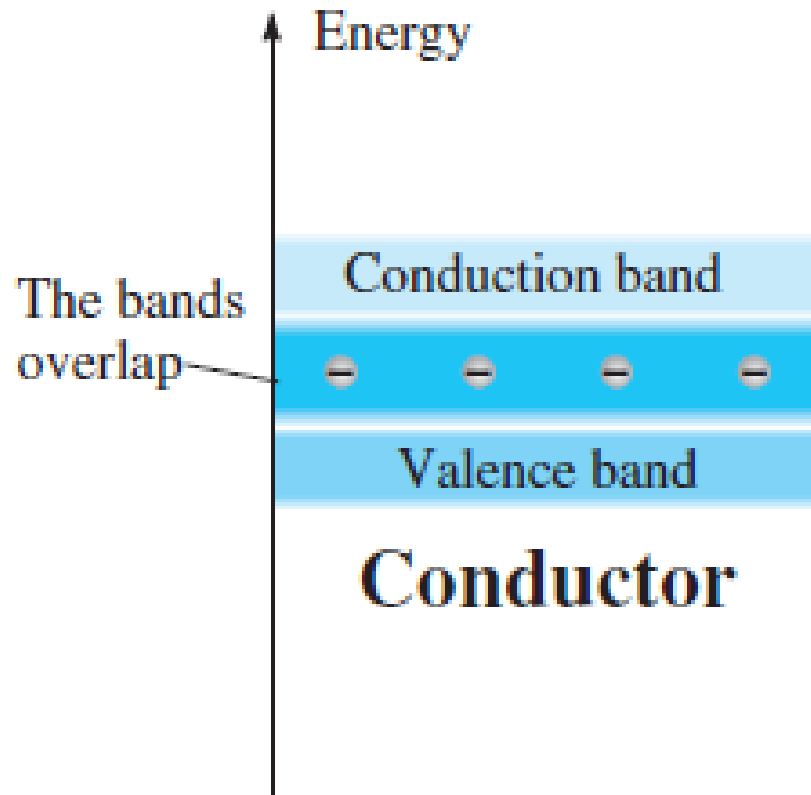
Energy band diagram



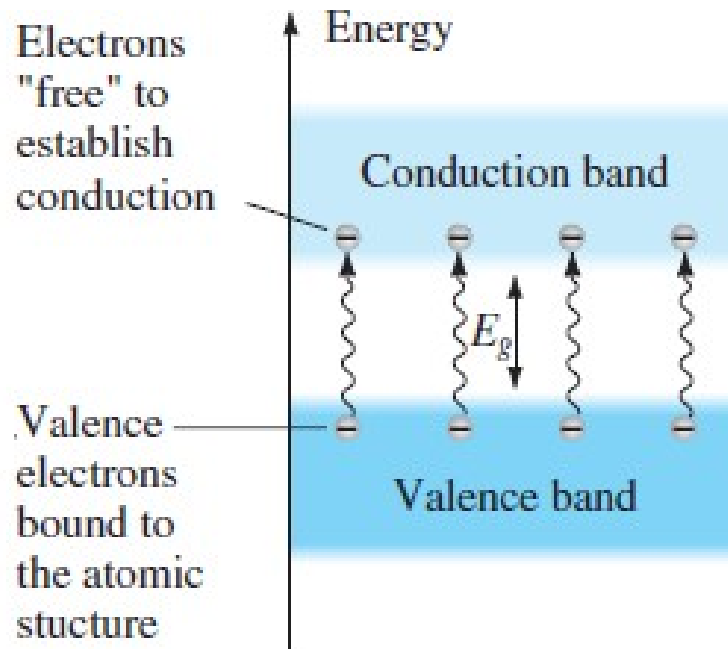
Classification of Solids based on Energy band diagram



Classification of Solids based on Energy band diagram



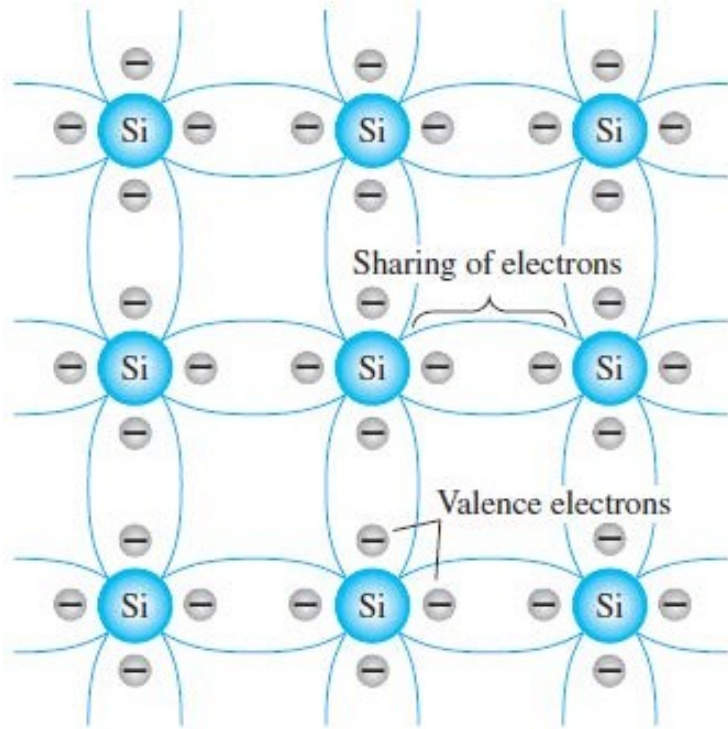
Classification of Solids based on Energy band diagram



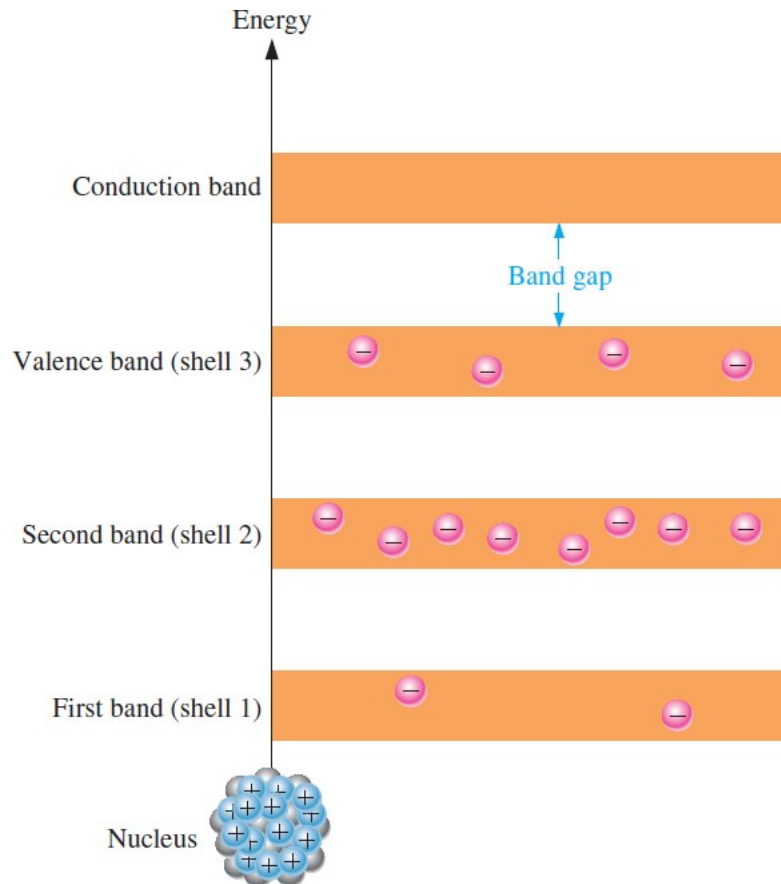
$$\begin{aligned} E_g &= 0.67 \text{ eV (Ge)} \\ E_g &= 1.1 \text{ eV (Si)} \\ E_g &= 1.43 \text{ eV (GaAs)} \end{aligned}$$

Semiconductor

Intrinsic Semiconductors

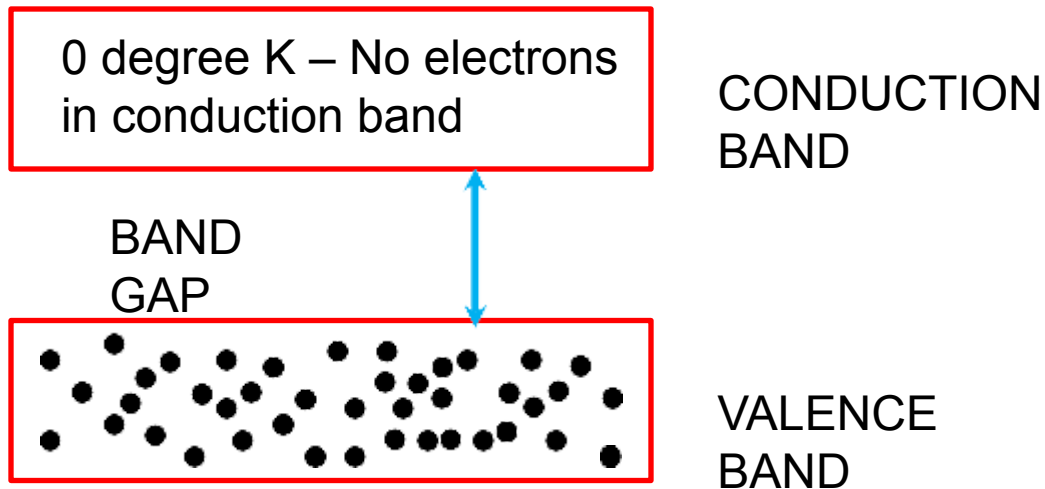


Intrinsic Semiconductors



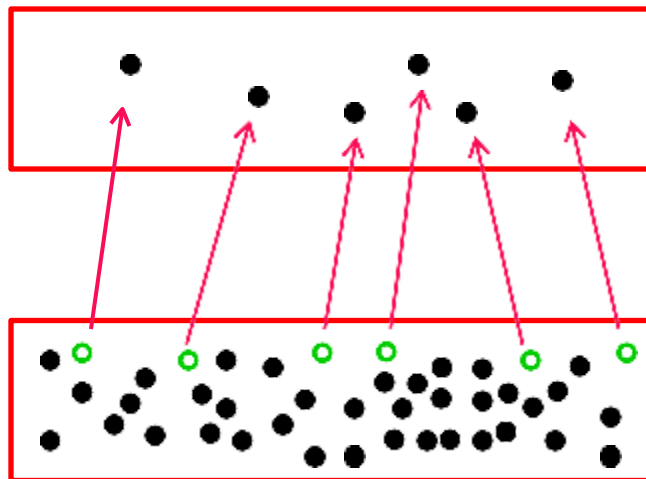
Energy band diagram for an atom in a pure (intrinsic) silicon crystal at its lowest energy state. There are no electrons in the conduction band at a temperature of 0 K.

Intrinsic Semiconductors



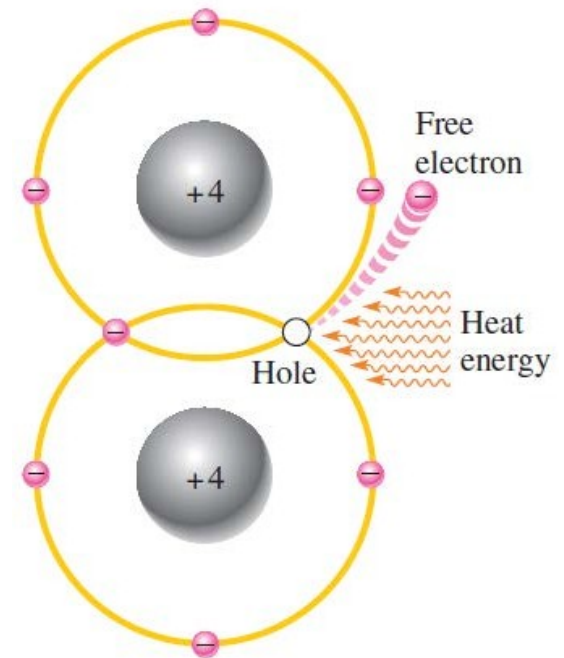
Intrinsic Semiconductors

300 degree K – Free
electrons in conduction
band



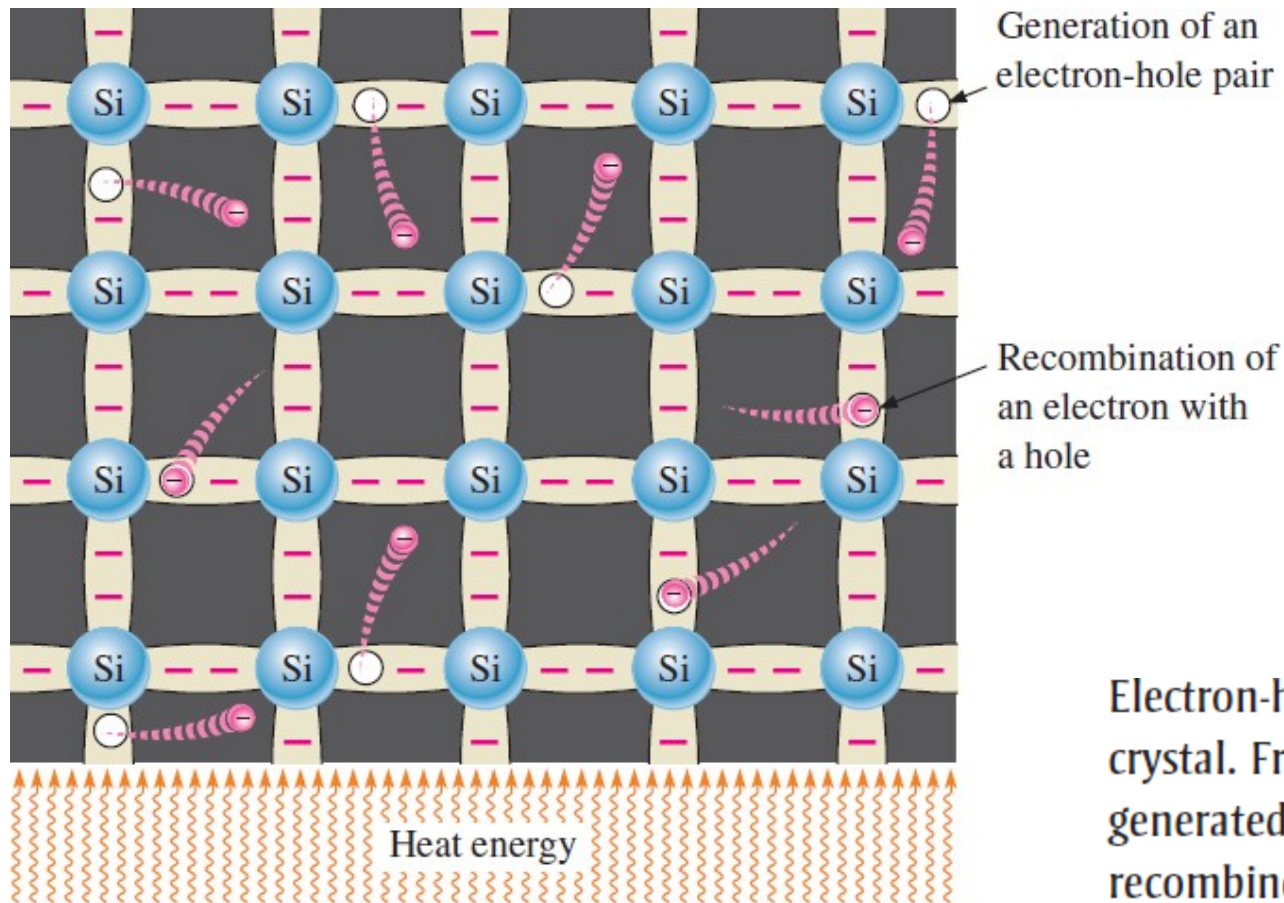
CONDUCTION
BAND

VALENCE
BAND



Creation of electron-hole pairs in a
silicon crystal. Electrons in the con-
duction band are free electrons.

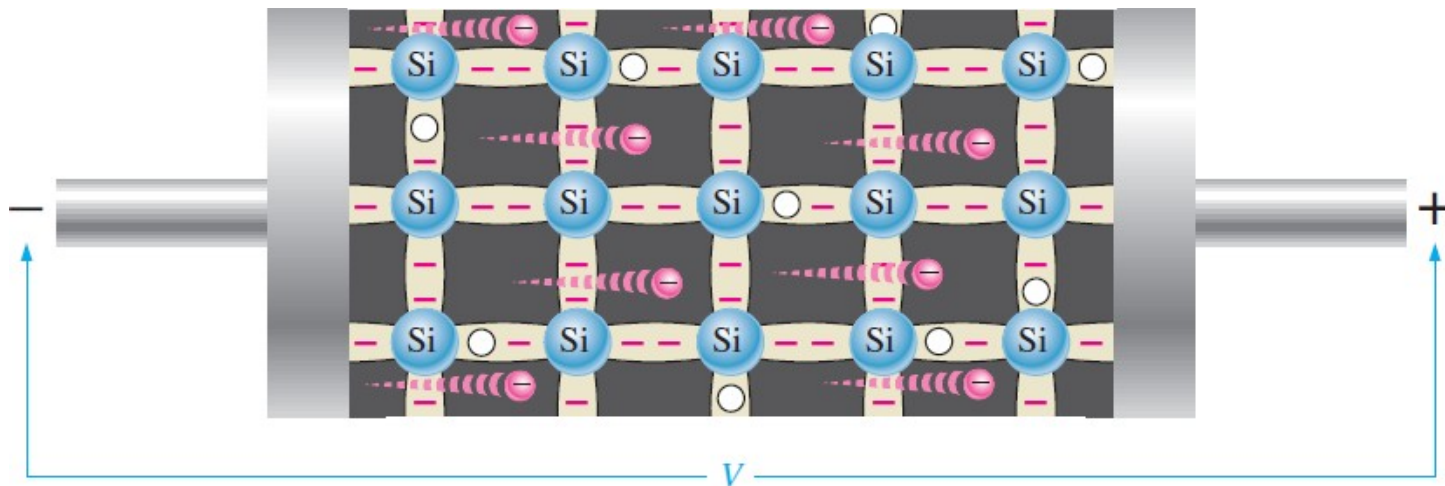
Intrinsic Semiconductors



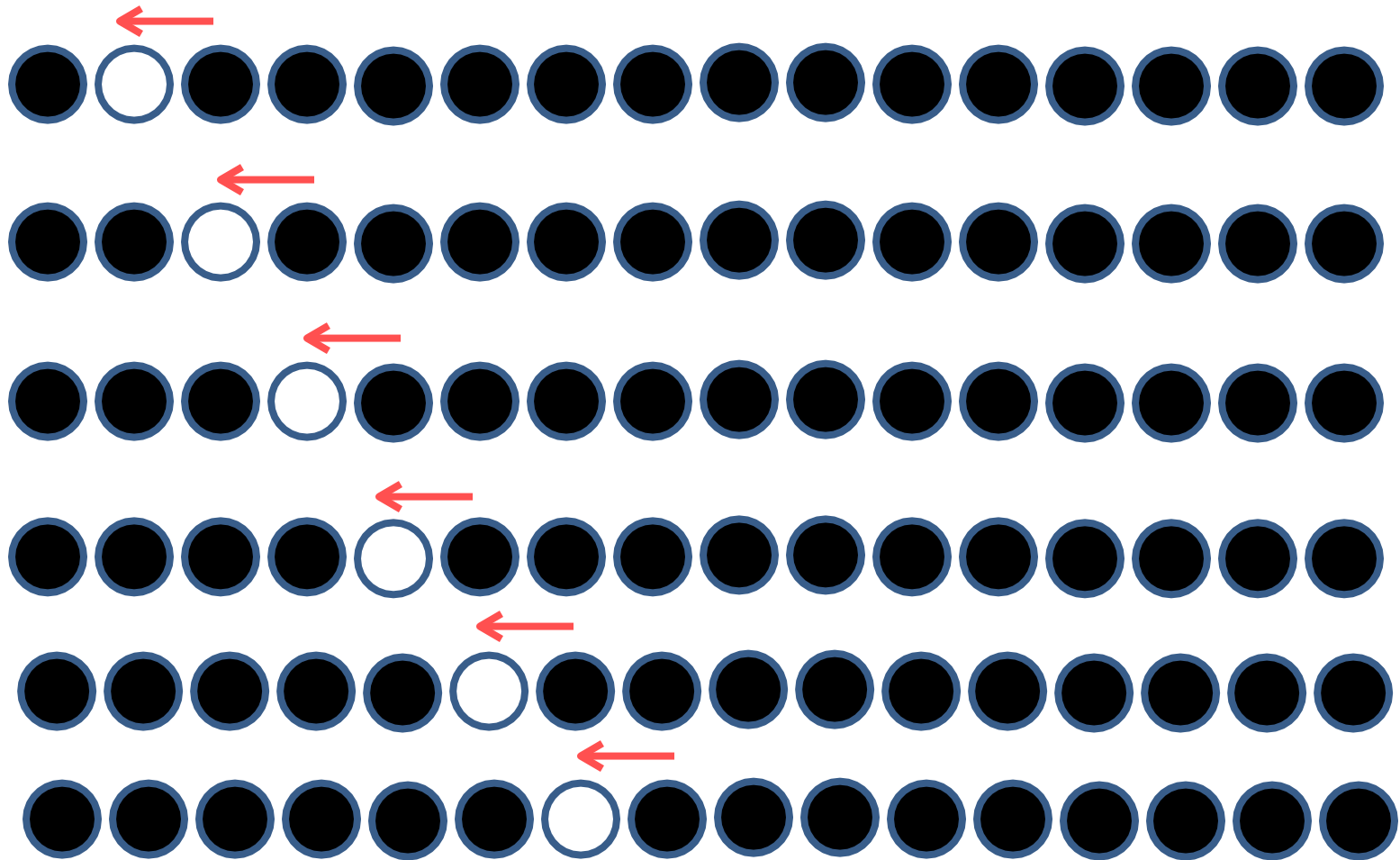
Electron-hole pairs in a silicon crystal. Free electrons are being generated continuously while some recombine with holes.

Intrinsic Semiconductors

Electron current in intrinsic silicon is produced by the movement of thermally generated free electrons.

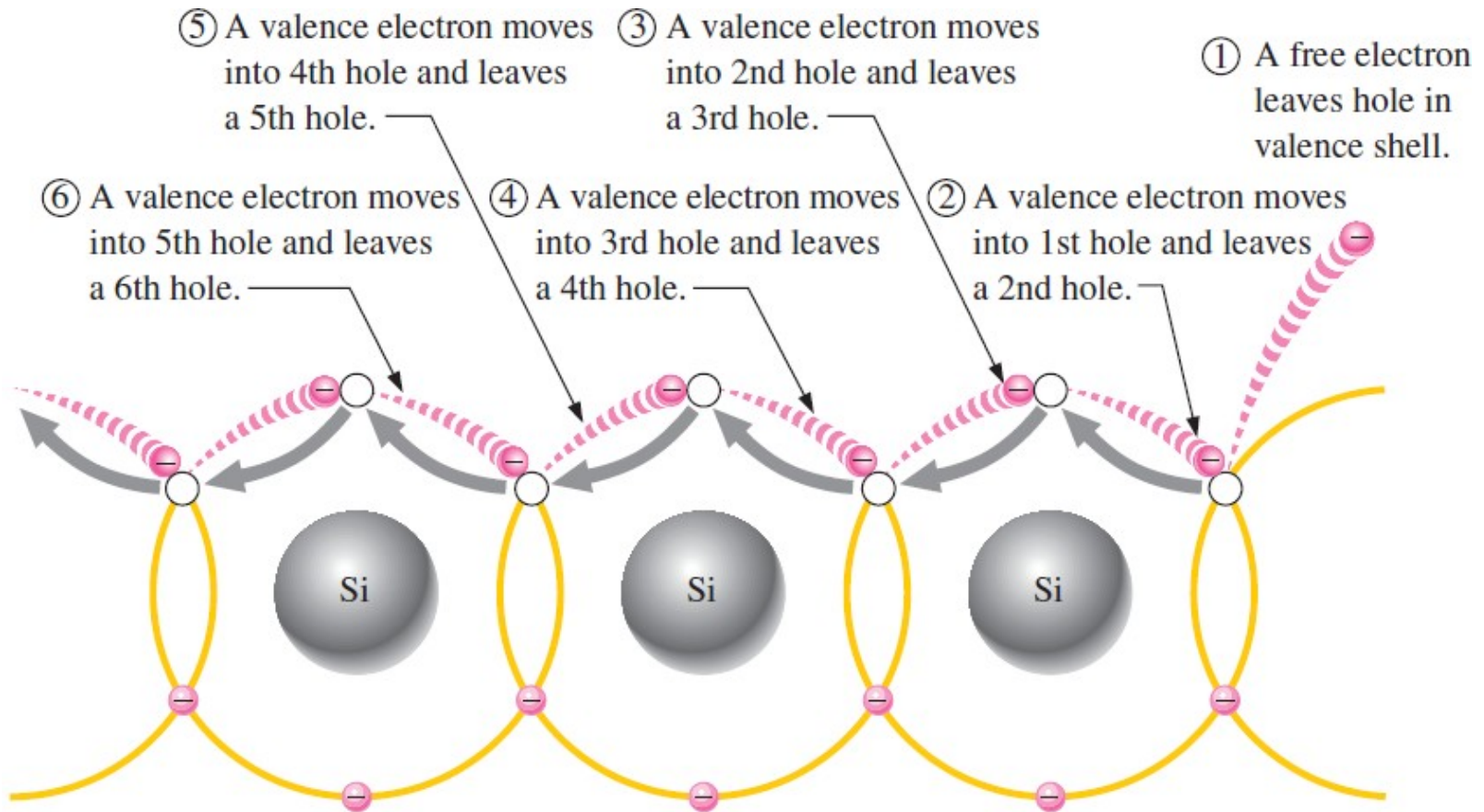


Intrinsic Semiconductors



The mechanism by which a hole contributes to the conductivity.

Intrinsic Semiconductors



When a valence electron moves left to right to fill a hole while leaving another hole behind, the hole has effectively moved from right to left. Gray arrows indicate effective movement of a hole.

Intrinsic Semiconductors

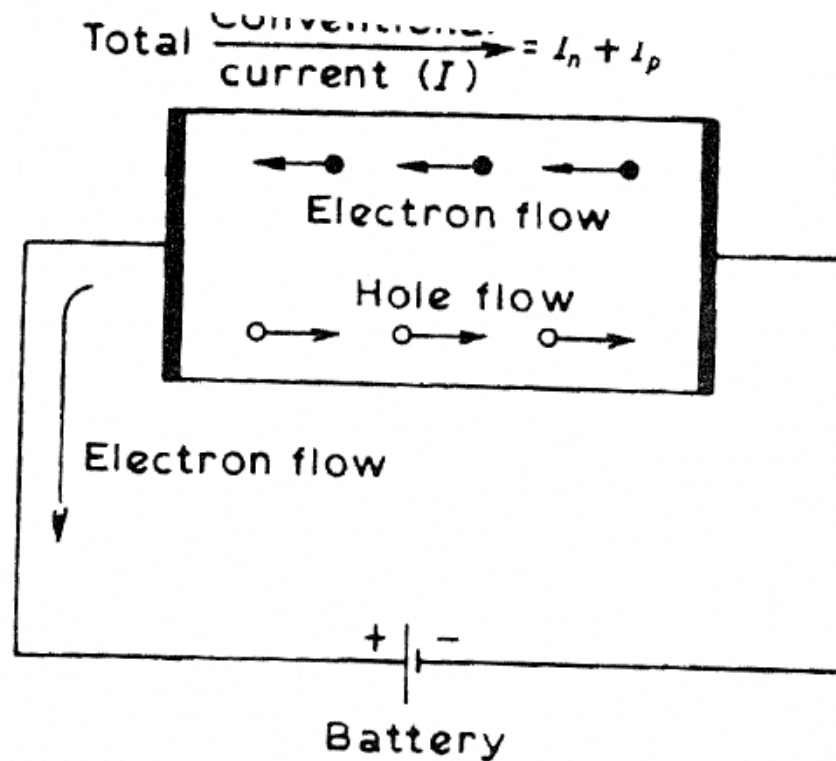


Fig. 3.10 Conduction of current in an intrinsic semiconductor

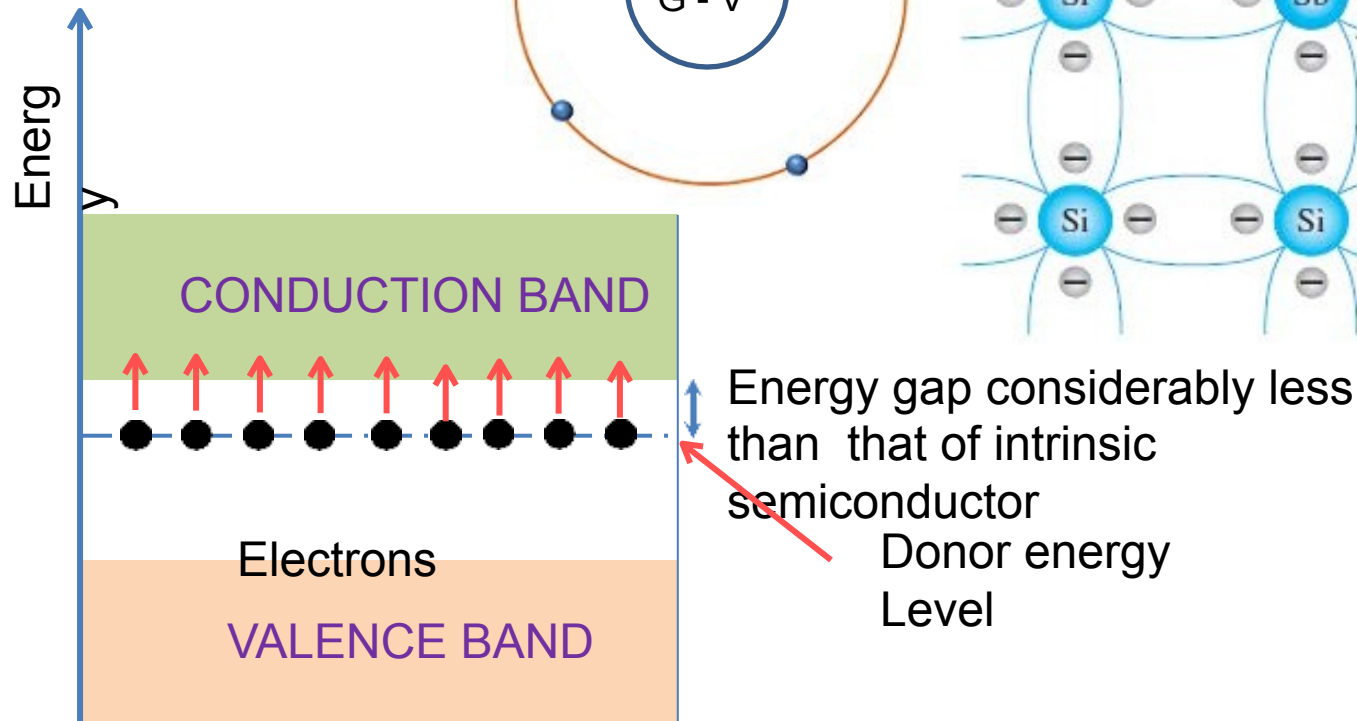
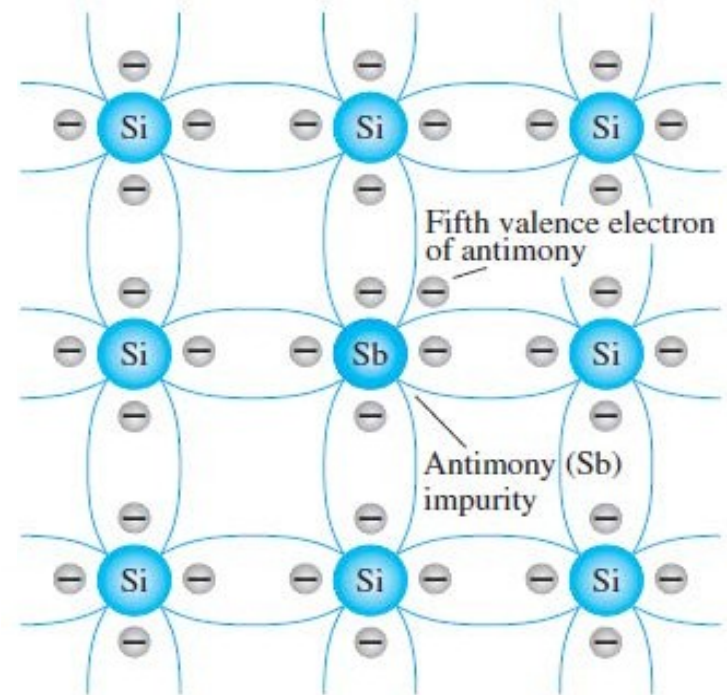
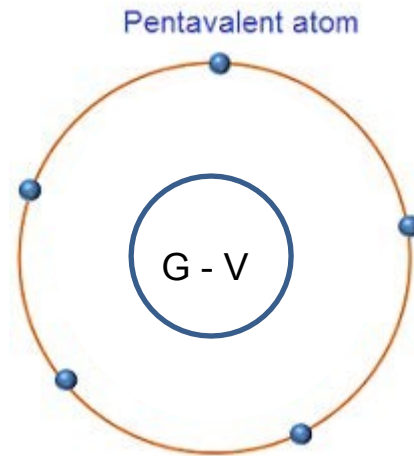
Semiconductor Materials

Group No.→	III	IV	V
	B (5)	C (6)	
	Al (13)	Si (14)	P (15)
	Ga (31)	Ge (32)	As (33)
	In (49)		Sb (51)

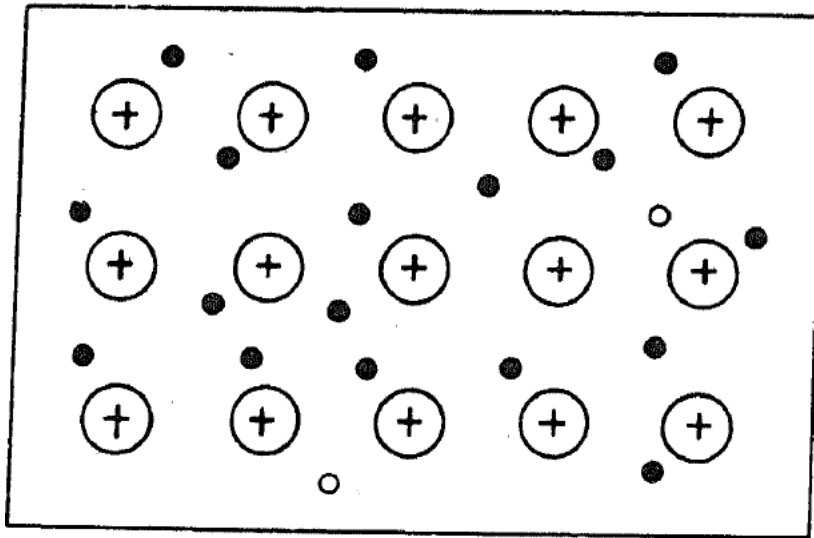
N-type Semiconductors (Extrinsic)

Pentavalent impurity - donor
(Group V elements)

1. Antimony (Sb)
2. Arsenic (As)
3. Phosphorous (P)
4. Bismuth (Bi)



N-type Semiconductors

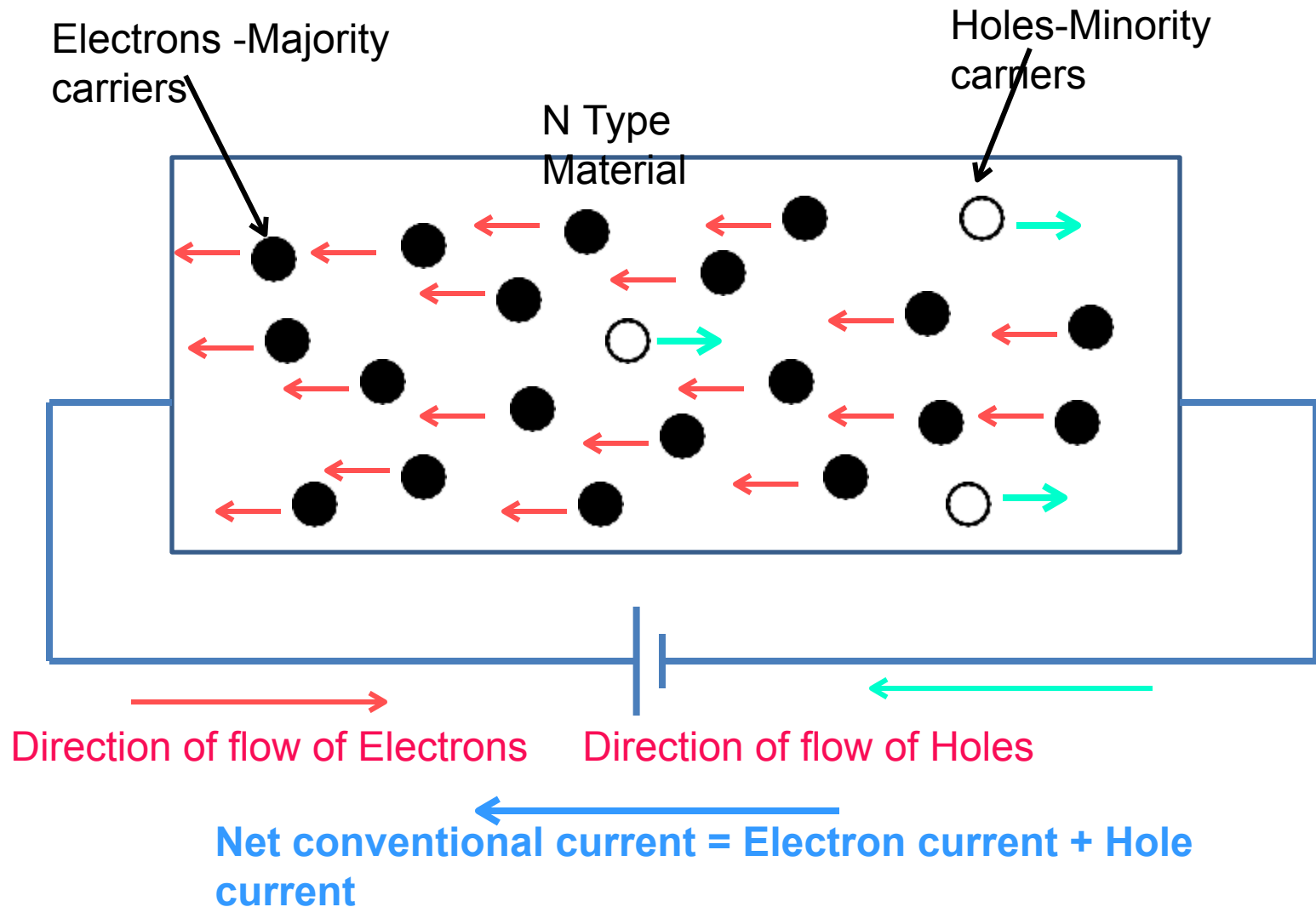


Legends :

- Free electron
(negative charge)
- Hole (positive charge)
- ⊕ Immobile ion
(positive charge)

Representation of an *N*-type semiconductor

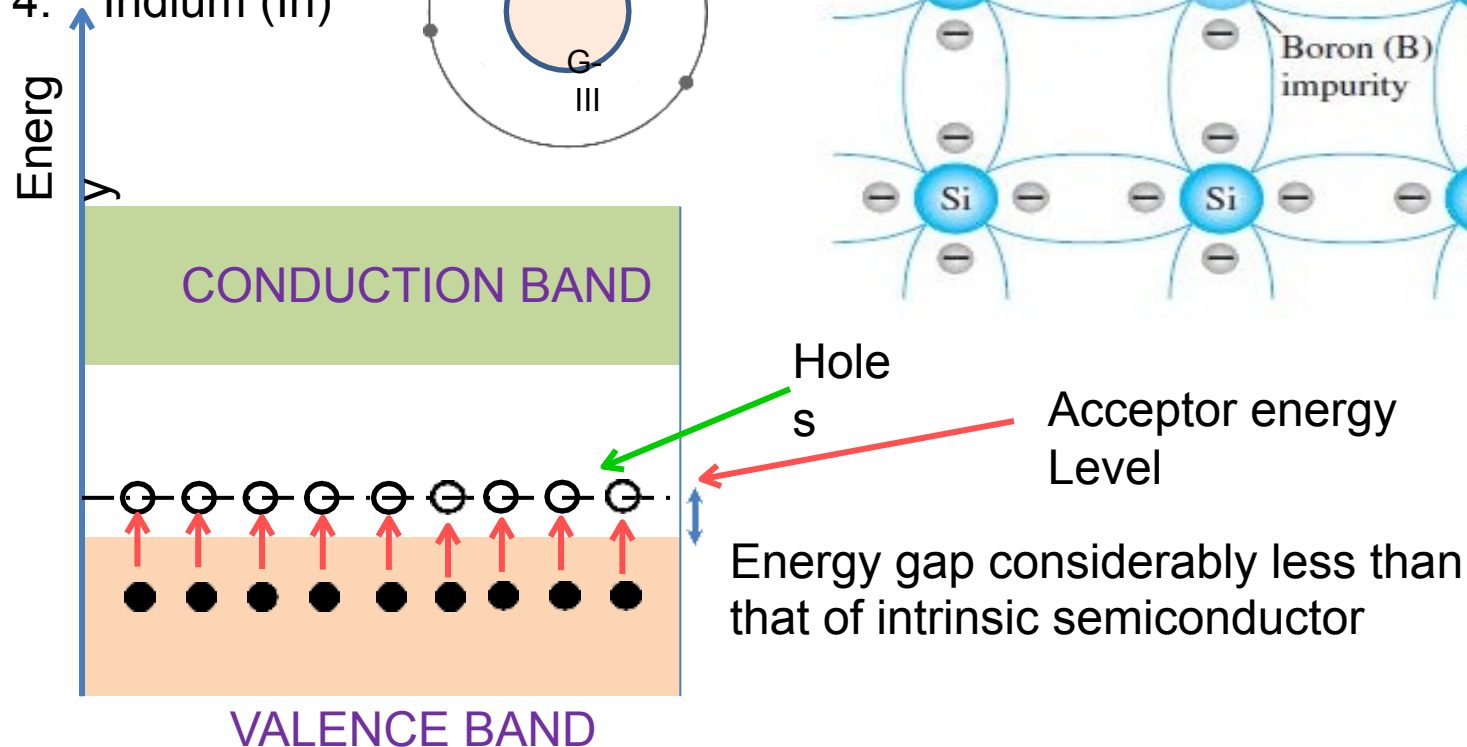
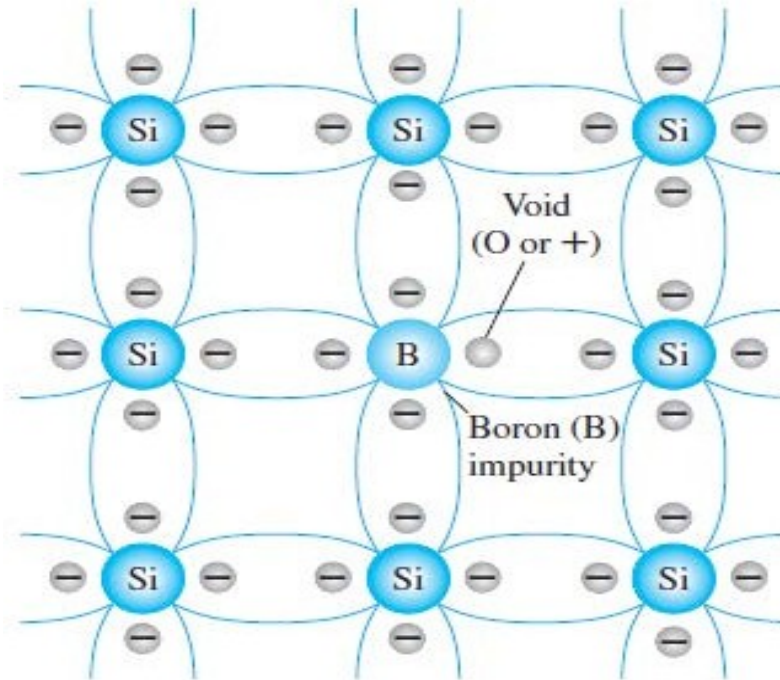
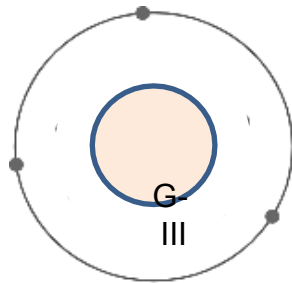
Current flow in N Type semiconductor



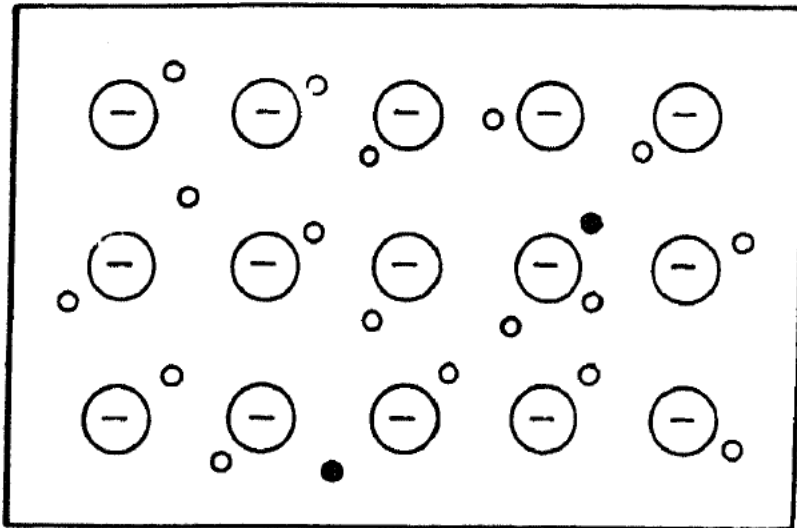
P-type Semiconductors (Extrinsic)

*Trivalent impurity -
acceptor (Group III
elements)*

1. Boron (B)
2. Aluminium (Al)
3. Gallium (Ga)
4. Indium (In)



P-type Semiconductors

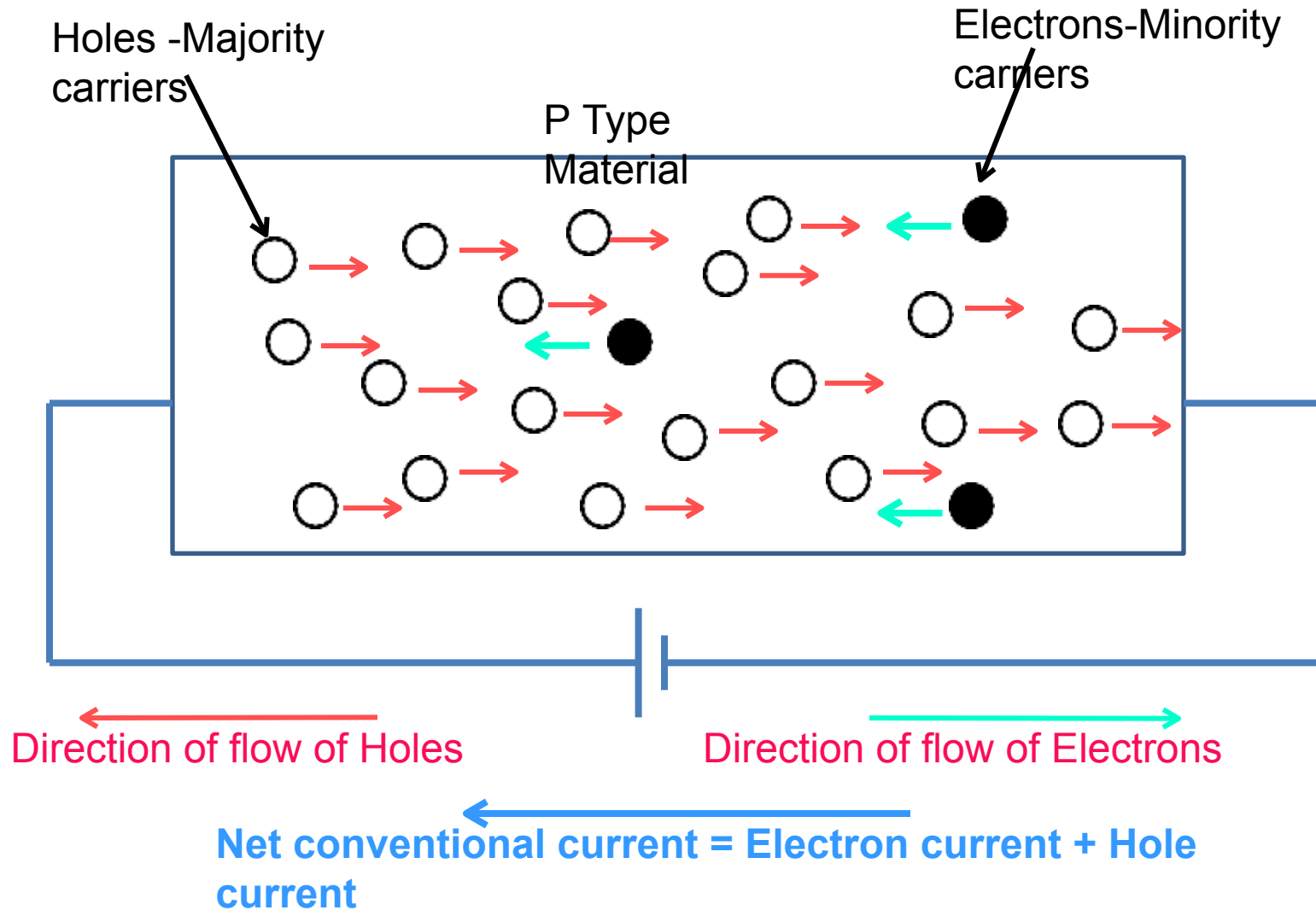


Legends:

- Hole (positive charge)
- Electron (negative charge)
- ⊖ Immobile ion (negative charge)

Representation of a *P*-type semiconductor

Current flow in P Type semiconductor



Topics in syllabus(for exam) starts from
slide number 34

Formation of P-N Junction

we use diffused semiconductor materials to make a P-N junction.

The processes that follow after forming a P-N junction are of two types – diffusion and drift. There is a difference in the concentration of holes and electrons at the two sides of a junction. The holes from the p-side diffuse to the n-side, and the electrons from the n-side diffuse to the p-side. These give rise to a diffusion current across the junction.

when an electron diffuses from the n-side to the p-side, an ionised donor is left behind on the n-side, which is immobile.

As the process goes on, a layer of positive charge is developed on the n-side of the junction.

Similarly, when a hole goes from the p-side to the n-side, an ionized acceptor is left behind on the p-side, resulting in the formation of a layer of negative charges in the p-side of the junction.

This region of positive charge and negative charge on either side of the junction is termed as the depletion region.

Due to this positive space charge region on either side of the junction, an electric field with the direction from a positive charge towards the negative charge is developed.

Due to this electric field, an electron on the p-side of the junction moves to the n-side of the junction.

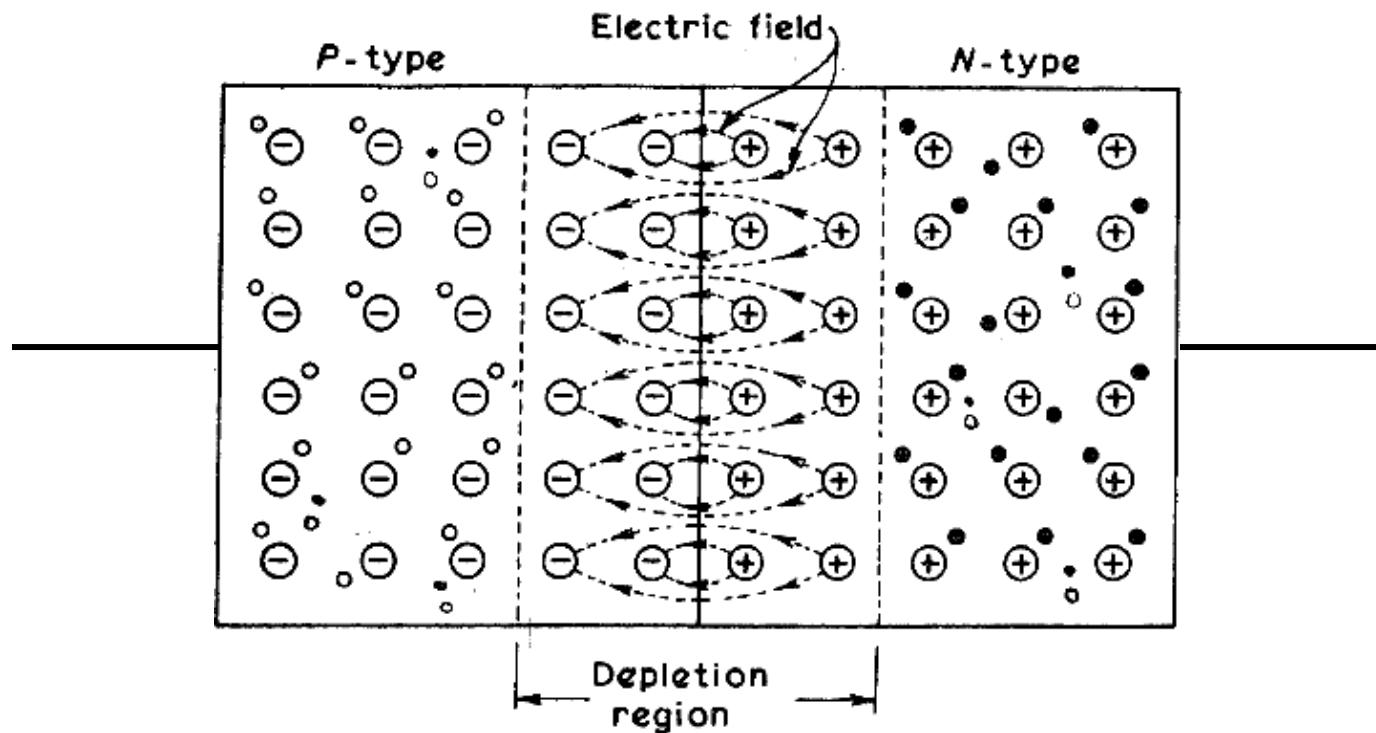
This motion is termed the drift. Here, we see that the direction of the drift current is opposite to that of the diffusion current.

PN Junction Diode

- (i) Holes from the P region diffuse into the N region. They then combine with the free electrons in the N region.
- (ii) Free electrons from the N region diffuse into the P region. These electrons combine with the holes.
- (iii) The diffusion of holes (from P region to N region) and electrons (from N region to P region) takes place because they move haphazardly due to thermal energy, and also because there is a difference in their concentrations in the two regions. The P region has more holes and the N region has more free electrons.
- (iv) The diffusion of holes and free electrons across the junction occurs for a very short time.

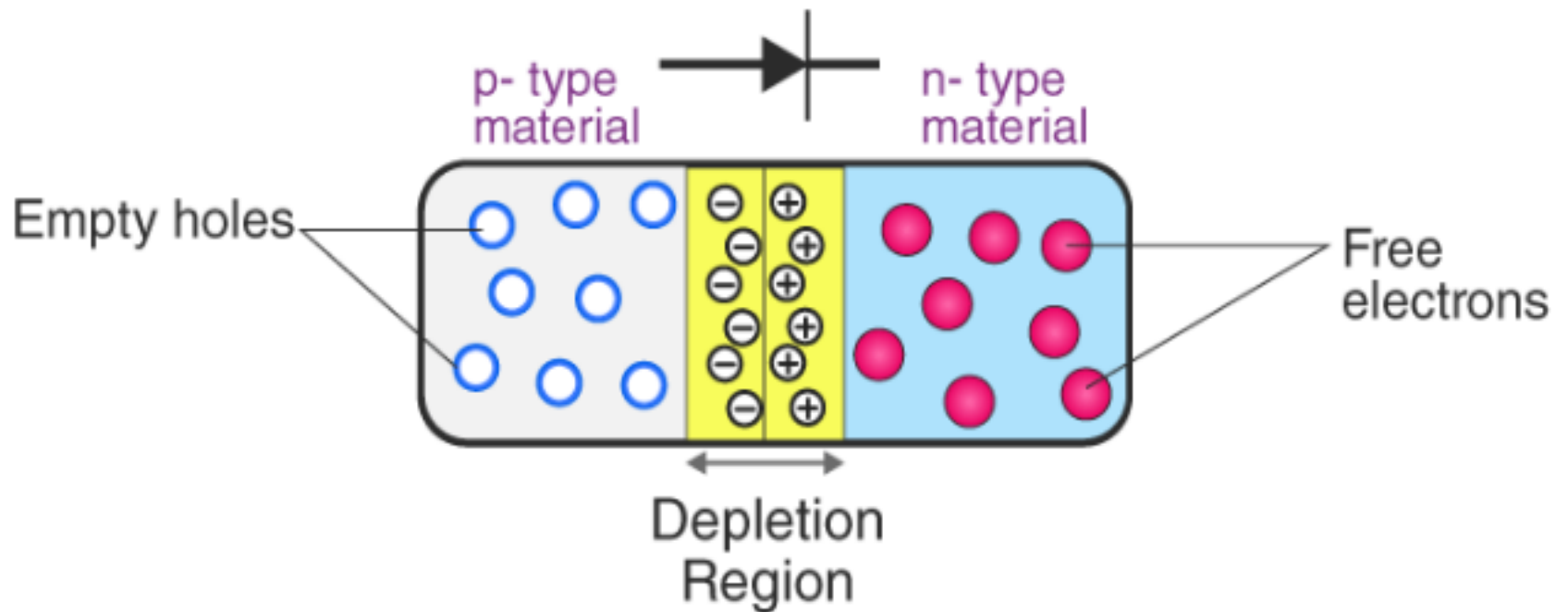
- (v) Some of the holes in the P region and some of the free electrons in N region diffuse towards each other and recombine. Each recombination eliminates a hole and a free electron. In this process, the negative acceptor ions in the P region and positive donor ions in the N region in the immediate neighbourhood of the junction are left uncompensated.
- (vi) The region containing the uncompensated acceptor and donor ions is called *depletion region*. That is, there is a depletion of mobile charges (holes and free electrons) in this region. Since this region has immobile (fixed) ions which are electrically charged it is also referred to as the *space-charge region*.

For a silicon PN -junction, the barrier potential is about 0.7 V, whereas for a germanium PN -junction it is approximately 0.3 V.



- (vii) The barrier discourages the diffusion of majority carriers across the junction. But what happens to the minority carriers? There are a few free electrons in the *P* region and a few holes in the *N* region. The barrier helps these minority carriers to drift across the junction.

UNBIASED P-N JUNCTION



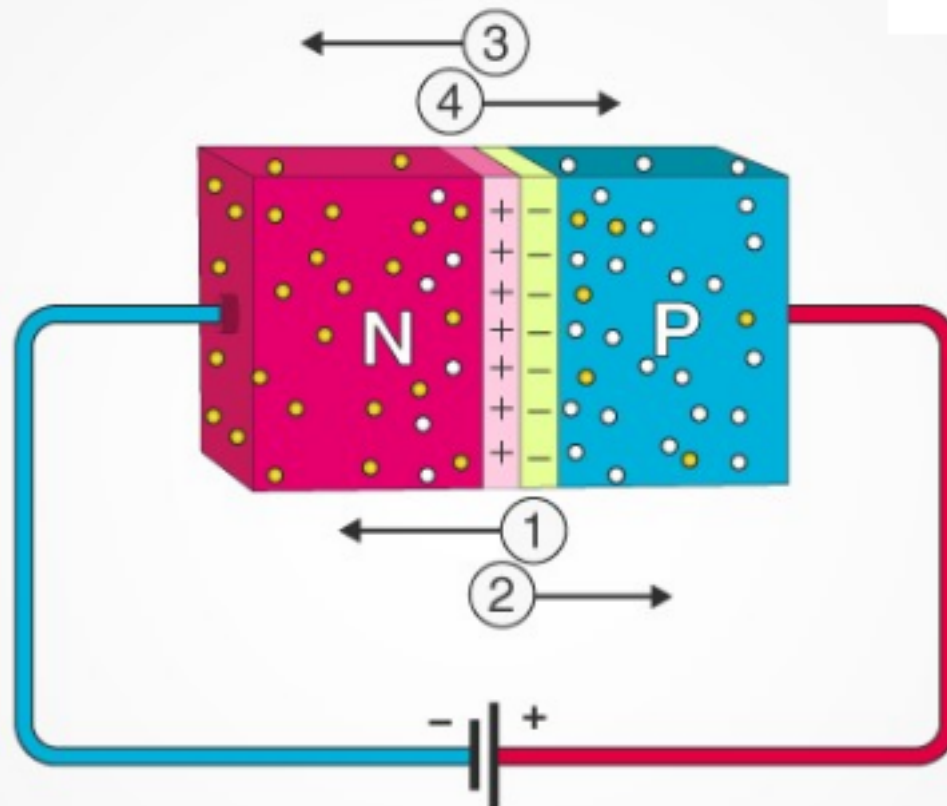
Biasing Conditions for the P-N Junction Diode

Zero bias: No external voltage is applied to the P-N junction diode.

Forward bias: The positive terminal of the voltage potential is connected to the p-type while the negative terminal is connected to the n-type.

Reverse bias: The negative terminal of the voltage potential is connected to the p-type and the positive is connected to the n-type.

FORWARD BIAS OF THE p-n JUNCTION

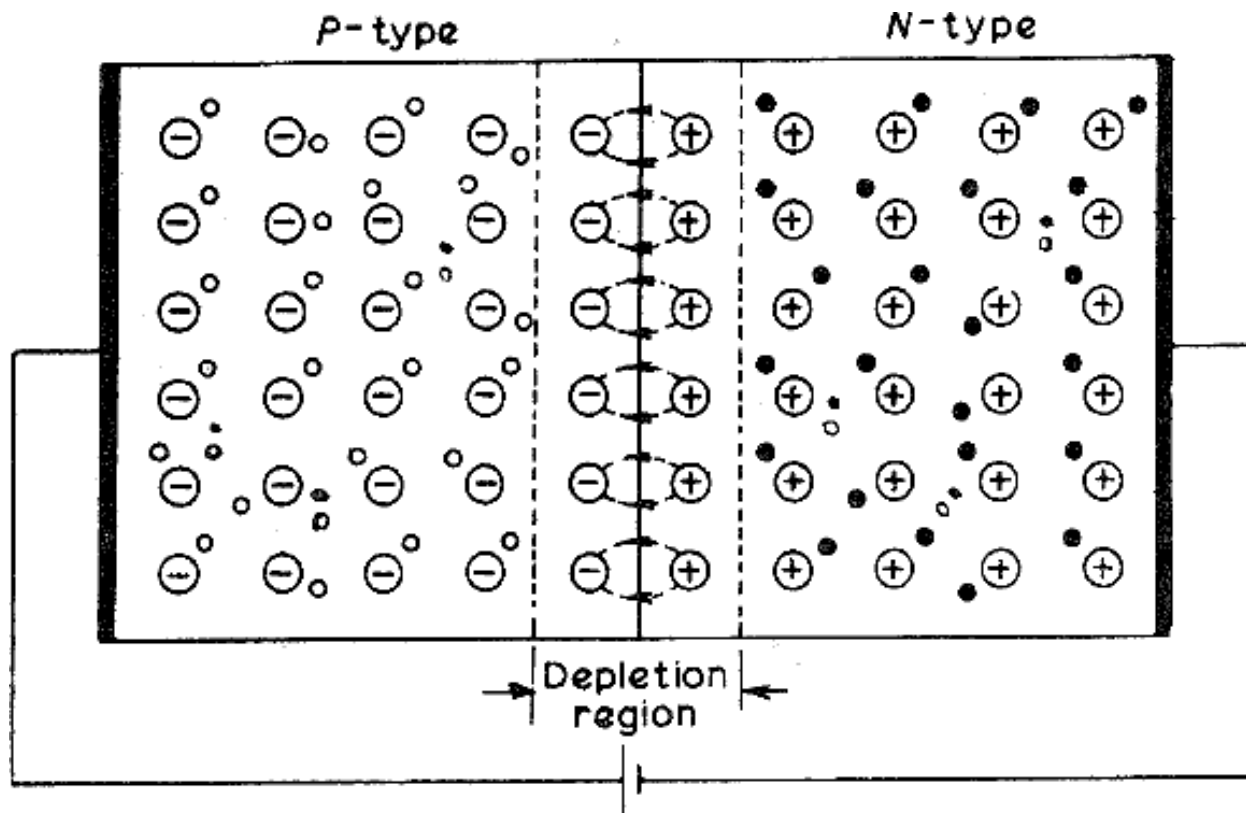


1 Battery induced electric field

2 Built-in electric field

3 Conventional current

4 Electron current



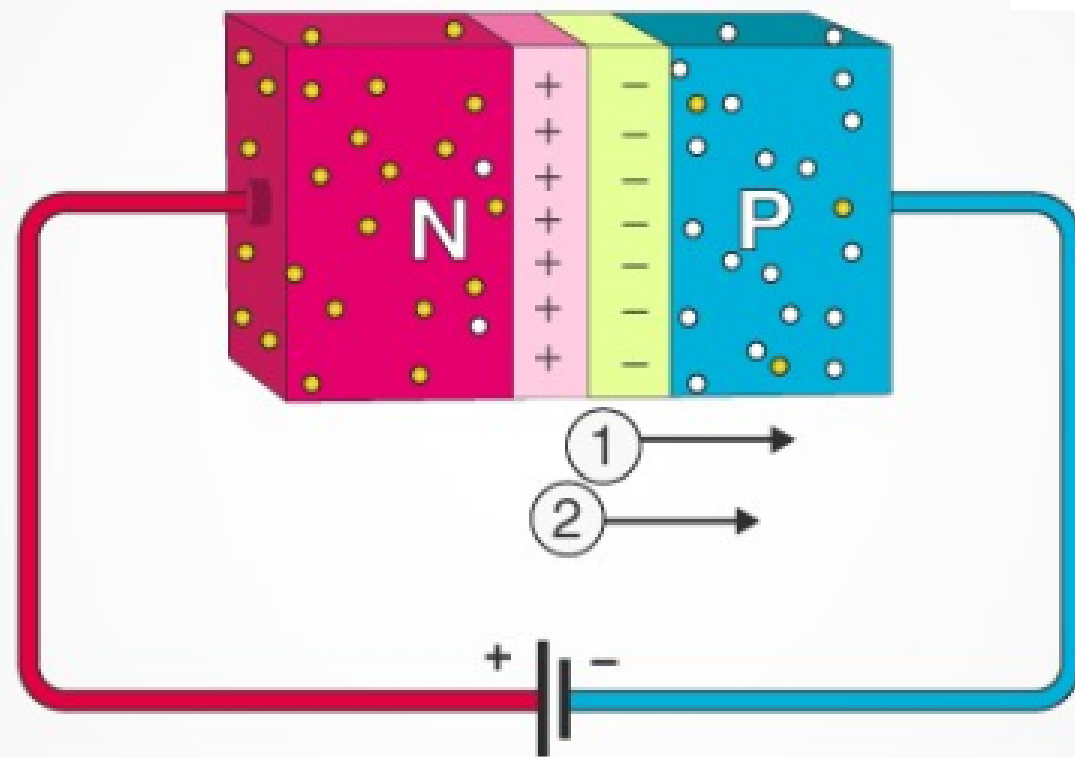
When the p-type is connected to the battery's positive terminal and the n-type to the negative terminal, then the P-N junction is said to be forward-biased. When the P-N junction is forward biased, the built-in electric field at the P-N junction and the applied electric field are in opposite directions. When both the electric fields add up, the resultant electric field has a magnitude lesser than the built-in electric field. This results in a less resistive and thinner depletion region. The depletion region's resistance becomes negligible when the applied voltage is large.

When the diode is in forward bias, the current increases slowly, and the curve obtained is non-linear as the voltage applied to the diode overcomes the potential barrier.

Once the diode overcomes the potential barrier, the diode behaves normally, and the curve rises sharply as the external voltage increases, and the curve obtained is linear.

REVERSE BIAS OF THE p-n JUNCTION

No current flow



1 Battery induced electric field

2 Built-in electric field

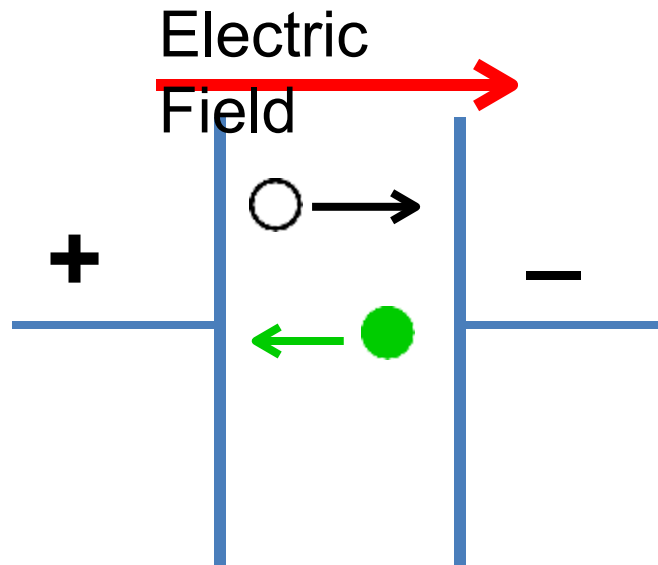
When the P-N junction diode is in reverse bias condition, the p-type sc is connected to the negative terminal while the n-type sc is connected to the positive terminal of the external voltage.

This results in an increase in the potential barrier. Reverse saturation current flows in the beginning as minority carriers are present in the junction.

When the applied voltage is increased, the minority charges will have increased kinetic energy which affects the majority charges.

This is the stage when the diode breaks down. This may also destroy the diode.

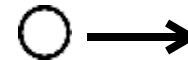
PN Junction Diode



Holes move along the electric field

Electrons move opposite to the electric field

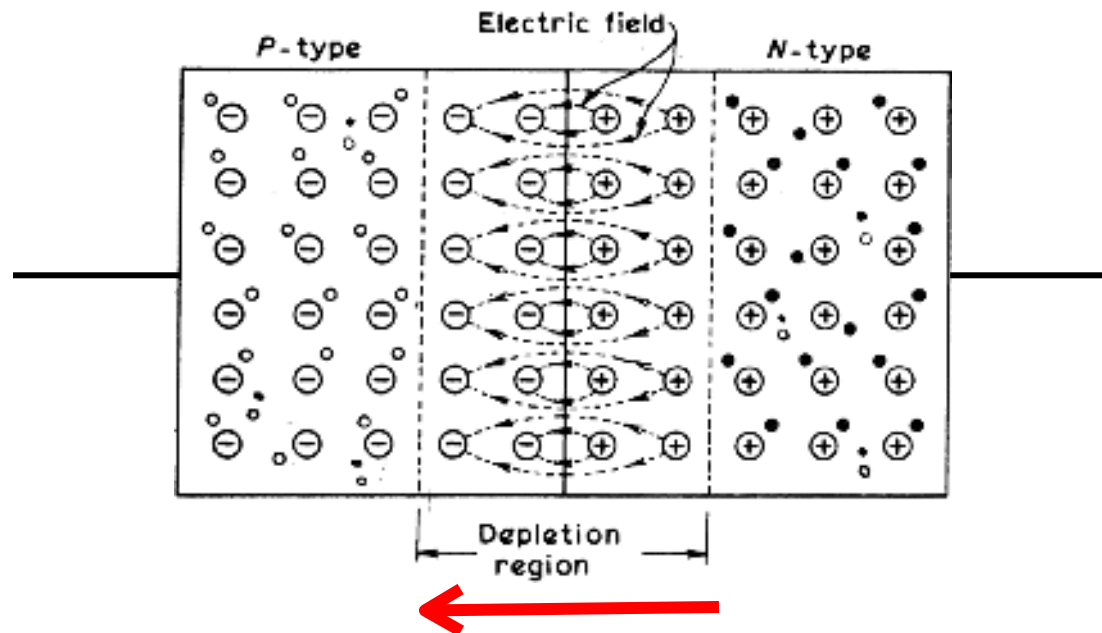
Electric Field



Electrons cannot move along the electric field Holes cannot
move opposite to the electric field



PN Junction Diode



Majority carriers cannot cross the junction unless external energy is supplied



Minority carriers can cross the junction easily

In reverse bias condition , the p-type of the pn junction is connected to the negative terminal and n-type is connected to the positive terminal of the external voltage.

This results in increased potential barrier at the junction.

Hence, the junction resistance becomes very high and as a result practically no current flows through the circuit.

However, a very small current of the order of μA , flows through the circuit in practice.

This is known as reverse saturation current(I_s) and it is due to the minority carriers in the junction.

As we already know, there are few free electrons in p-type material and few holes in n-type material. These free electrons in p-type and holes in n-type are called minority carriers .

The reverse bias applied to the pn junction acts as forward bias to there minority carriers and hence, small current flows in the reverse direction.

If the applied reverse voltage is increased continuously, the kinetic energy of the minority carriers may become high enough to knock out electrons from the semiconductor atom.

At this stage breakdown of the junction may occur. This is characterized by a sudden increase of reverse current and a sudden fall of the resistance of barrier region. This may destroy the junction permanently.

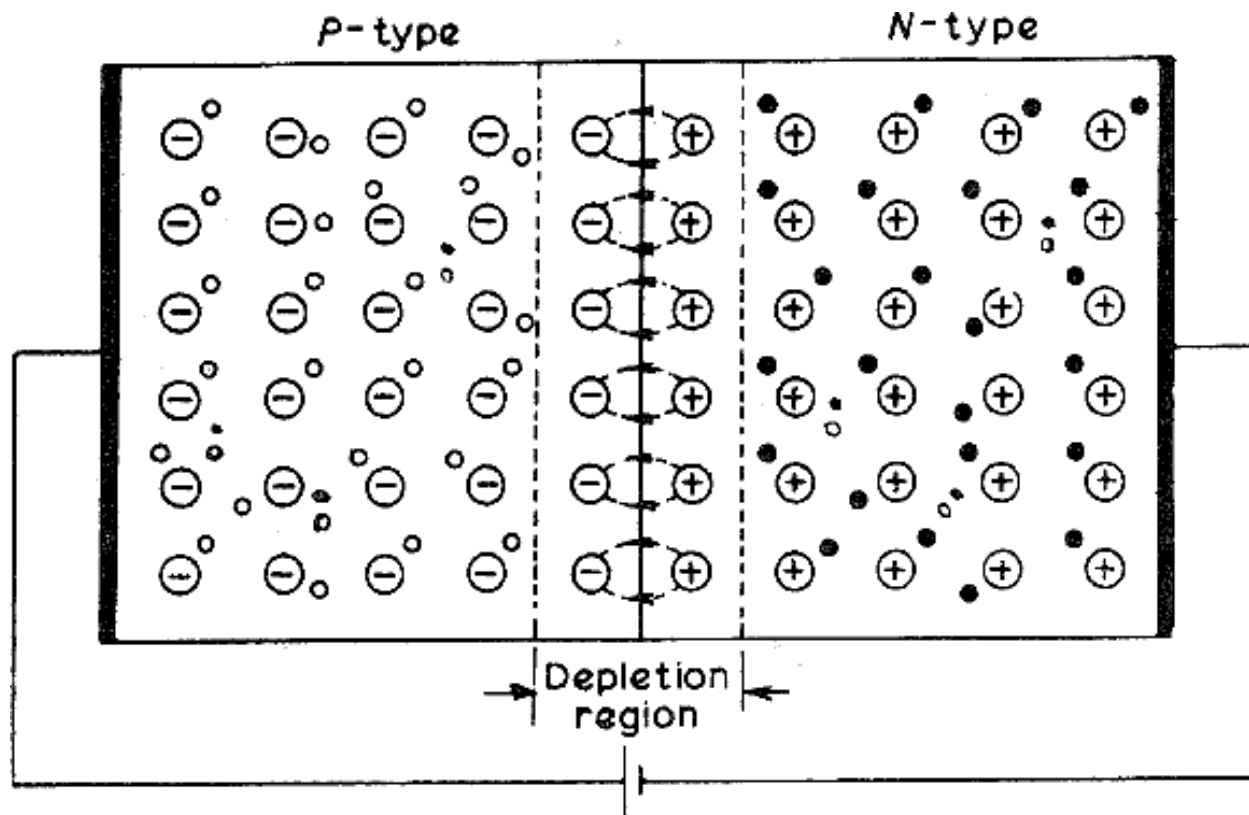
PN Junction Diode

The drift of minority carriers across the junction is counterbalanced by the diffusion of the same number of majority carriers across the junction. These few majority carriers have sufficiently high kinetic energy* to overcome the barrier and cross the junction. In fact, the barrier height adjusts itself so that the flow of minority carriers is exactly balanced by the flow of majority carriers across the junction.

Thus, we conclude that a barrier voltage is developed across the PN-junction even if no external battery is connected.

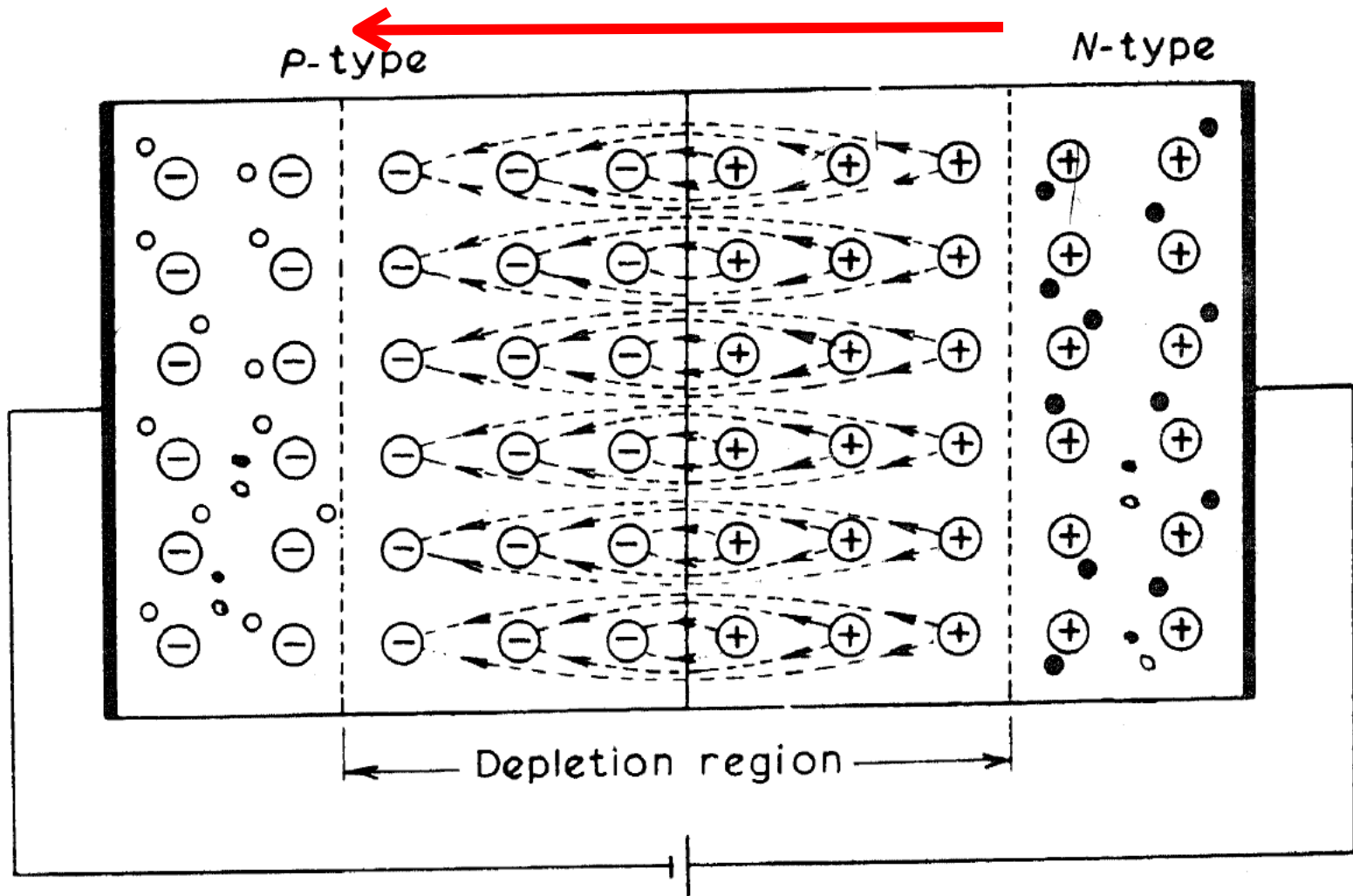
PN Junction Diode

PN-Junction with Forward Bias

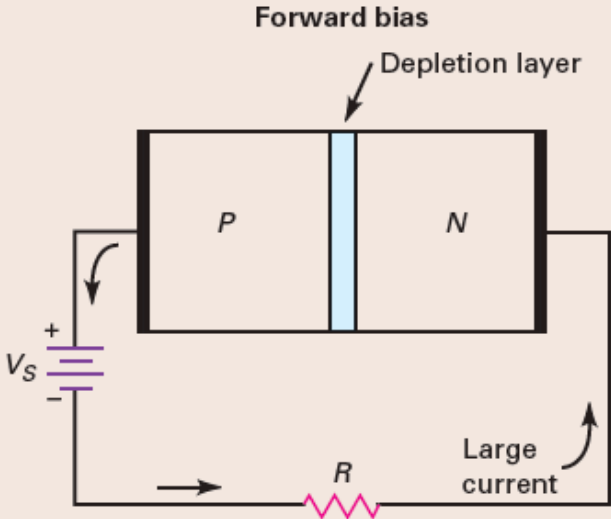


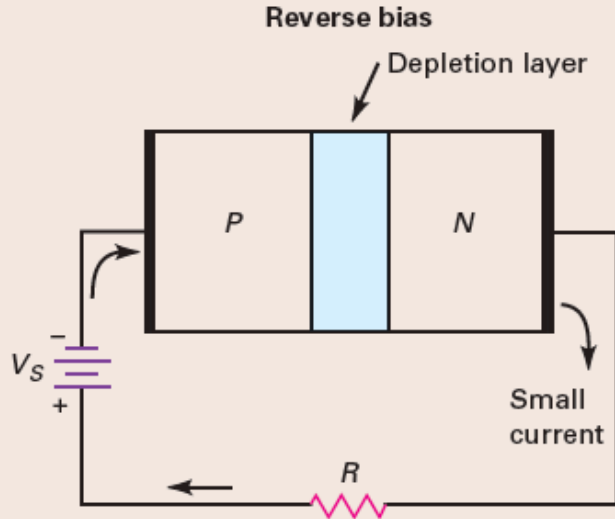
PN Junction Diode

PN-Junction with Reverse Bias

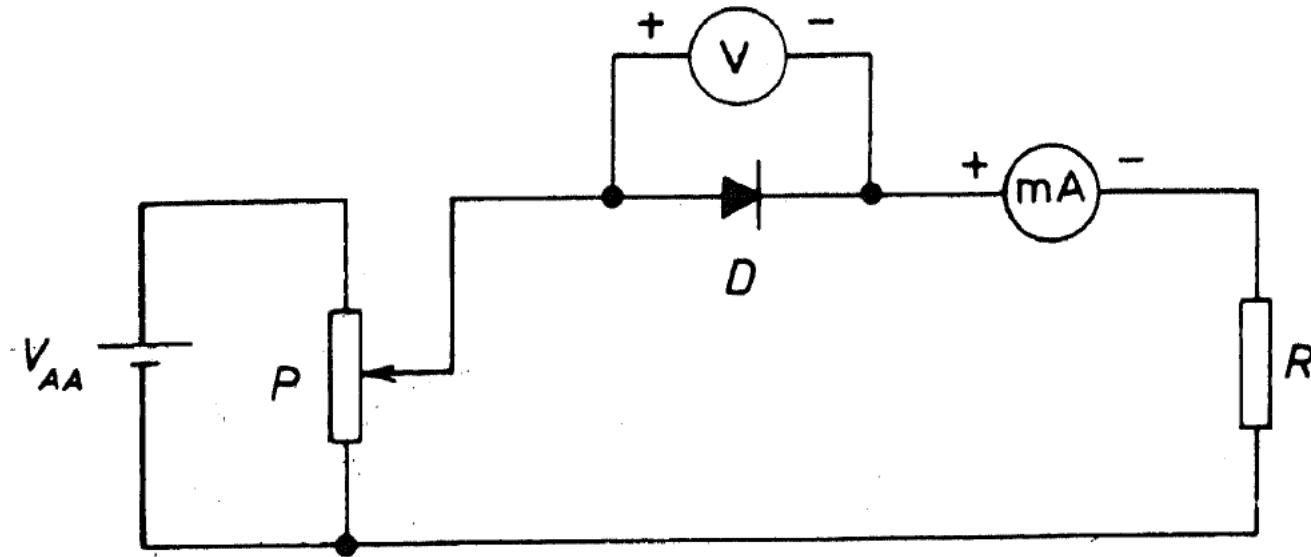


PN Junction Diode

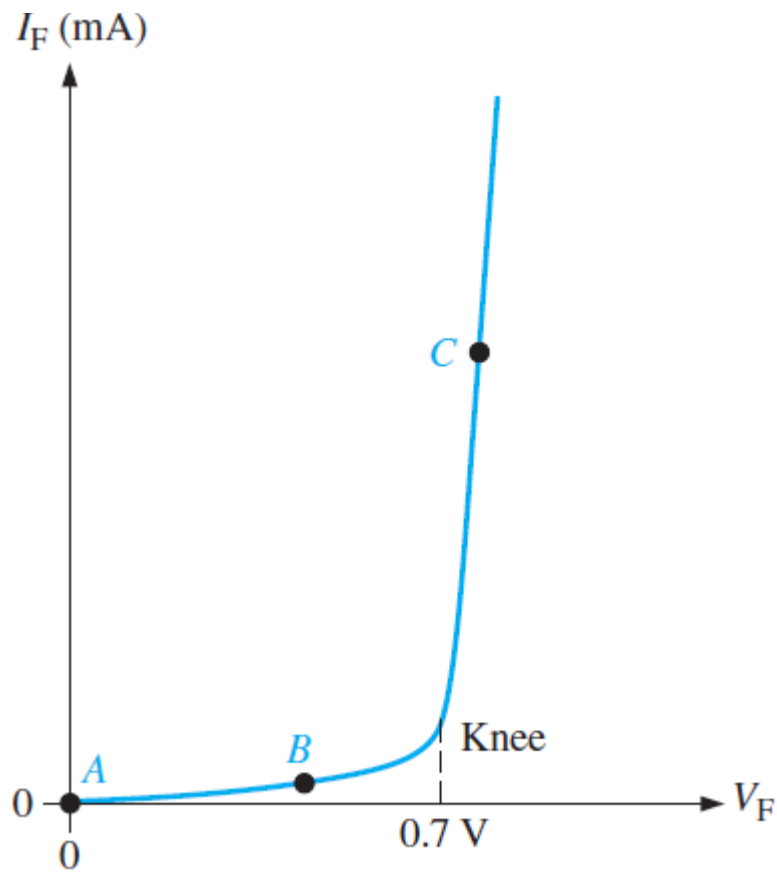
Forward bias  Depletion layer P N V_s R Large current	
V_s polarity	(+) to P material (-) to N material
Current flow	Large forward current if $V_s > 0.7\text{ V}$
Depletion layer	Narrow

Reverse bias  Depletion layer P N V_s R Small current	
V_s polarity	(-) to P materials (+) to N material
Current flow	Small reverse current (saturation current and surface leakage current) if $V_s < \text{breakdown voltage}$
Depletion layer	Wide

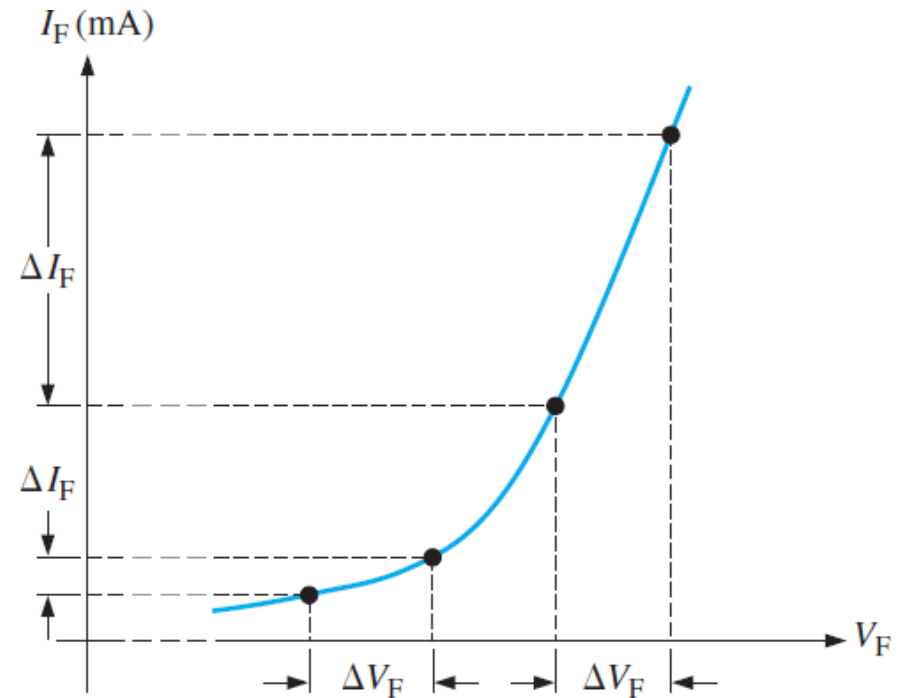
PN Junction Diode Characteristics



PN Junction Diode Characteristics

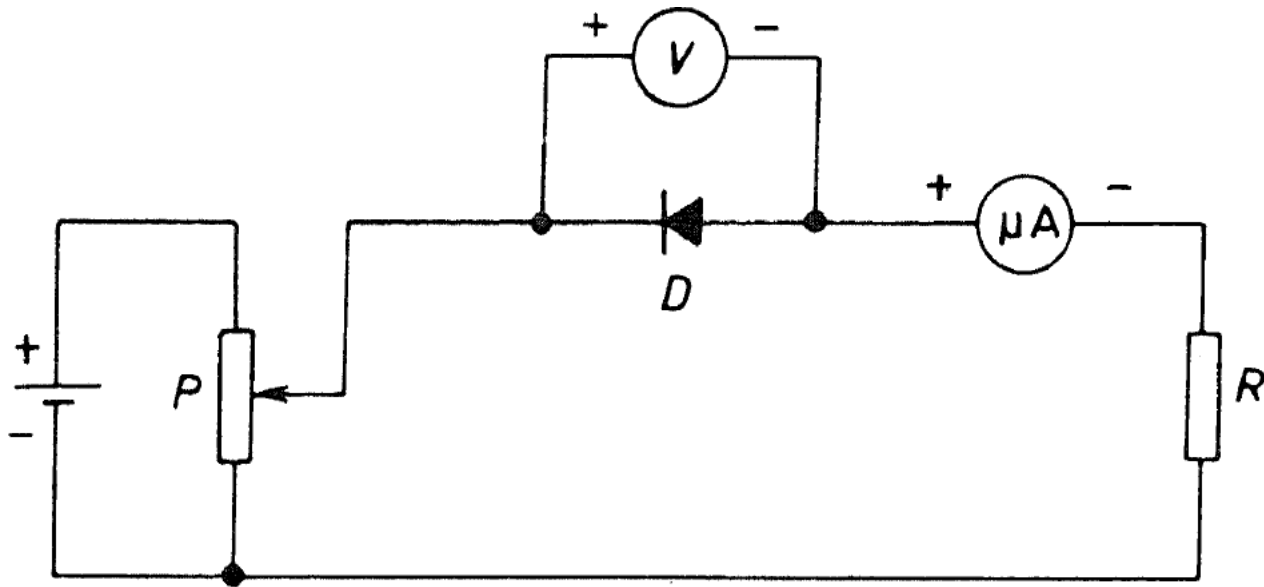


(a) V - I characteristic curve for forward bias.

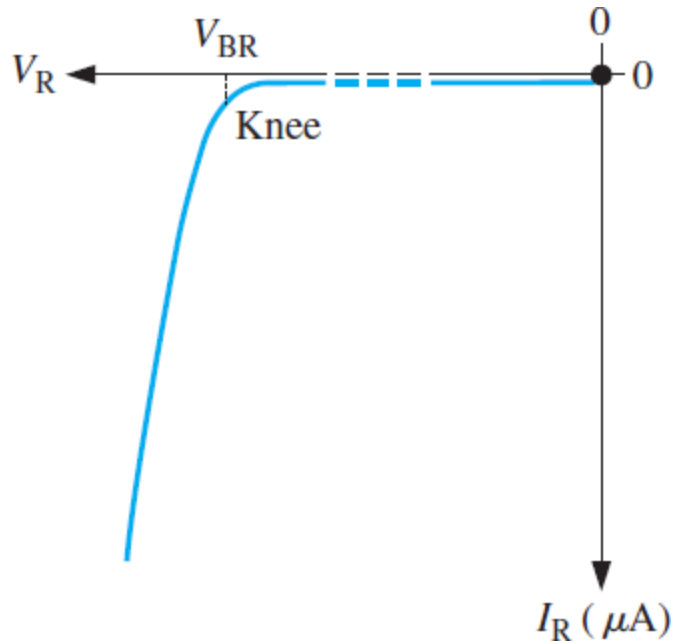


(b) Expanded view of a portion of the curve in part (a). The dynamic resistance r'_d decreases as you move up the curve, as indicated by the decrease in the value of $\Delta V_F / \Delta I_F$.

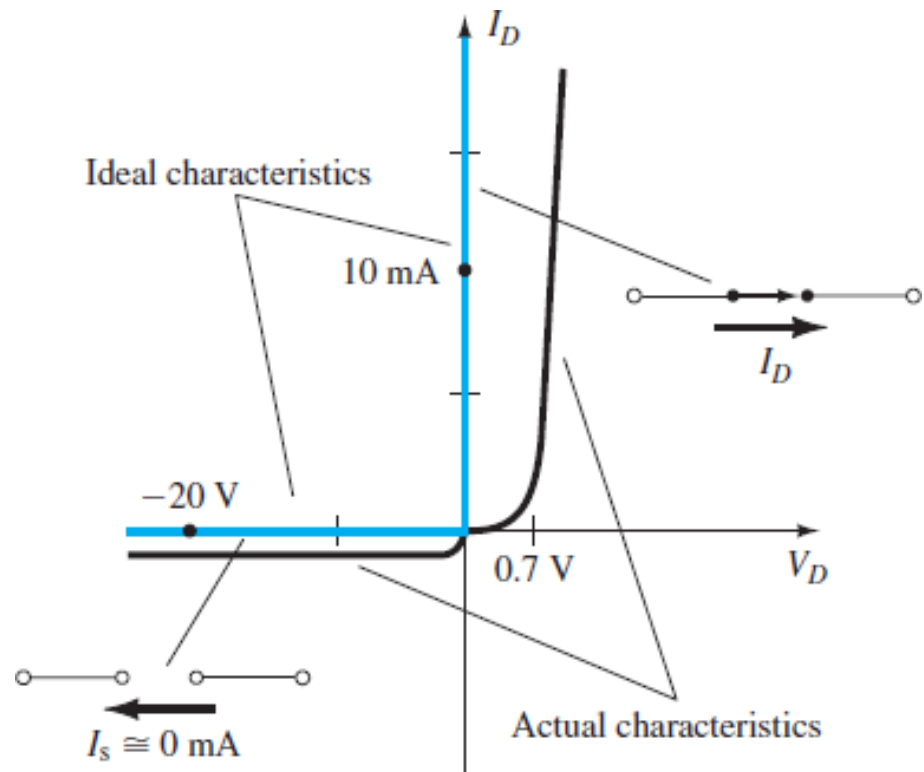
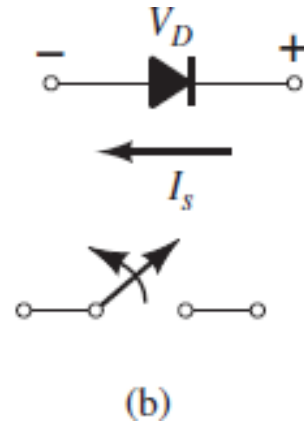
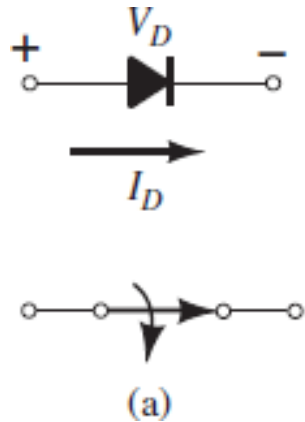
PN Junction Diode Characteristics



PN Junction Diode Characteristics



PN Junction Diode ideal Characteristics



Avalanche Breakdown :

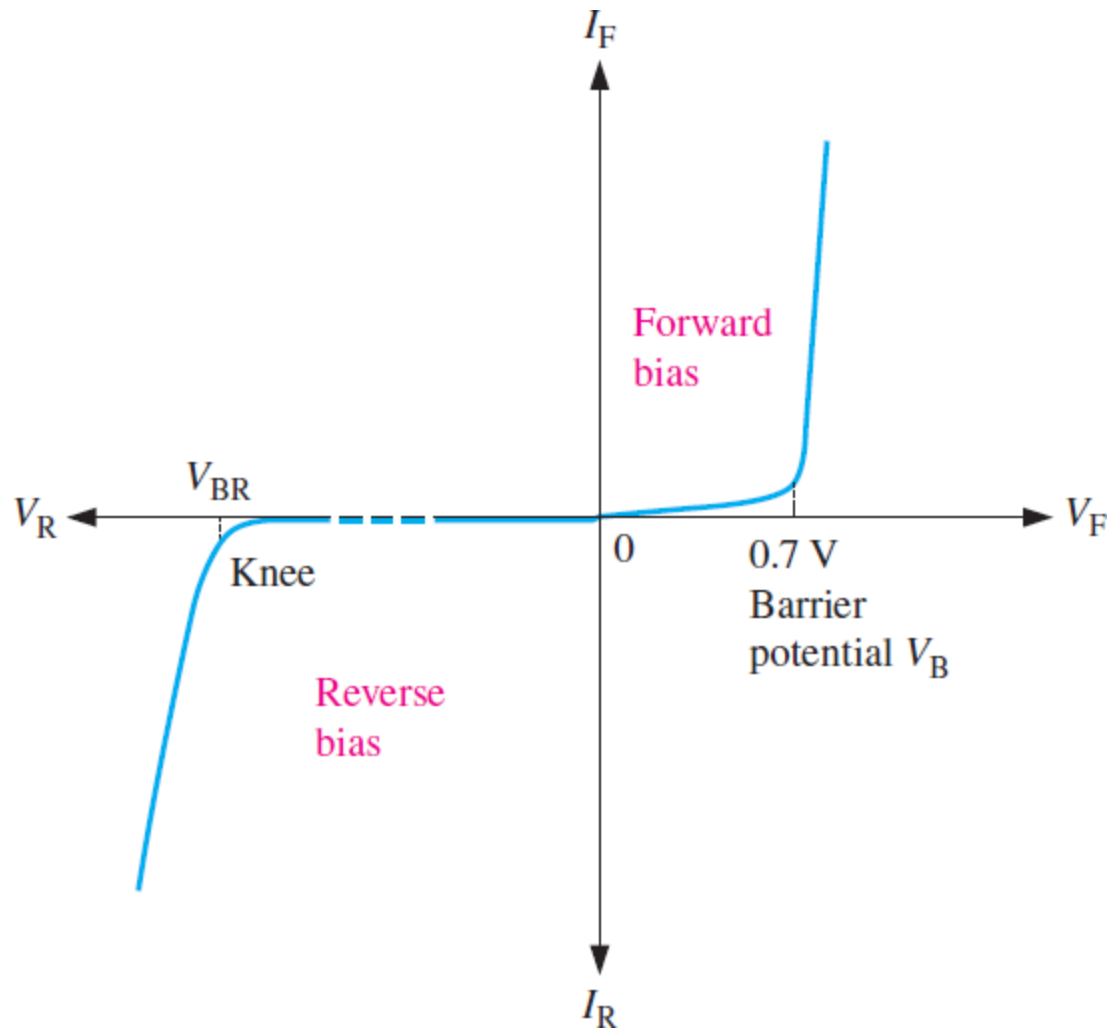
Avalanche Breakdown Definition: Avalanche Breakdown is a phenomenon in a p-n junction where a rapid increase in reverse current occurs due to carrier multiplication.

Carrier Multiplication: Fast-moving electrons collide with atoms, releasing more electrons and significantly increasing the current flow.

Avalanche Breakdown Voltage: This is the specific reverse voltage at which Avalanche Breakdown occurs, denoted as VBR.

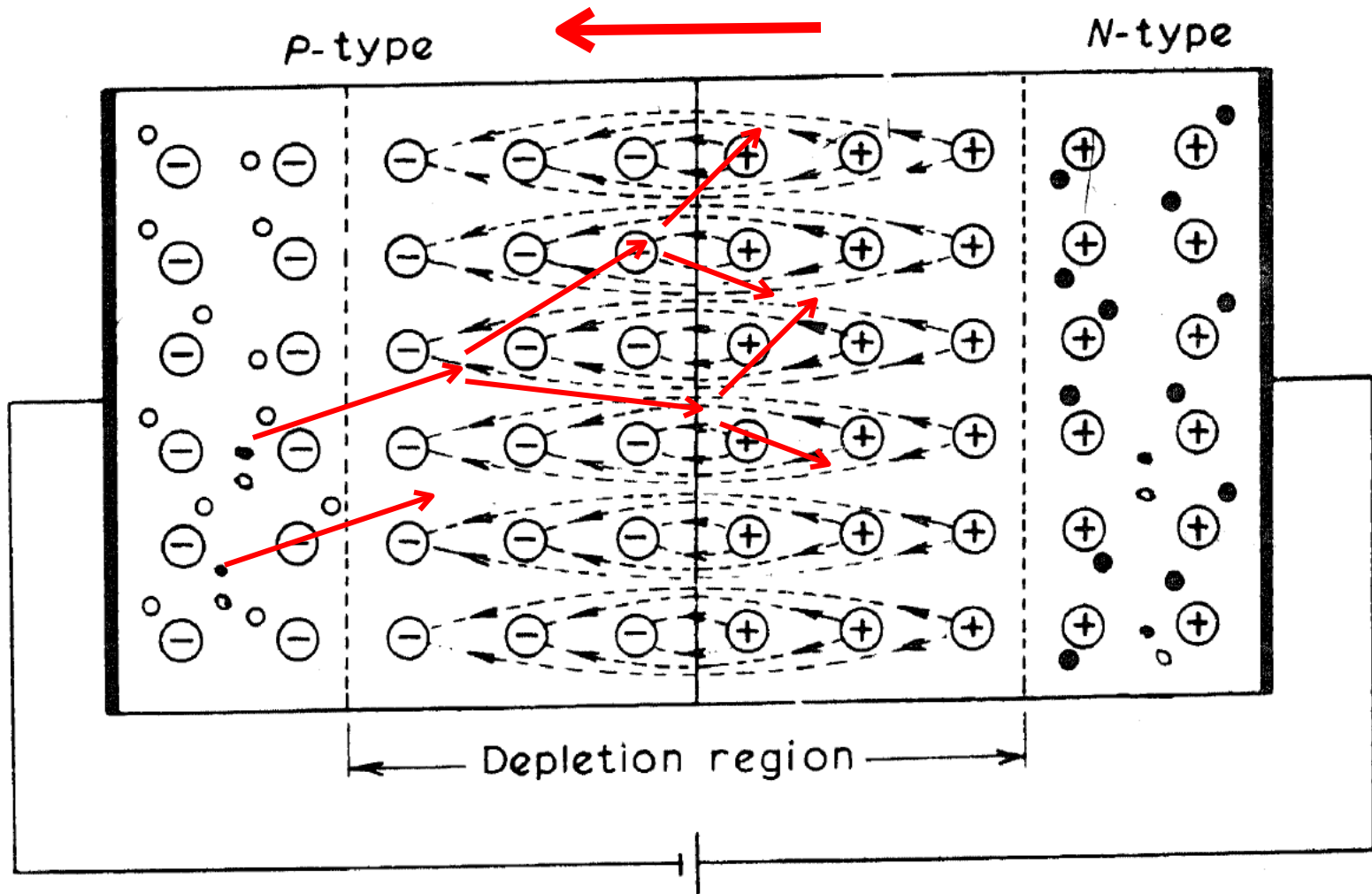
Conditions for Avalanche Breakdown: It typically occurs in lightly doped p-n junctions when the reverse voltage exceeds 5 V.

PN Junction Diode Characteristics



Avalanche breakdown

PN-Junction with Reverse Bias



The avalanche breakdown occurs when a high reverse voltage is applied across the diode.

As we increase the applied reverse voltage, the electric field across the junction increases.

This electric field exerts a force on the electrons at the junction and frees them from covalent bonds.

These free electrons start moving with high velocity across the junction and collide with the other atoms, thus creating more free electrons.

This results in a rapid increase in net current. Both these breakdowns occur in Zener diodes.

Zener diode

A Zener diode is a heavily doped semiconductor device that is designed to operate in the reverse direction.

Zener diodes are manufactured with a great variety of Zener voltages (V_z), and some are even made variable.

Zener diode operates just like a normal diode when it is forward-biased.

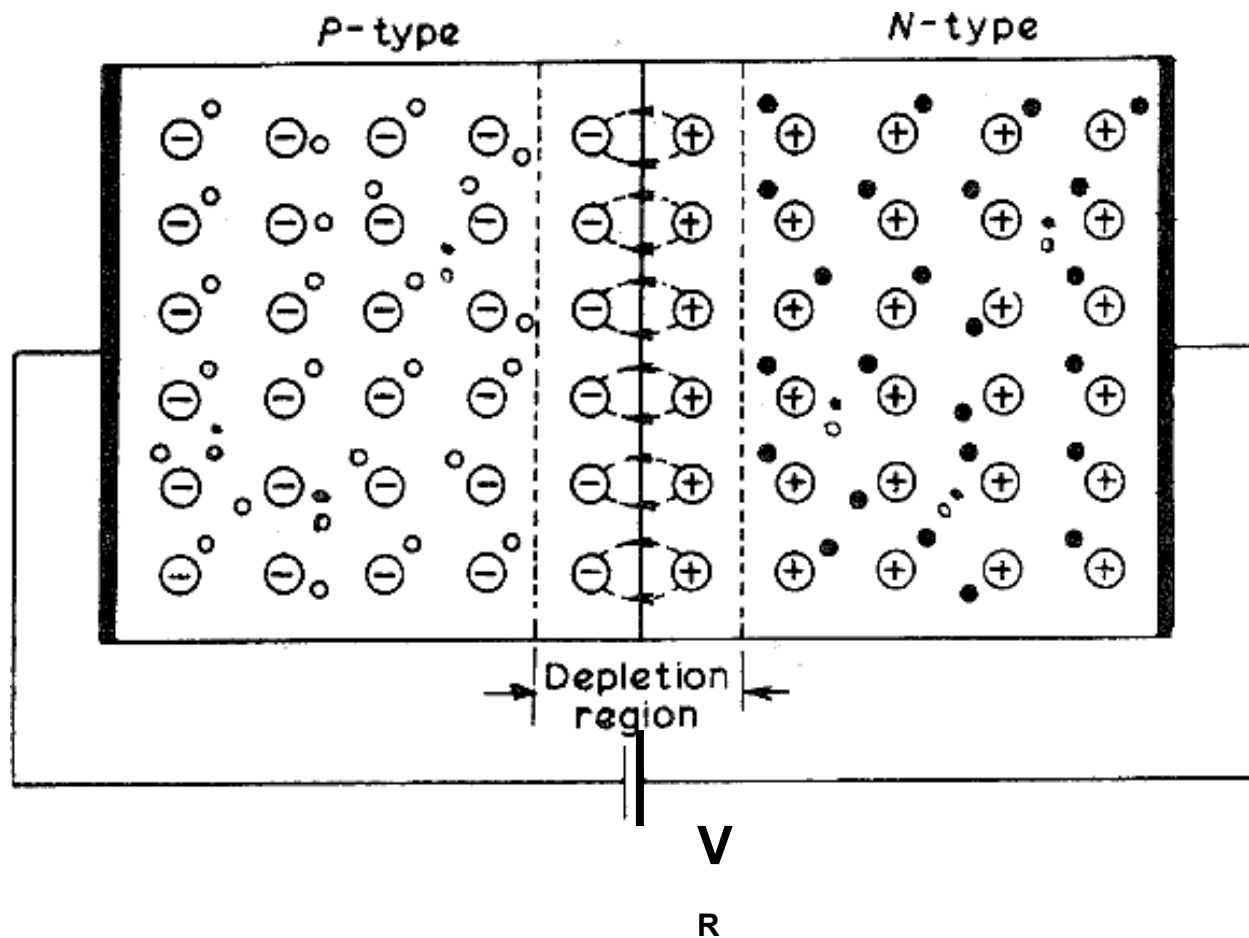
However, a small leakage current flows through the diode when connected in reverse-biased mode.

As the reverse voltage increases to the predetermined breakdown voltage (V_z), current starts flowing through the diode.

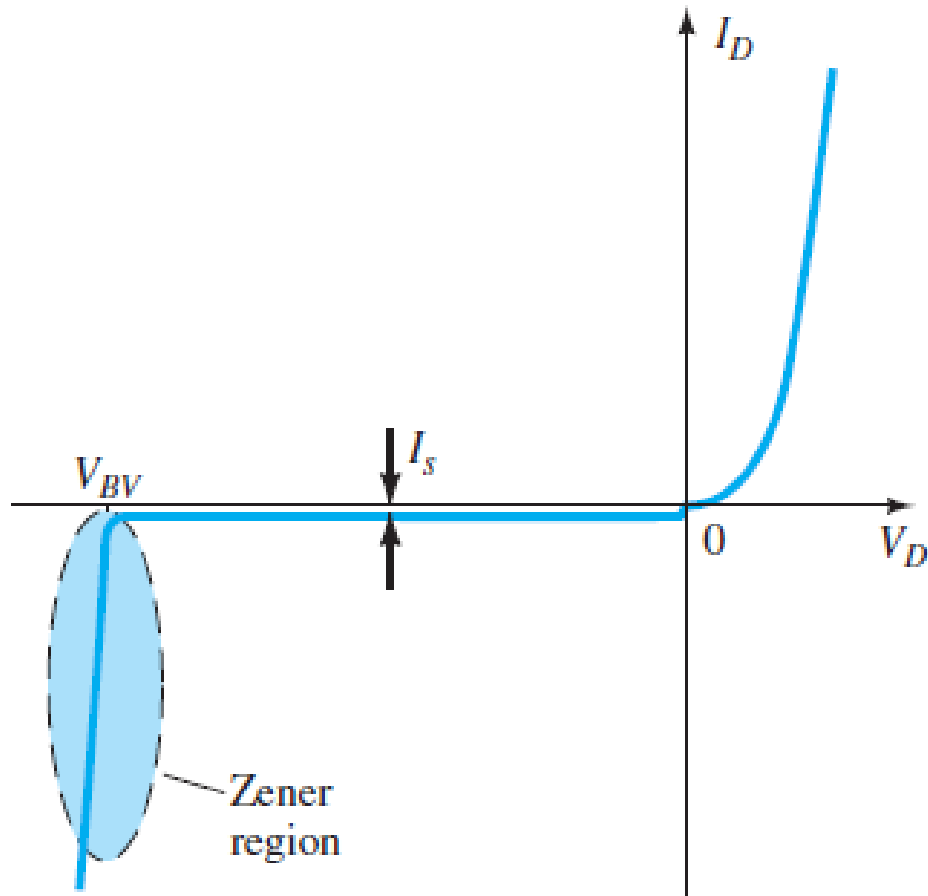
The current increases to a maximum, which is determined by the series resistor, after which it stabilizes and remains constant over a wide range of applied voltage.

Avalanche breakdown occurs in the normal diode and Zener Diode at high reverse voltage. When a high value of reverse voltage is applied to the PN junction, the free electrons gain sufficient energy and accelerate at high velocities. These free electrons moving at high velocity collide with other atoms and knock off more electrons. Due to this continuous collision, a large number of free electrons are generated as a result of [electric current](#) in the diode rapidly increases. This sudden increase in electric current may permanently destroy the normal diode. However, a Zener diode is designed to operate under avalanche breakdown and can sustain the sudden spike of current. Avalanche breakdown occurs in Zener diodes with applied Zener voltage (V_z) greater than (V_z).

Zener Break down



Zenar Diode break down



When the diode is reverse biased,
the kinetic energy of the electrons increases, and
they move at a high velocity.

The high-velocity electrons collide with other
atoms and give rise to free electrons.

These free electrons, in turn, give rise to a high
value of reverse saturation current.

This is known as Zener breakdown.

Zener Breakdown	Avalanche Breakdown
The process in which the electrons move across the barrier from the valence band of p-type material to the conduction band of n-type material is known as Zener breakdown.	The process of applying high voltage and increasing the free electrons or electric current in semiconductors and insulating materials is called an avalanche breakdown.
This is observed in Zener diodes having a Zener breakdown voltage V_z of 5 to 8 volts.	This is observed in Zener diode having a Zener breakdown voltage V_z greater than 8 volts.
The valence electrons are pulled into conduction due to the high electric field in the narrow depletion region.	The valence electrons are pushed to conduction due to the energy imparted by accelerated electrons, which gain their velocity due to their collision with other atoms.
The increase in temperature decreases the breakdown voltage.	The increase in temperature increases the breakdown voltage.
The VI characteristics of a Zener breakdown has a sharp curve.	The VI characteristic curve of the avalanche breakdown is not as sharp as the Zener breakdown.
It occurs in diodes that are highly doped.	It occurs in diodes that are lightly doped.

Diode current

$$i_D = I_o(e^{v_D/\eta V_T} - 1) \quad \text{A}$$

where $V_T = kT/q$; V

V_D diode terminal voltage, Volts

I_o temperature-dependent saturation current, μA

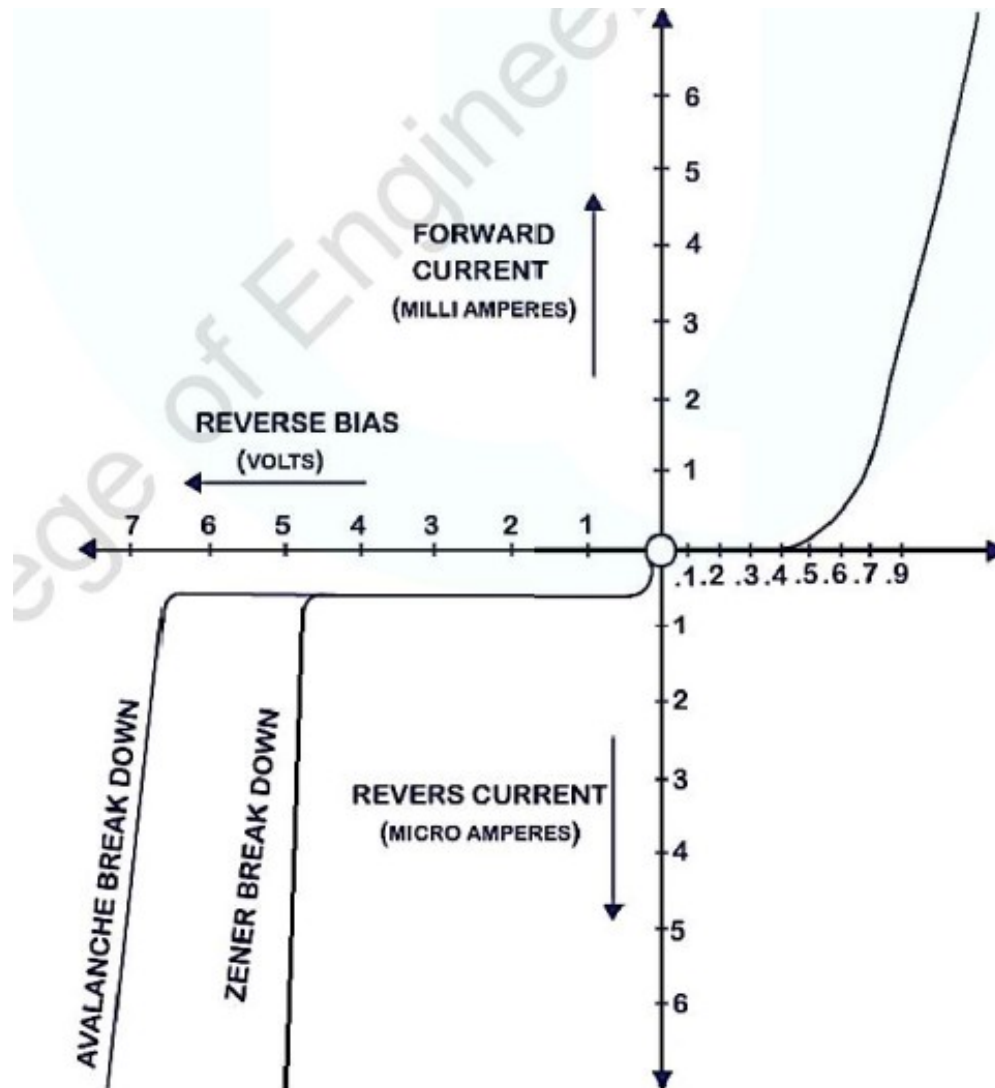
T absolute temperature of p-n junction, K

k Boltzmann's constant $1.38 \times 10^{-23} \text{ J/K}$

q electron charge $1.6 \times 10^{-19} \text{ C}$

η = empirical constant, 1 for Ge and 2 for Si

V I Characteristics of zener diode

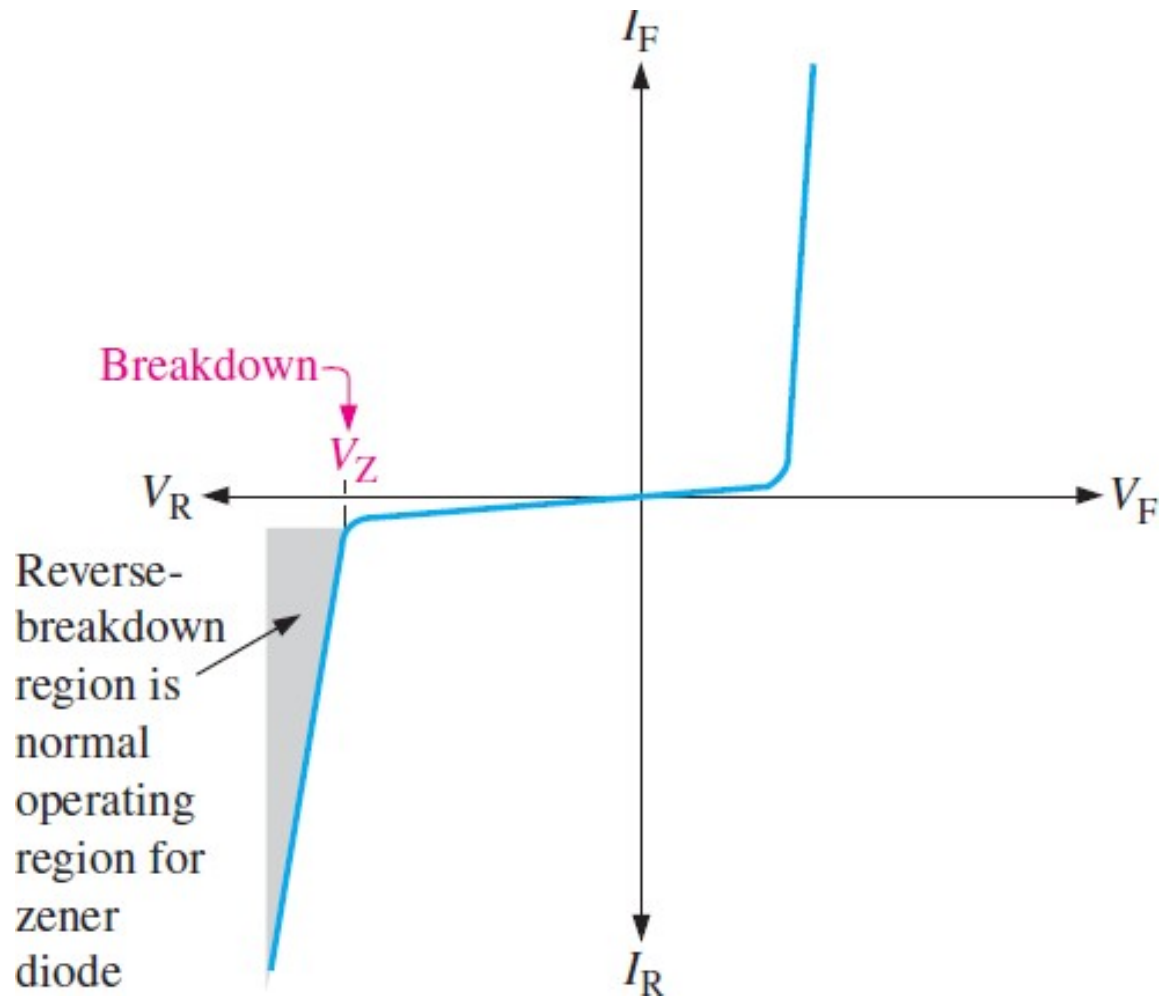


When the applied reverse bias voltage reaches closer to the Zener voltage, the electric field in the depletion region gets strong enough to pull electrons from their valence band.

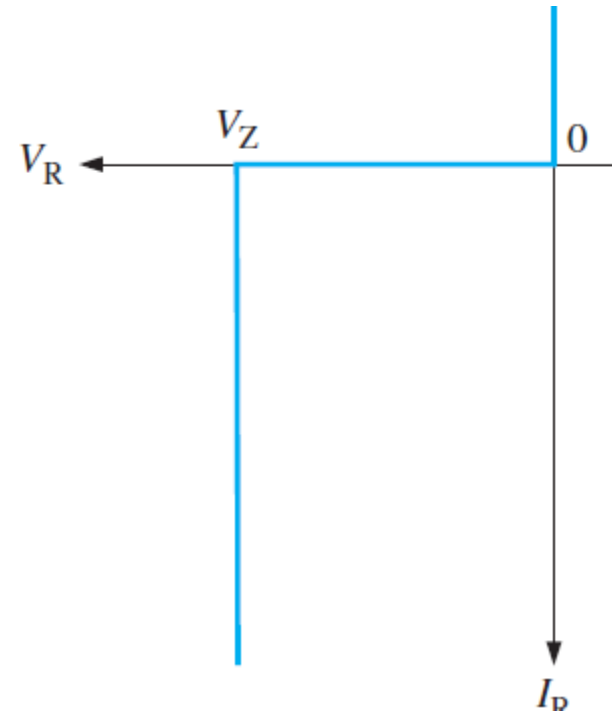
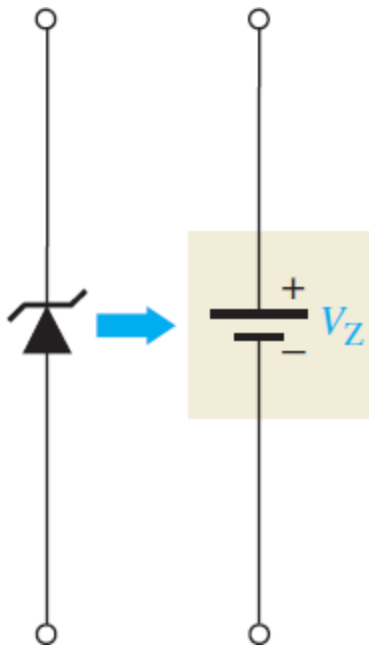
The valence electrons that gain sufficient energy from the strong electric field of the depletion region break free from the parent atom.

At the Zener breakdown region, a small increase in the voltage results in the rapid increase of the electric current.

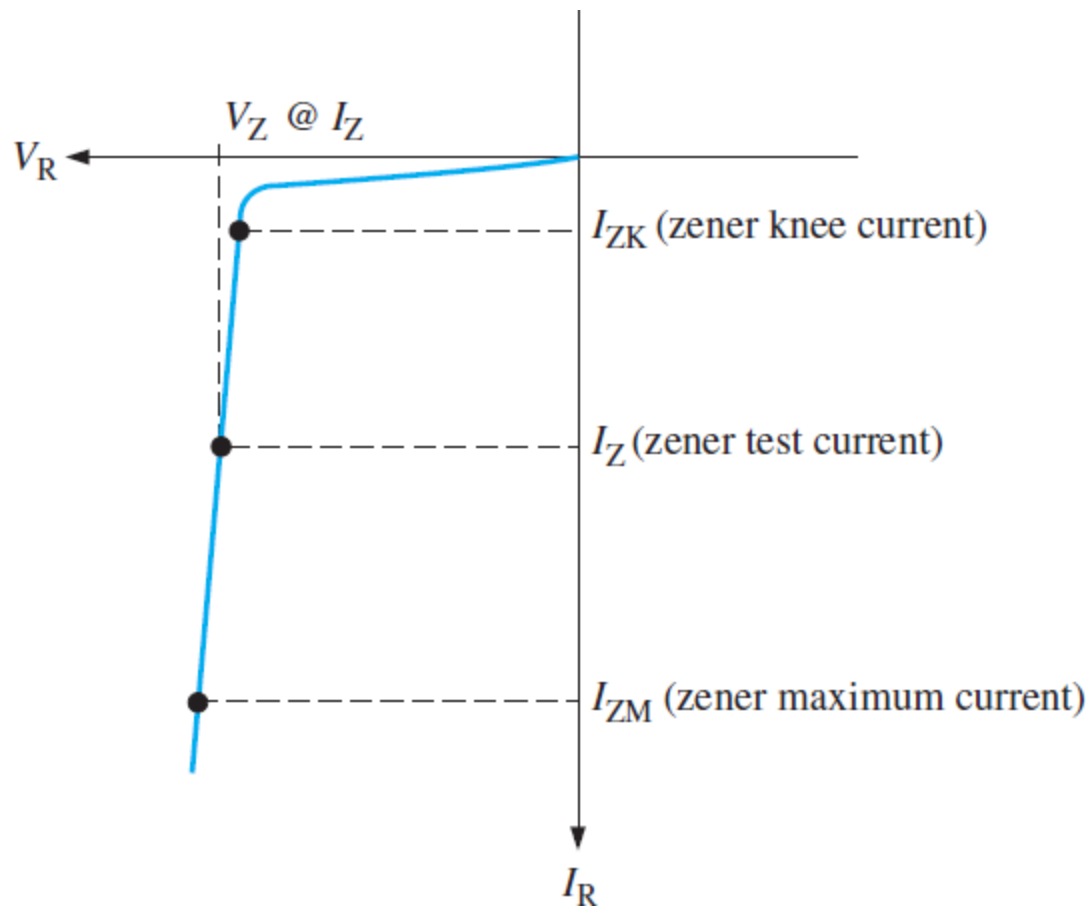
Zener diode



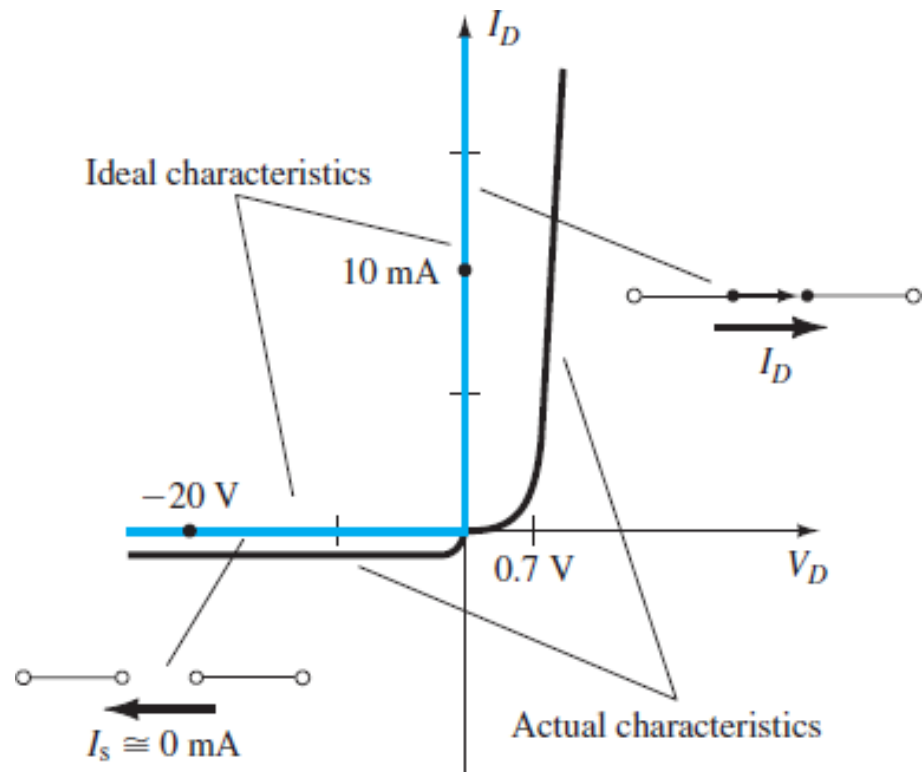
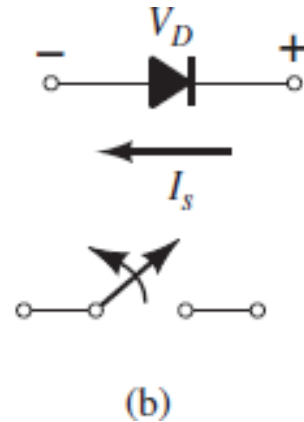
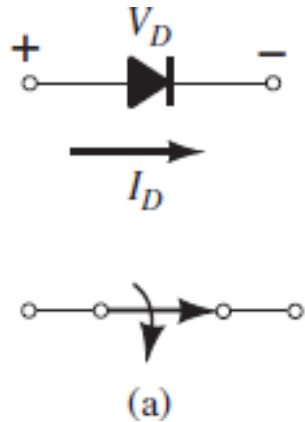
Zener diode



Zener diode



PN Junction Diode ideal Characteristics



Zener diode

- A **zener diode** is a **silicon** diode that the manufacturer has optimized for operation in the breakdown region.
- The zener diode is the backbone of voltage regulators, circuits that hold the load voltage almost constant despite large changes in line voltage and load resistance.
- By varying the doping level of silicon diodes, a manufacturer can produce zener diodes with breakdown voltages from about 2 to over 1000 V.
- These diodes can operate in any of three regions:

Zener diode

- Zener diodes are designed to operate in reverse breakdown.
- Two types of reverse breakdown in a zener diode are *avalanche and zener*.
- *The avalanche effect, occurs in both rectifier and zener diodes at a sufficiently high reverse voltage.*
- **Zener breakdown** occurs in a zener diode at low reverse voltages. A zener diode is heavily doped to reduce the breakdown voltage.
- This causes a very thin depletion region. As a result, an intense electric field exists within the depletion region. Near the zener breakdown voltage (V_Z) the field is intense enough to pull

Zener diode

- Zener diodes with breakdown voltages of less than approximately 5 V operate predominately in zener breakdown.
- Those with breakdown voltages greater than approximately 5 V operate predominately in avalanche breakdown.
- Both types, however, are called *zener diodes*.
Zeners are commercially available with breakdown voltages from less than 1 V to more than 250 V with specified tolerances from 1% to 20%.

Zener diode

- In the forward region, it starts conducting around 0.7 V, just like an ordinary silicon diode. In the leakage region (between zero and breakdown), it has only a small reverse current.
- In a zener diode, the breakdown has a very sharp knee, followed by an almost vertical increase in current.
- Note that the voltage is almost constant, approximately equal to V_Z over most of the breakdown region.
- Data sheets usually specify the value of V_Z at a particular test current I_{ZT} .

Zenar Diodes N N B

Zenar diodes are designed to operate in break down region with out damage.

BY varying the doping level ,it is possible to produce zenar diode with break down voltages from 2 volt to 200volt.

When a diode is heavily doped, the depletion layer is very narrow.

When the reverse voltage across the diode is increased the electric field across the depletion layer become very intense.

When the field is about 3×10^7 v/m electrons are pulled from the covalent bonds. A large number of electron holes pairs are thus produced and the reverse current sharply increases. This is known as the **zenar effect**.

Avalanche effect occurs because of a cumulative action.

The external applied voltage accelerates the minority carriers in the depletion region.

They attain sufficient kinetic energy to ionize atoms by collision.

This creates new electrons which are again accelerated to high enough velocities to ionize more atoms.

This way **an avalanche of free electrons** is obtained.

The reverse current sharply increases.

The zener effect is predominant for breakdown voltages less than about 4 volt.

The avalanche breakdown is predominant for voltages greater than 6 volt.

Between 4 and 6 volt both effects are present.

It is the zener effect that was first discovered.

And the term "zener diode" is in wide use for a breakdown diode whether it uses zener effect or avalanche effect or both.

If the applied reverse voltage exceeds the breakdown voltage.

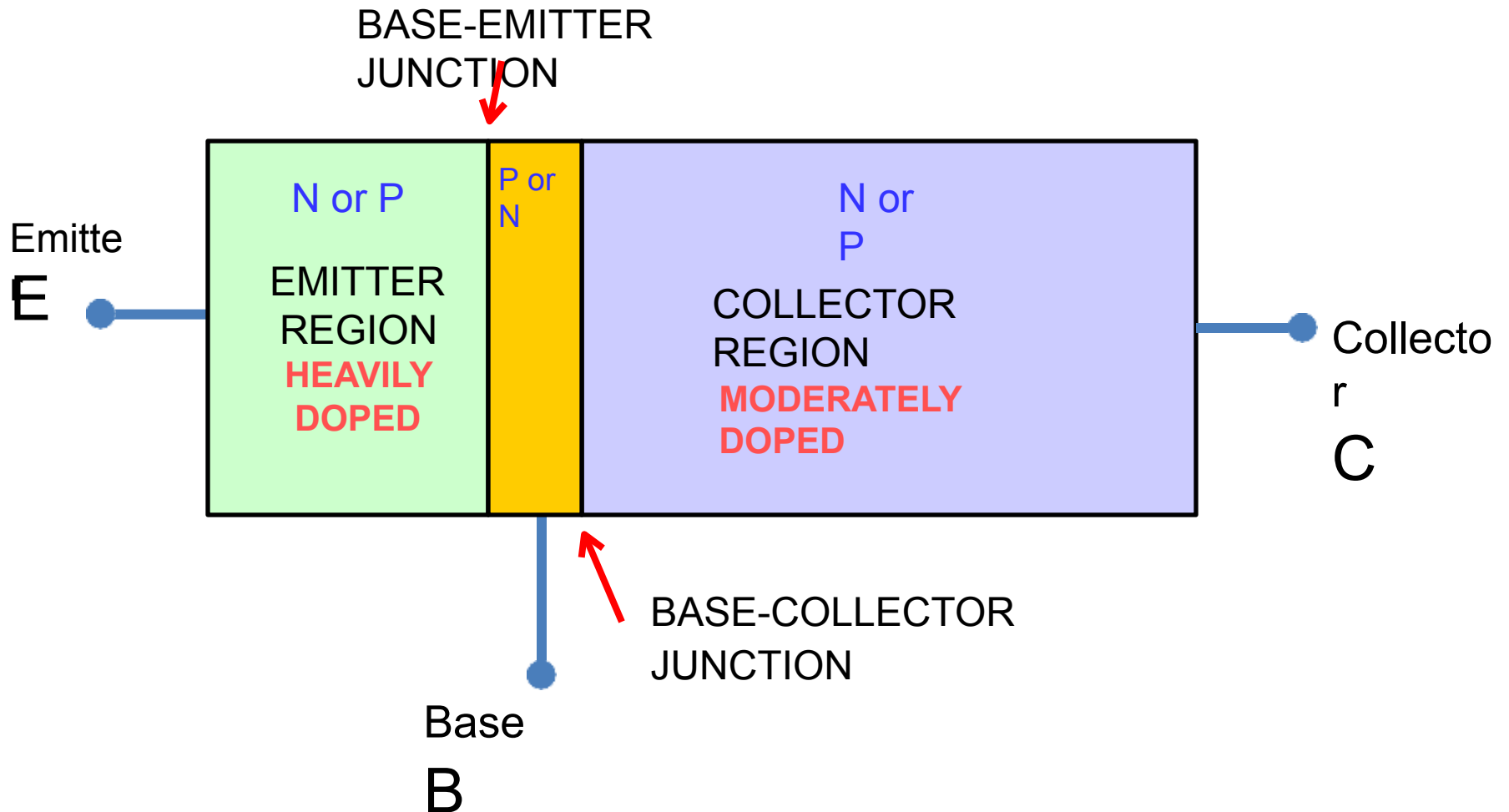
A zener diode acts like a constant voltage source.

For this reason a zener diode is also called voltage reference diode

Transistor

The transistor was invented in 1948 by John Bardeen, Walter Brattain and William Shockley at Bell Laboratory in America. They were awarded the Nobel Prize in recognition of their contributions to Physics. This invention completely revolutionized the electronic industry. Since then, there has been a rapidly expanding effort to utilize and develop many types of semiconductor devices such as FET, MOSFET, UJT, SCR, etc.

Transistor



Transistor

- Emitter
 - Region that supplies carriers – Electrons or holes
 - Heavily doped
 - Medium in physical size
 - Same type as collector

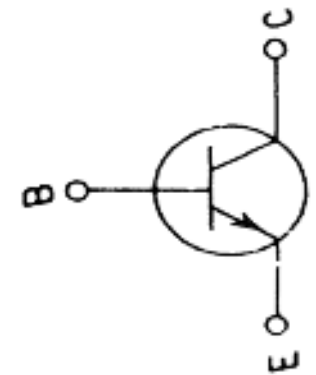
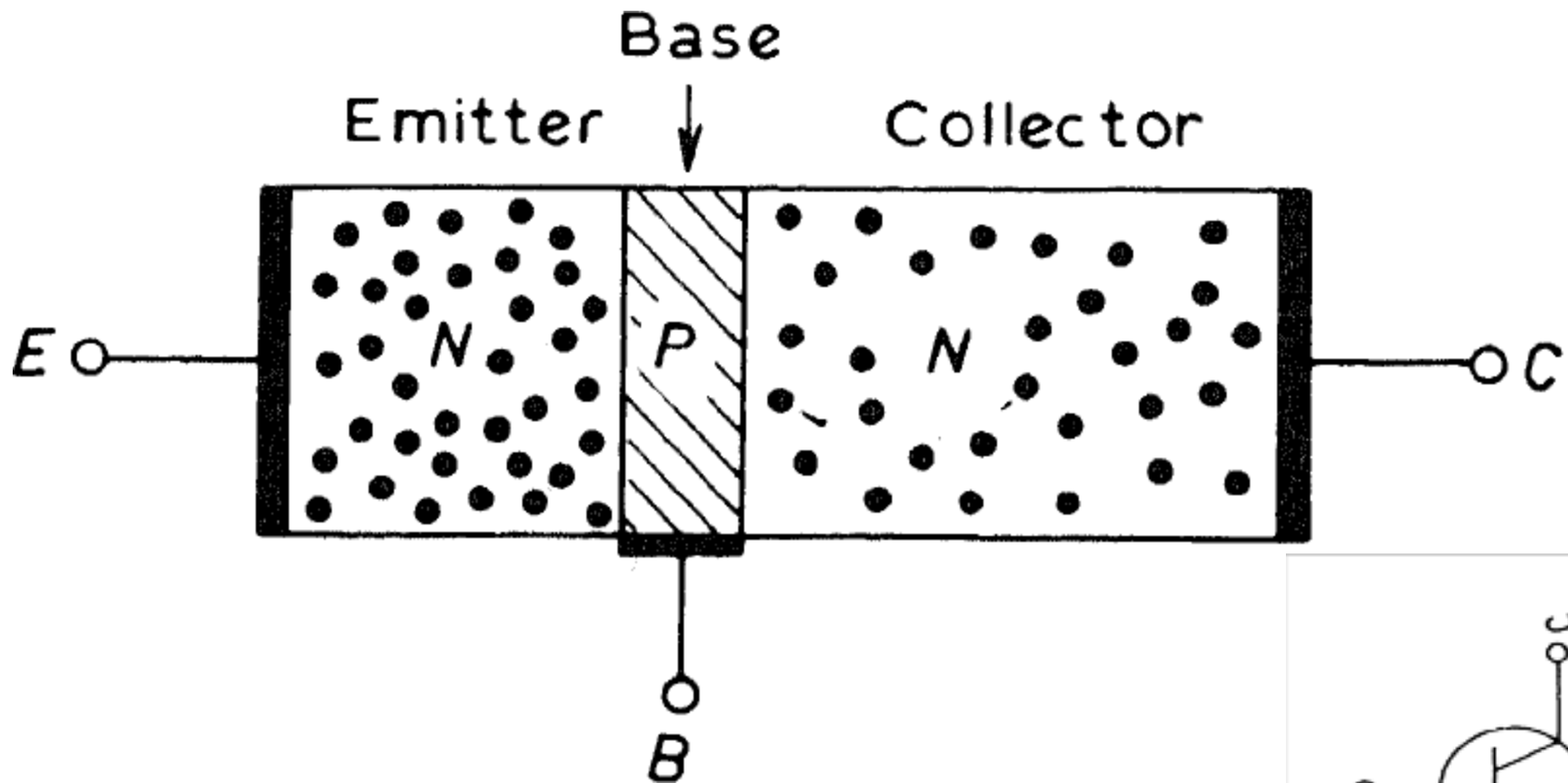
Transistor

- Base
 - Sandwiched between Emitter and collector
 - Lightly doped
 - Smallest in physical size
 - Different type from emitter and collector
 - Controls the flow of current from emitter to collector

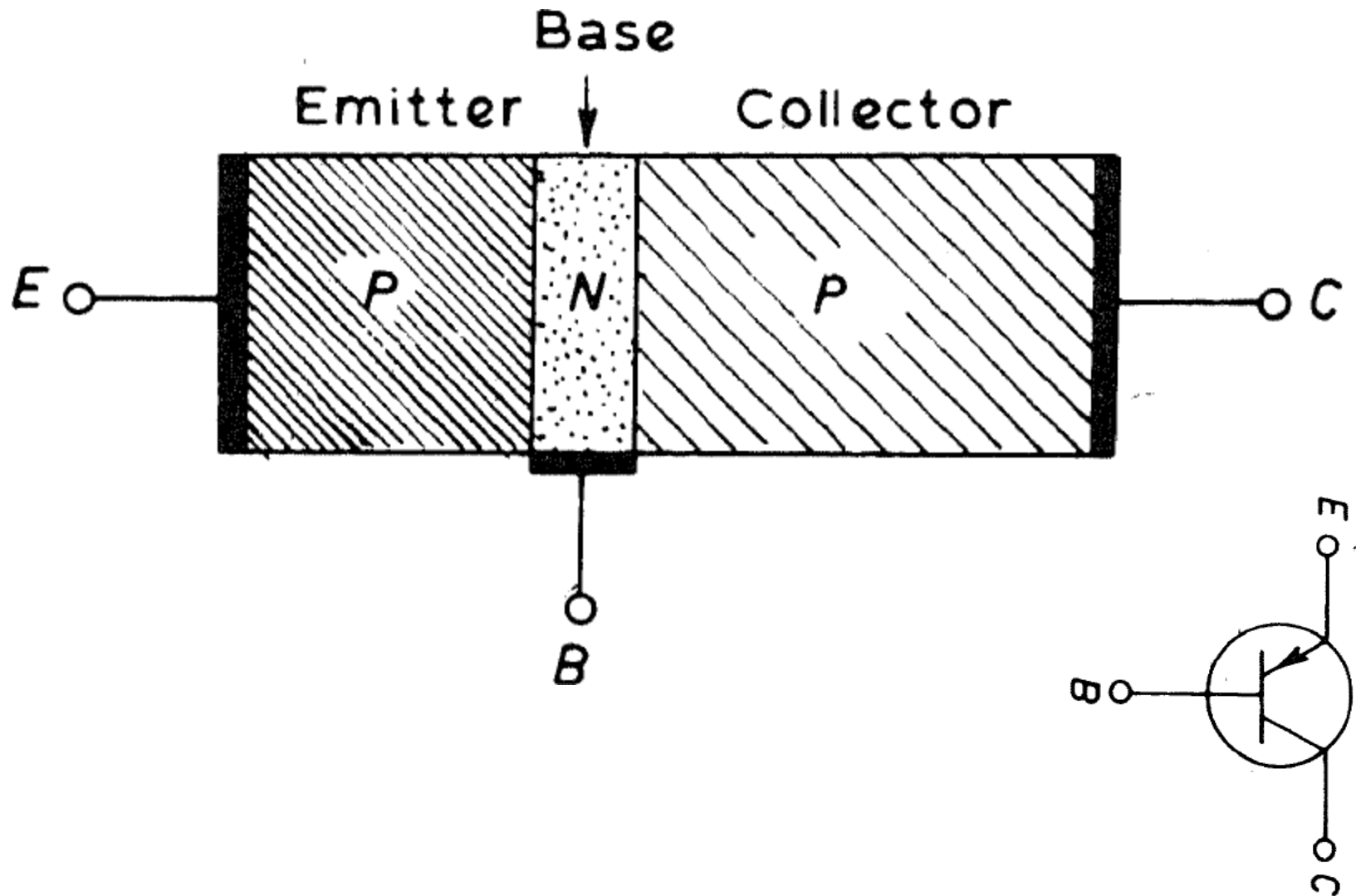
Transistor

- Collector
 - Region that collects carriers – Electrons or holes
 - Medium doped
 - Largest in physical size
 - Same type as emitter

NPN Transistor



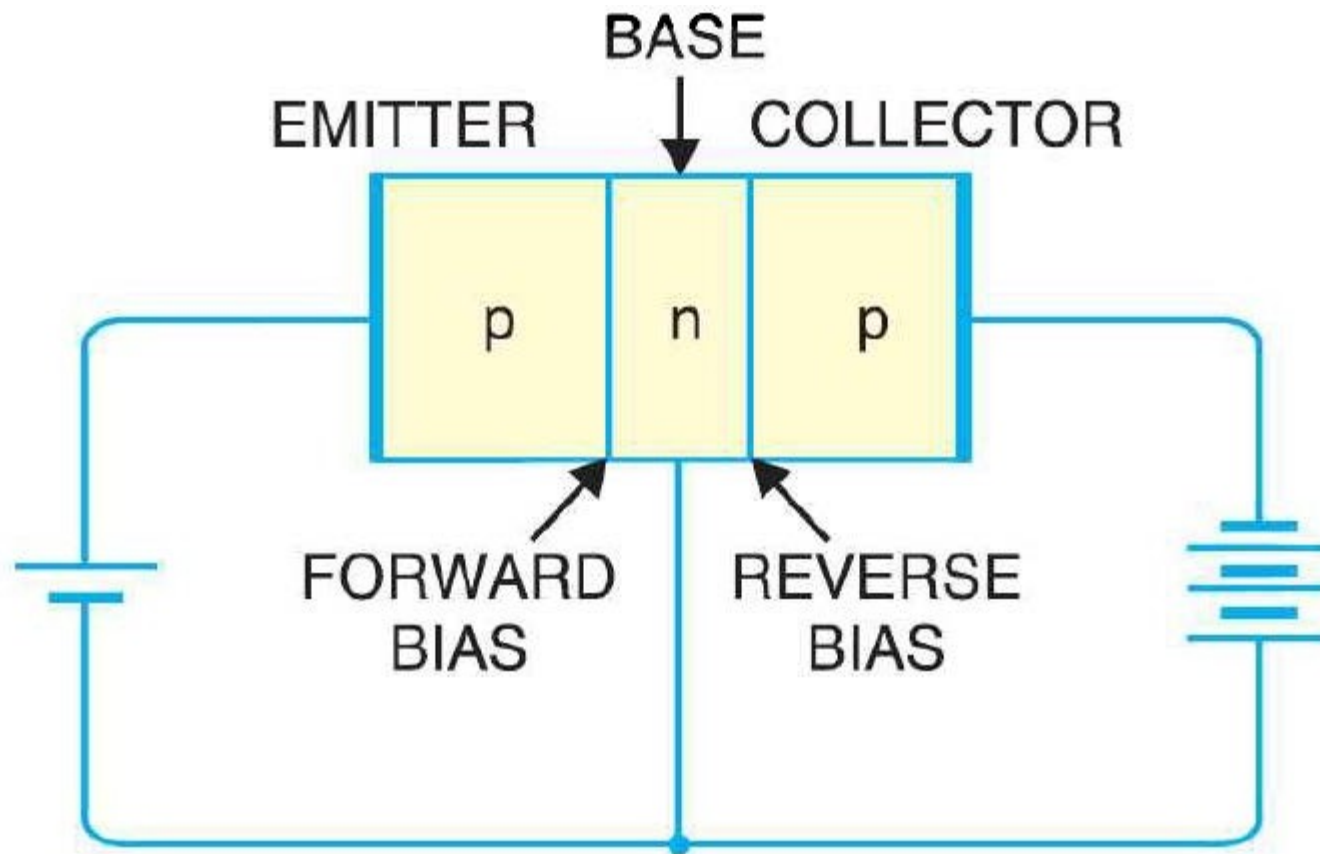
PNP Transistor



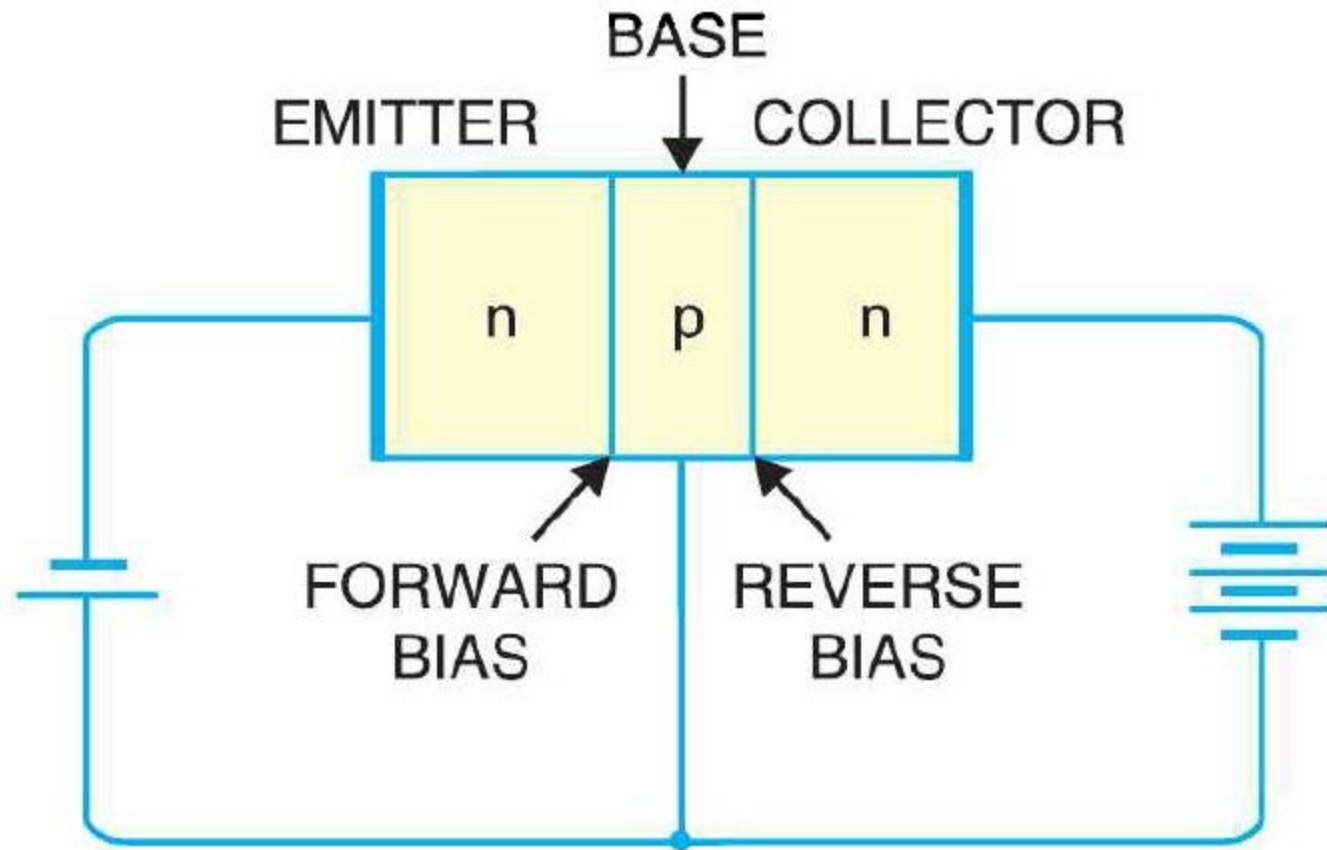
Transistor – Modes of operation

<i>Condition</i>		<i>Emitter junction</i>	<i>Collector junction</i>	<i>Region of operation</i>
I.	FR	Forward-biased	Reverse-biased	Active
II.	FF	Forward-biased	Forward-biased	Saturation
III.	RR	Reverse-biased	Reverse-biased	Cutoff
IV.	RF	Reverse-biased	Forward-biased	Inverted

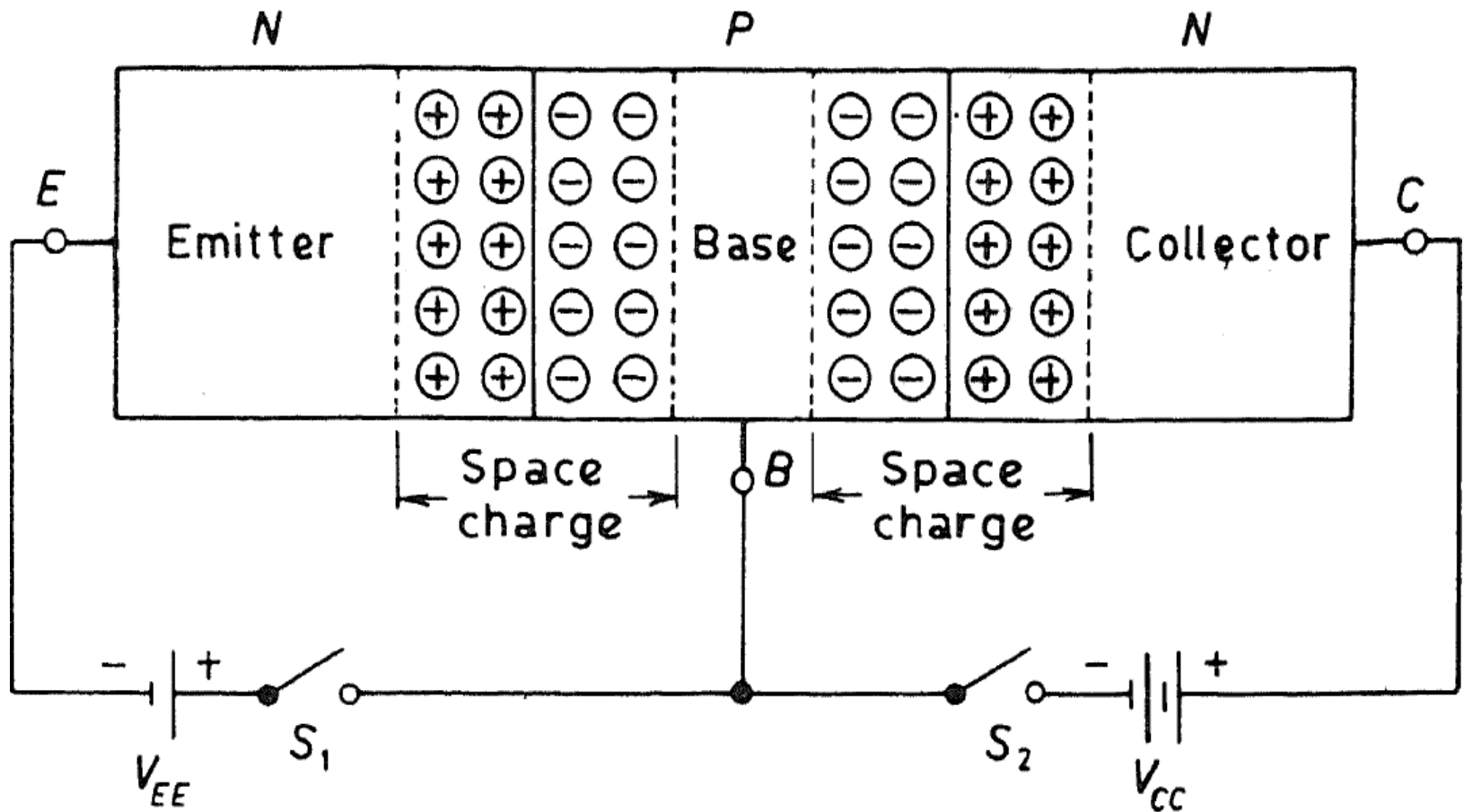
PNP Transistor – FR Mode of operation



NPN Transistor – FR Mode of operation

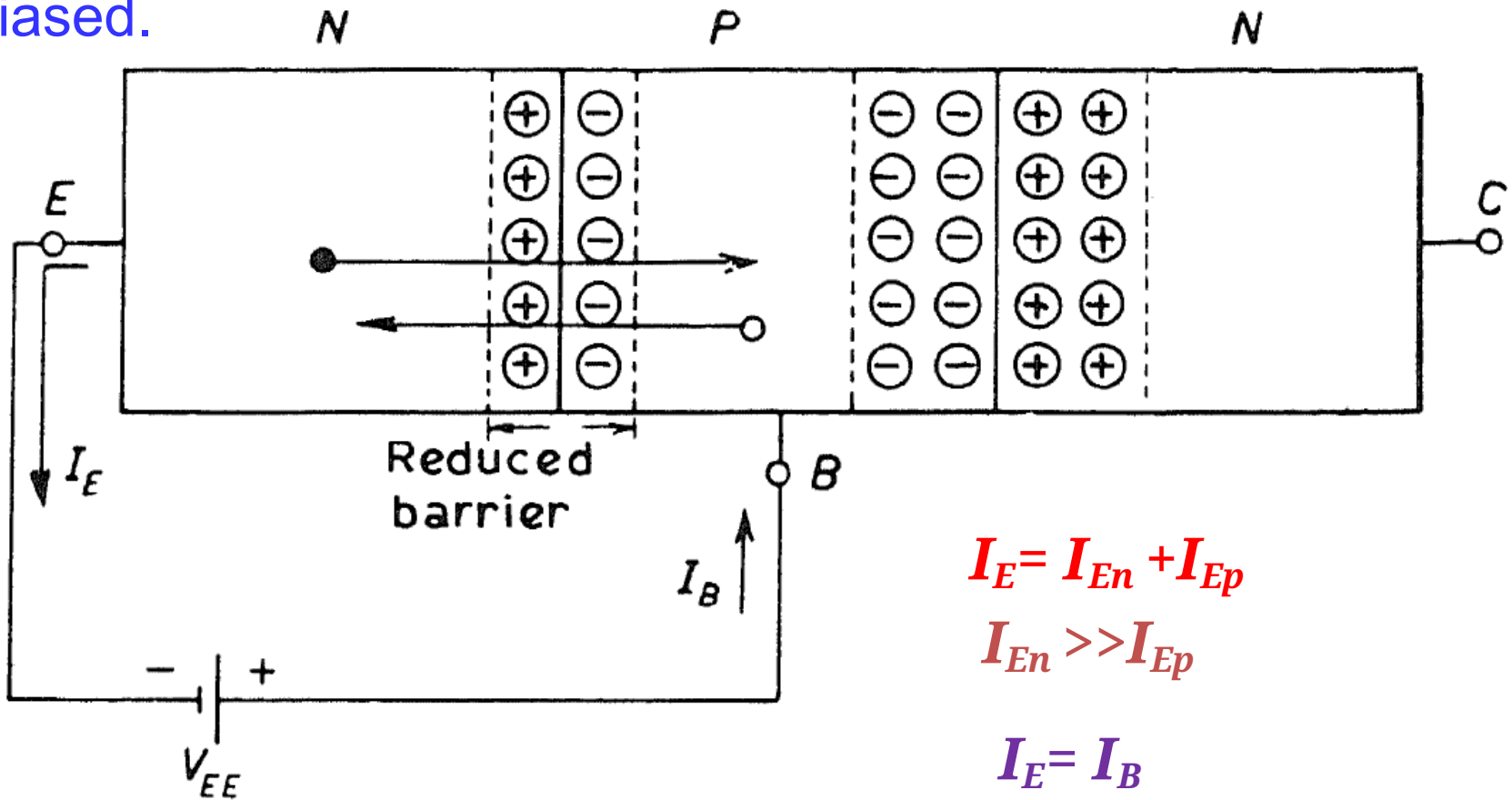


Transistor Action



Transistor Action

Case I: Only Emitter junction is forward biased.



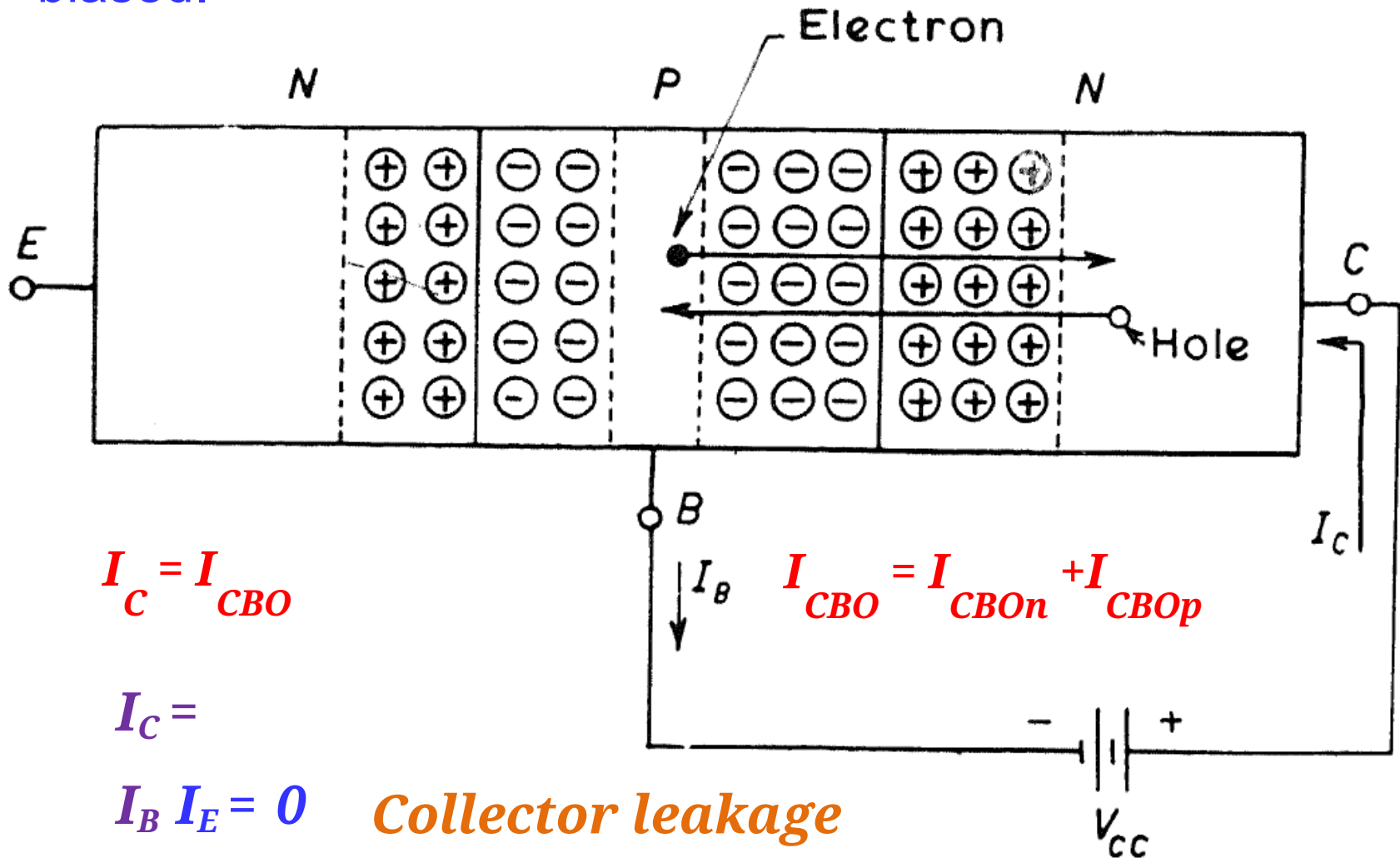
$$I_E = I_{En} + I_{Ep}$$

$$I_{En} \gg I_{Ep}$$

$$I_E = I_B$$

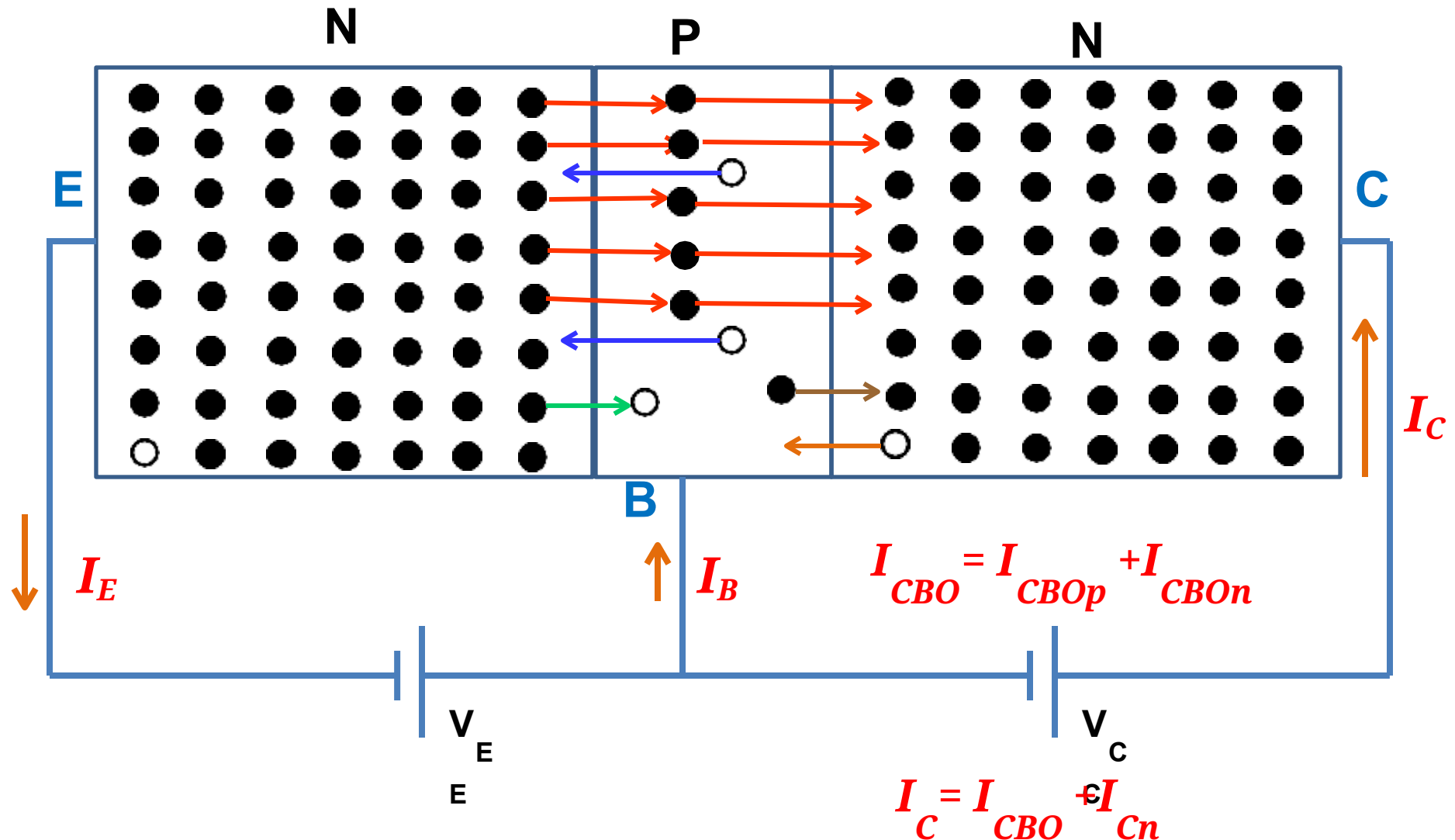
Transistor Action

Case I: Only Collector junction is reversed biased.

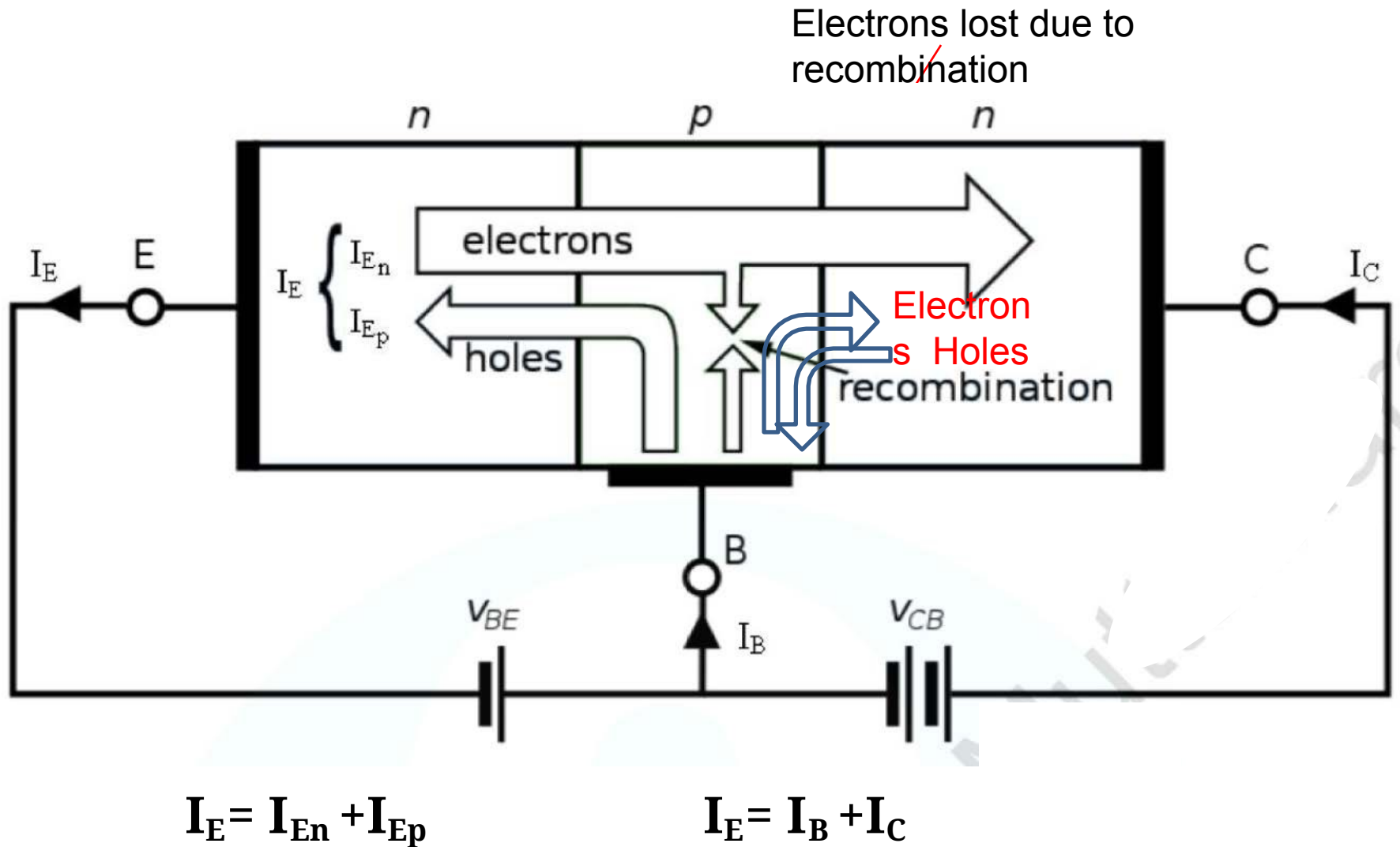


Transistor Action

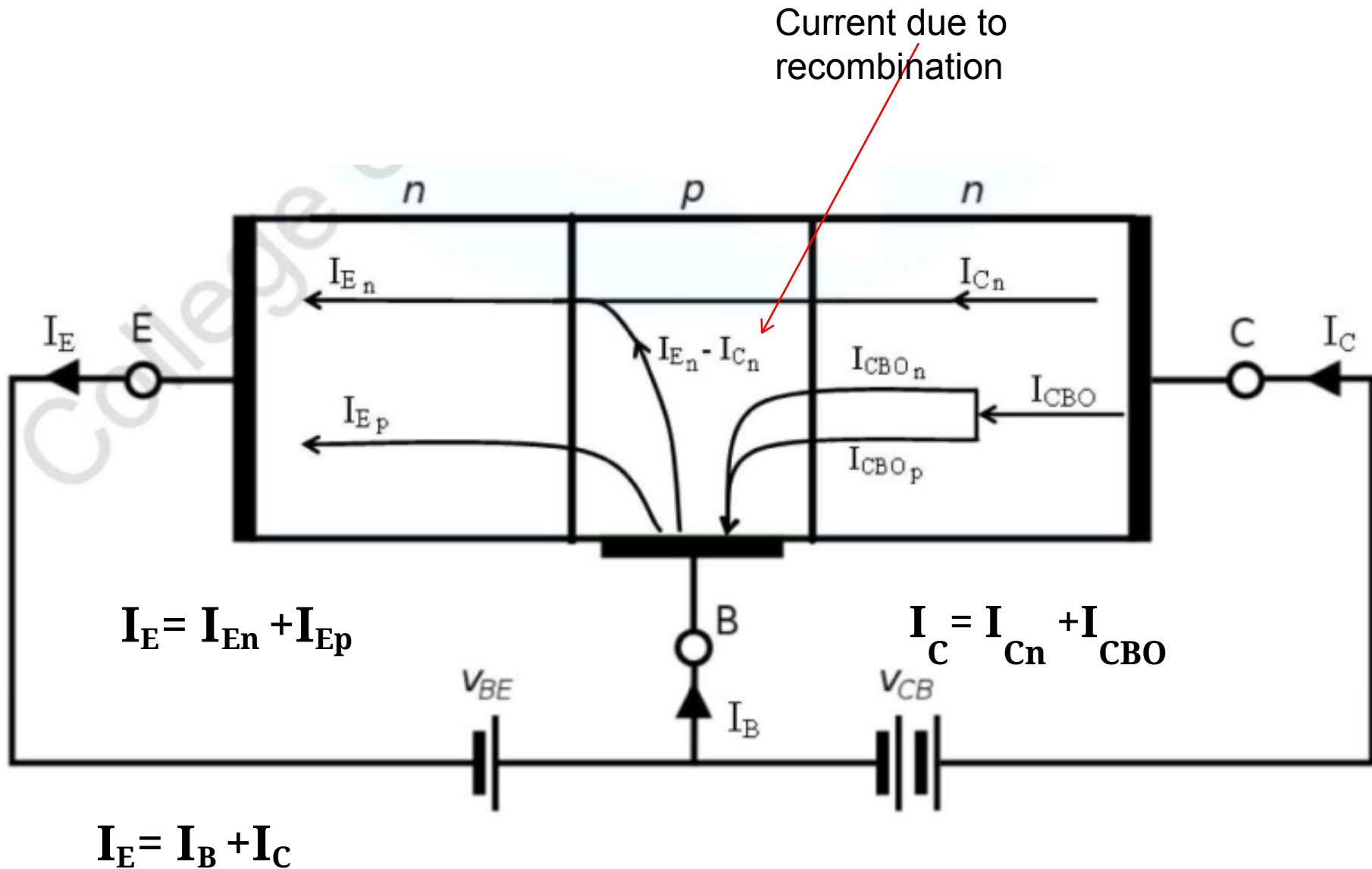
$$I_E = I_{En} + I_{Ep}$$



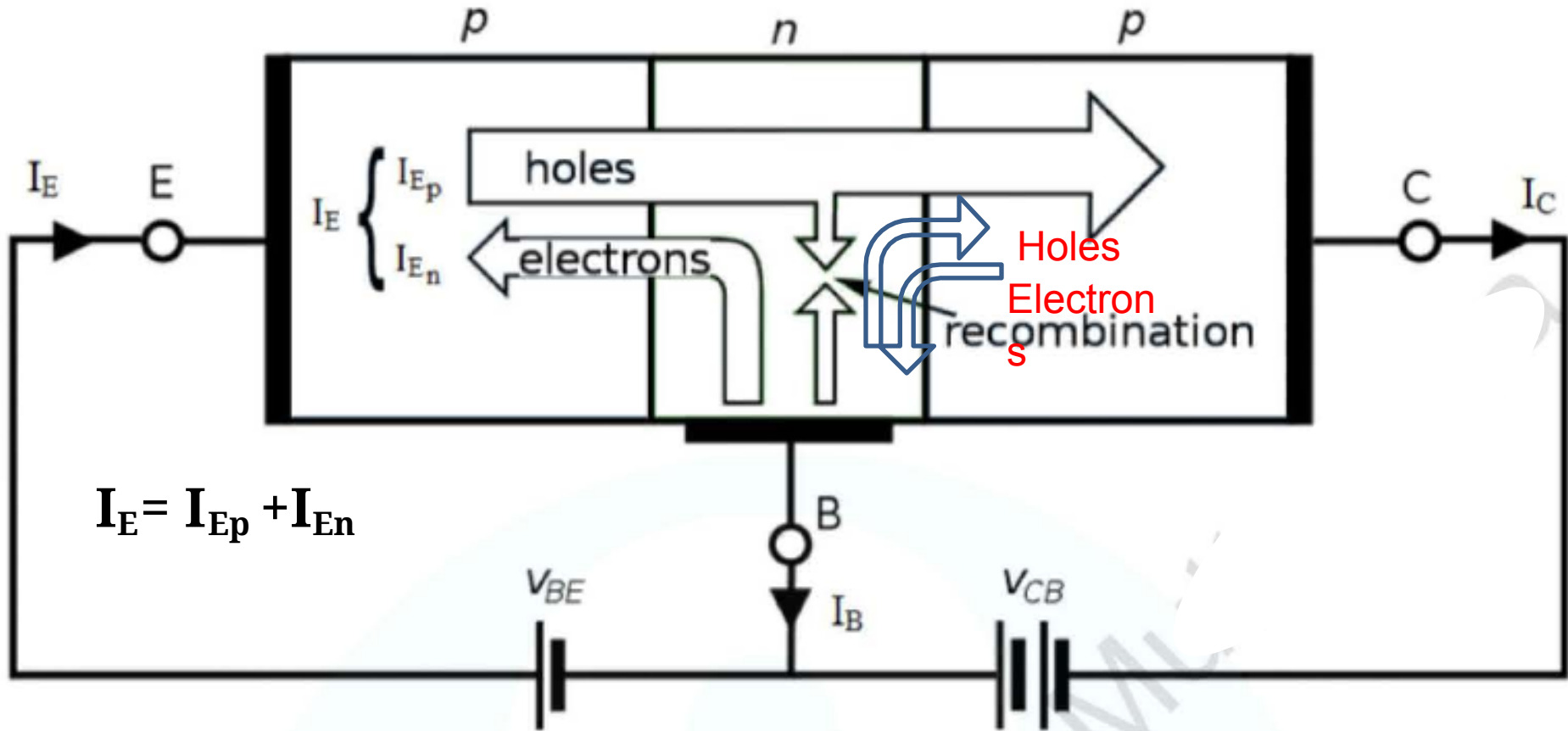
NPN Transistor Current components



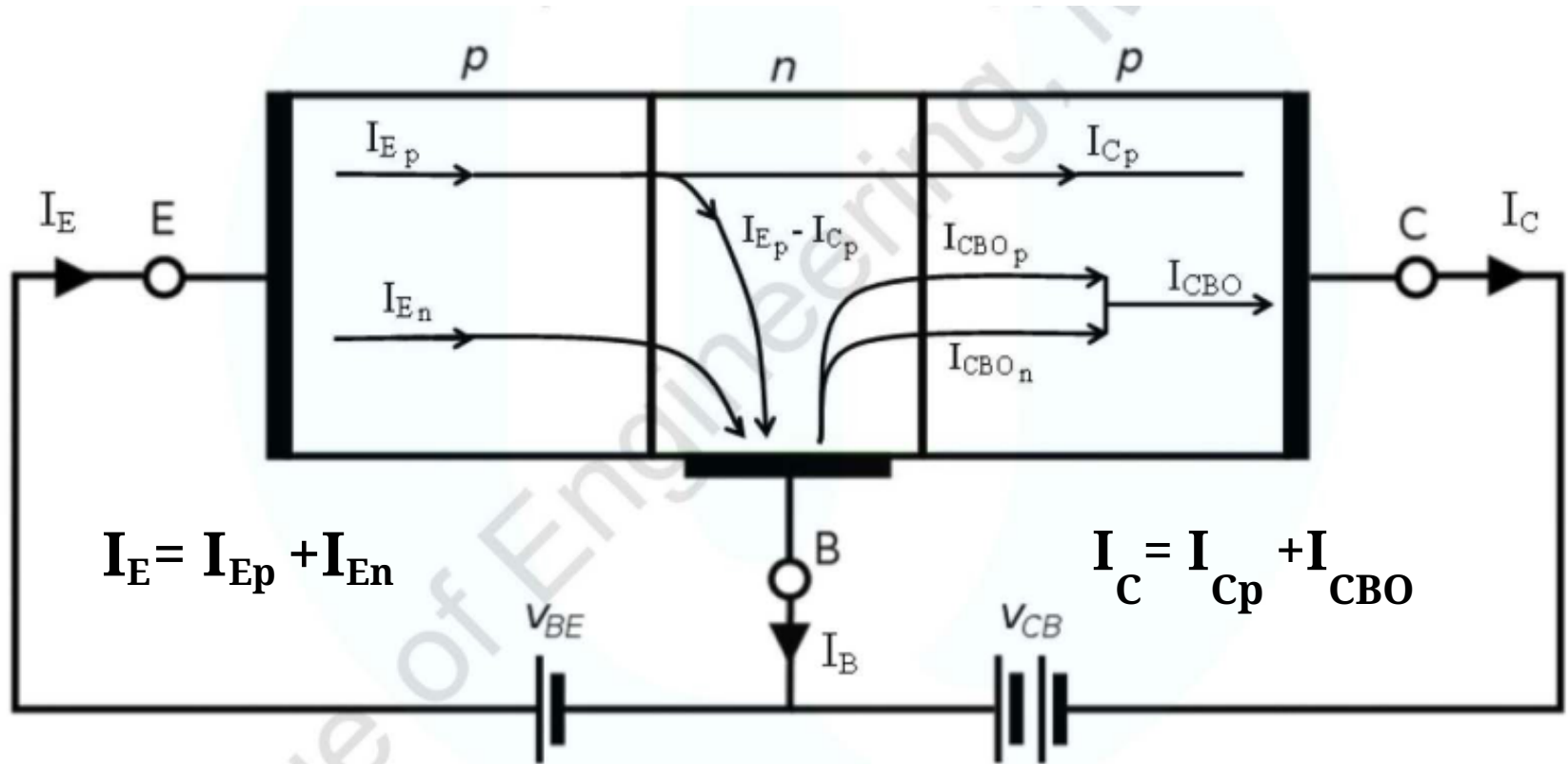
NPN Transistor Current components



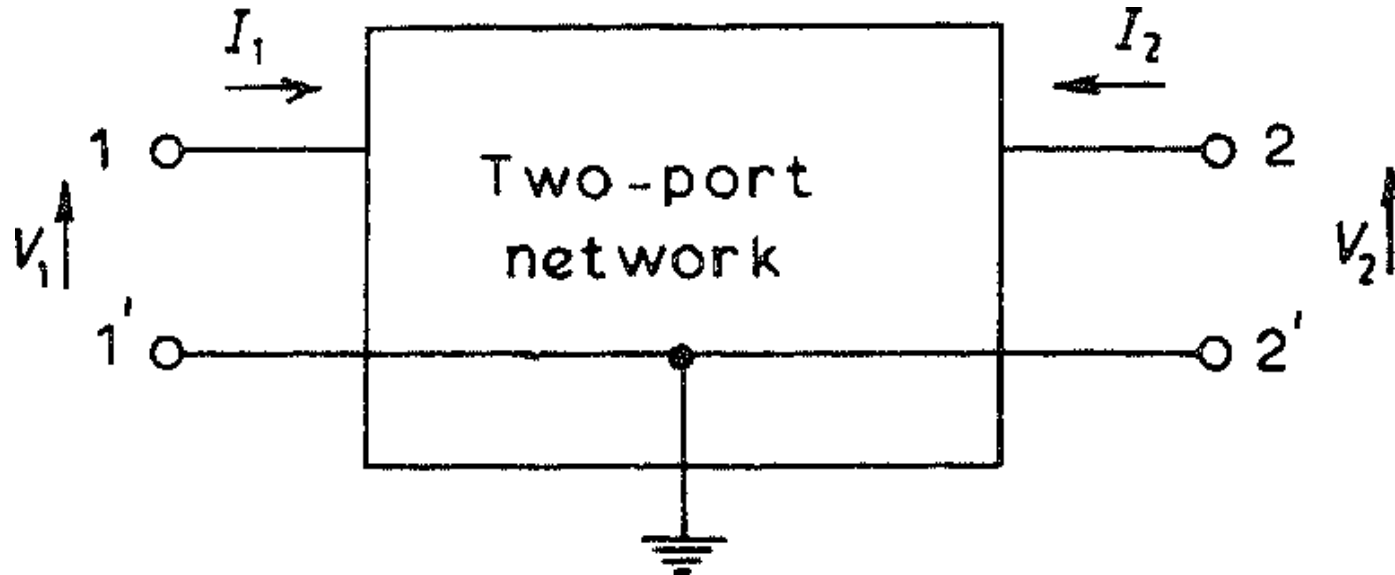
PNP Transistor Current components



PNP Transistor Current components

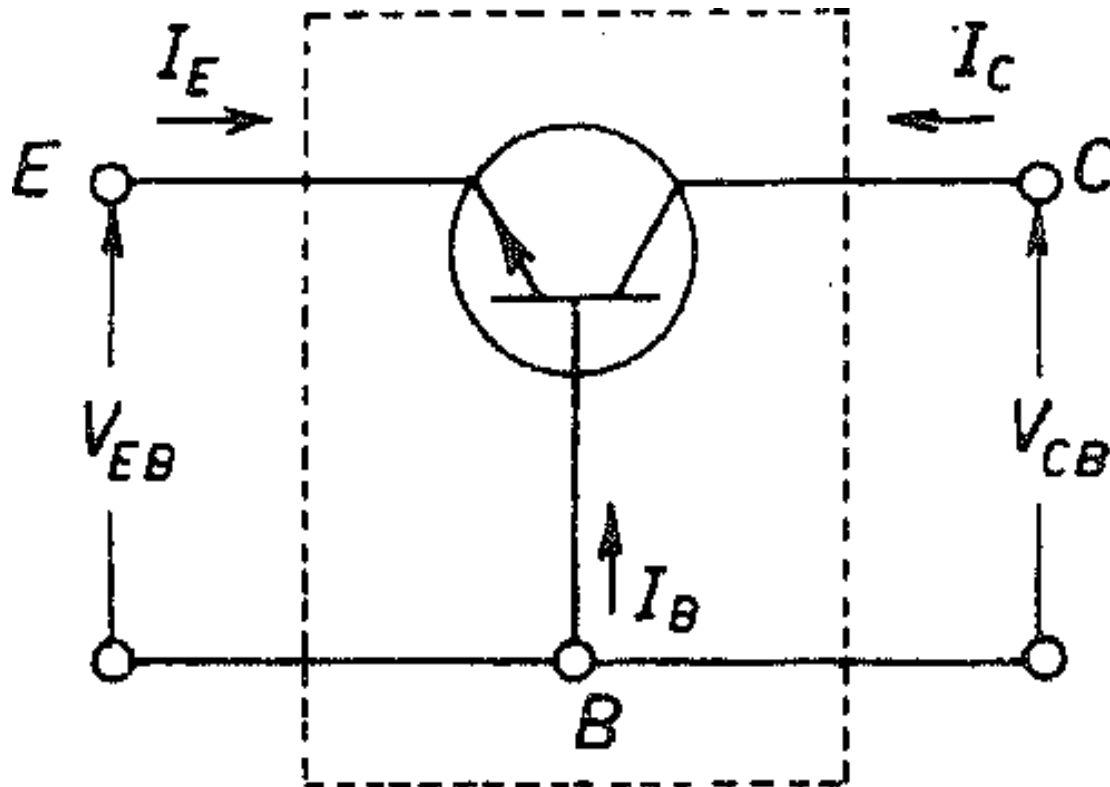


Transistor Configurations



3 terminal device connected as a two port circuit requires one terminal common for input and output

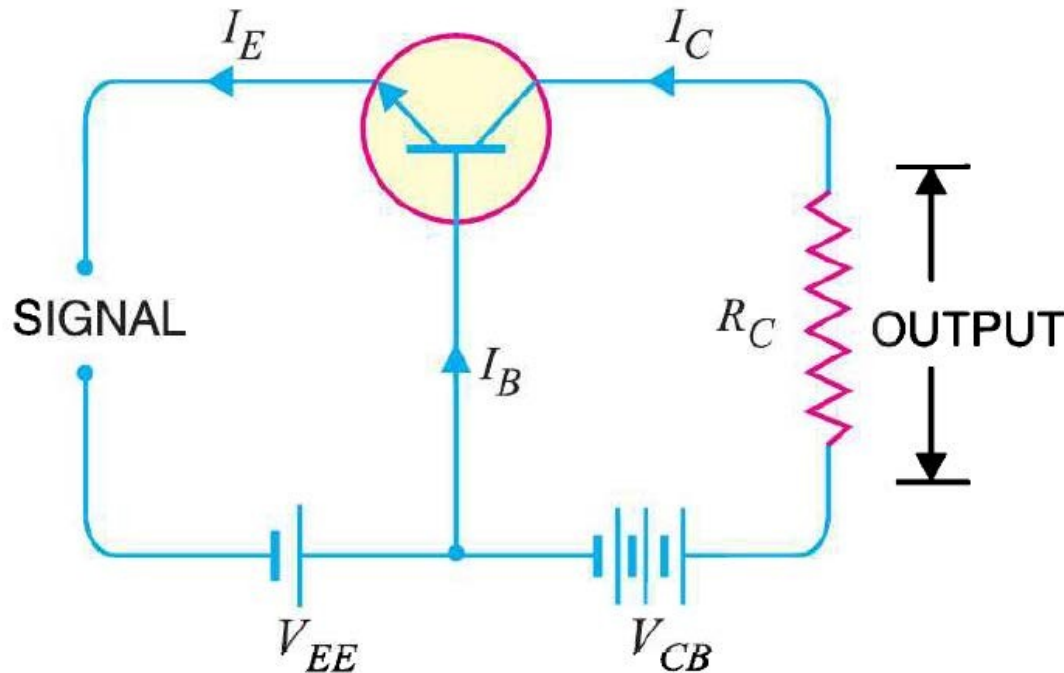
Transistor Configurations



Transistor Configurations

- Common base configuration
- Common emitter configuration
- Common collector configuration

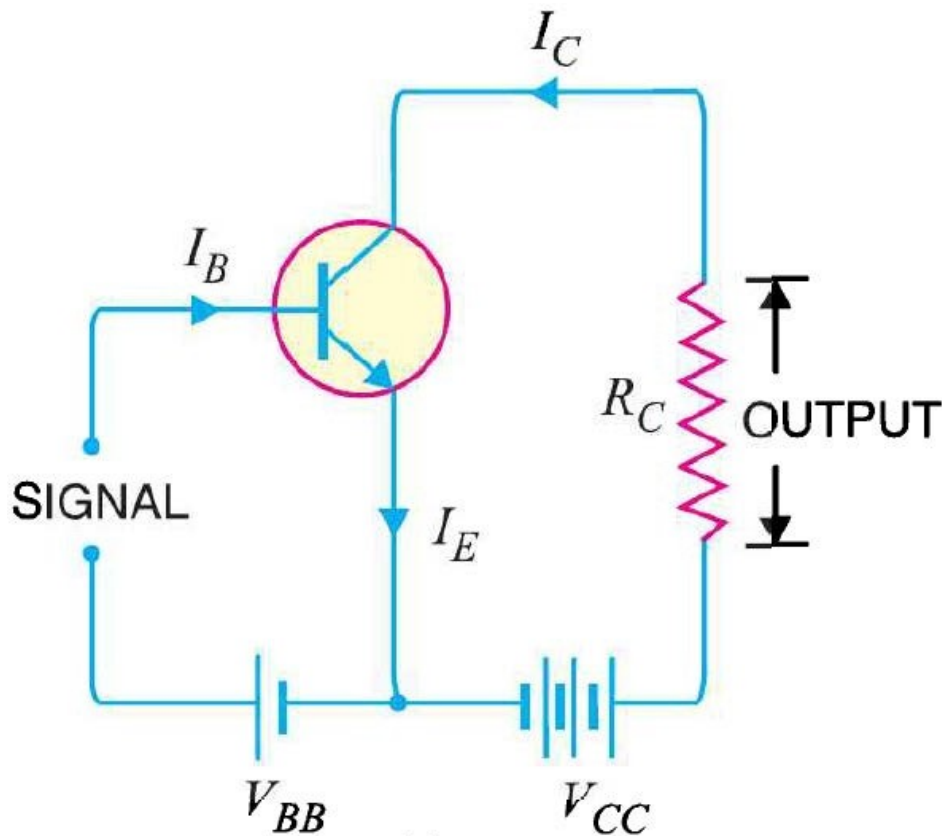
Common base configuration



Current gain = $\frac{\text{Output Current}}{\text{Input Current}}$

Current gain $\alpha = \frac{I_C}{I_E}$

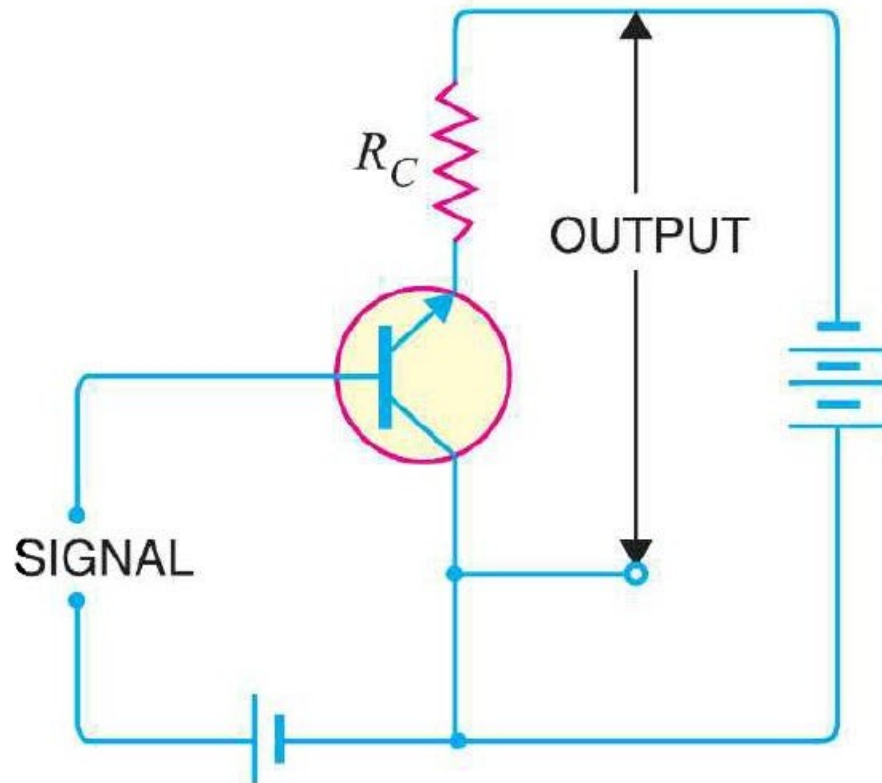
Common emitter configuration



Current gain = $\frac{\text{Output Current}}{\text{Input Current}}$

$$\text{Current gain} = \beta = \frac{I_C}{I_B}$$

Common collector configuration



Current gain = $\frac{\text{Output Current}}{\text{Input Current}}$

$$= I_E = \frac{I_C}{I_B}$$

Relation between current gains

$$\text{Current gain } \alpha = \frac{I_C}{I_E} \quad \text{Current gain } \beta = \frac{I_C}{I_B}$$

$$\text{Current gain } \gamma = \frac{I_E}{I_B}$$

$$\text{Current gain } \alpha = \frac{I_C}{I_C + I_B}$$

$$I_E = I_C + I_B$$

$$\frac{1}{\alpha} = \frac{I_C + I_B}{I_C} = 1 + \frac{I_B}{I_C} = 1 + \frac{1}{\beta}$$

$$\frac{1}{\alpha} = \frac{\beta + 1}{\beta}$$

$$\alpha = \frac{\beta}{\beta + 1}$$

$$\beta = \frac{\alpha}{1 - \alpha}$$

Relation between current gains

$$\text{Current gain } \beta = \frac{I_C}{I_B}$$

$$\text{Current gain } \gamma = \frac{I_E}{I_B}$$

$$\beta = \frac{I_E - I_B}{I_B}$$

$$I_C = I_E - I_B$$

$$I_E = I_C + I_B$$

$$\beta = \frac{I_E}{I_B} - 1$$

$$\frac{I_E}{I_B} = \beta + 1$$

$$\gamma = \beta + 1$$

$$\gamma = \beta + 1$$

$$\beta = \gamma - 1$$

Relation between current gains

Current gain $\alpha = \frac{I_C}{I_E}$

$$I_E = I_C + I_B$$

$$I_C = I_E - I_B$$

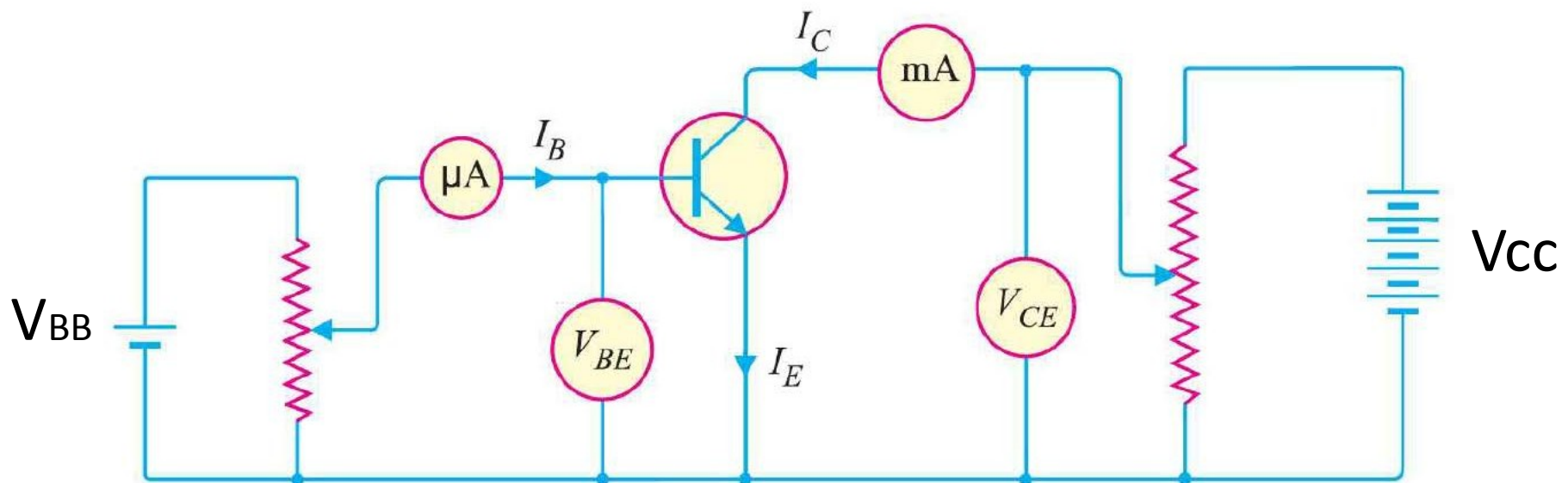
Current gain $\alpha = \frac{I_C}{I_E} = \frac{I_E - I_B}{I_E}$

$$\alpha = 1 - \frac{I_B}{I_E} = 1 - \frac{1}{\beta} = \frac{\beta - 1}{\beta}$$

$$\alpha = \frac{\beta - 1}{\beta}$$

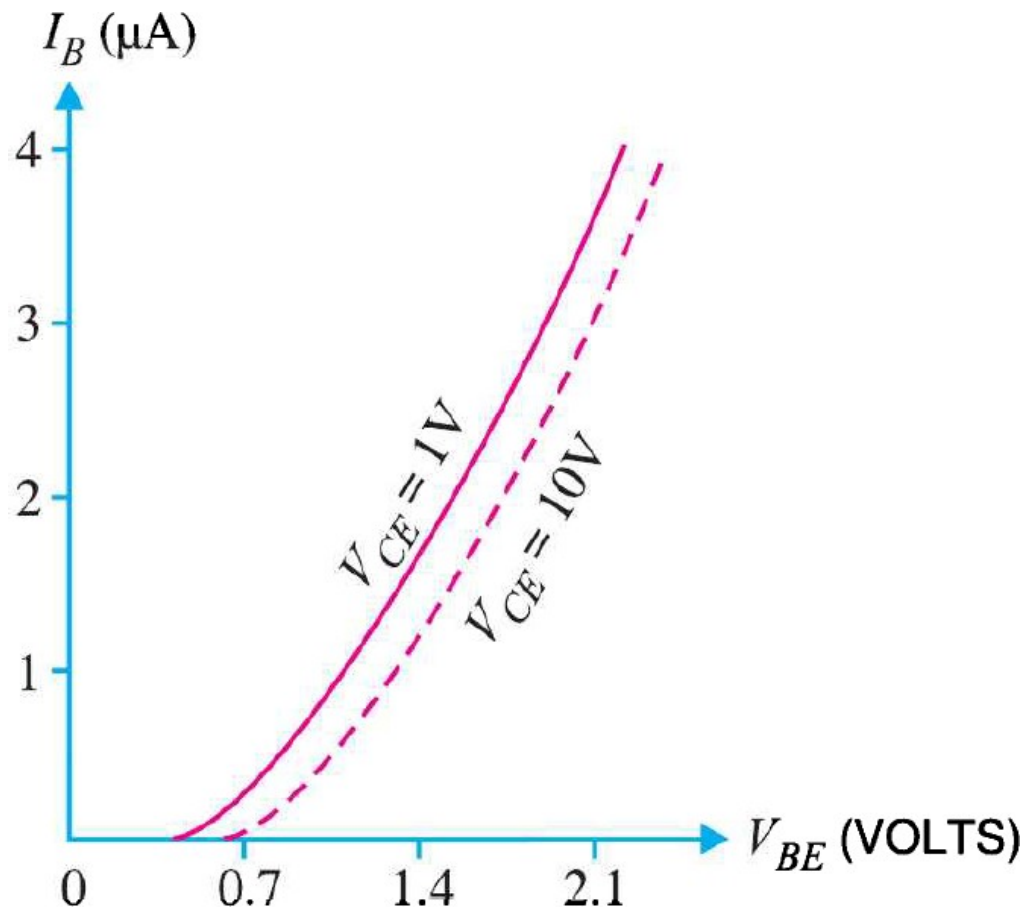
$$\beta = \frac{1}{1 - \alpha}$$

Characteristics of CE configuration



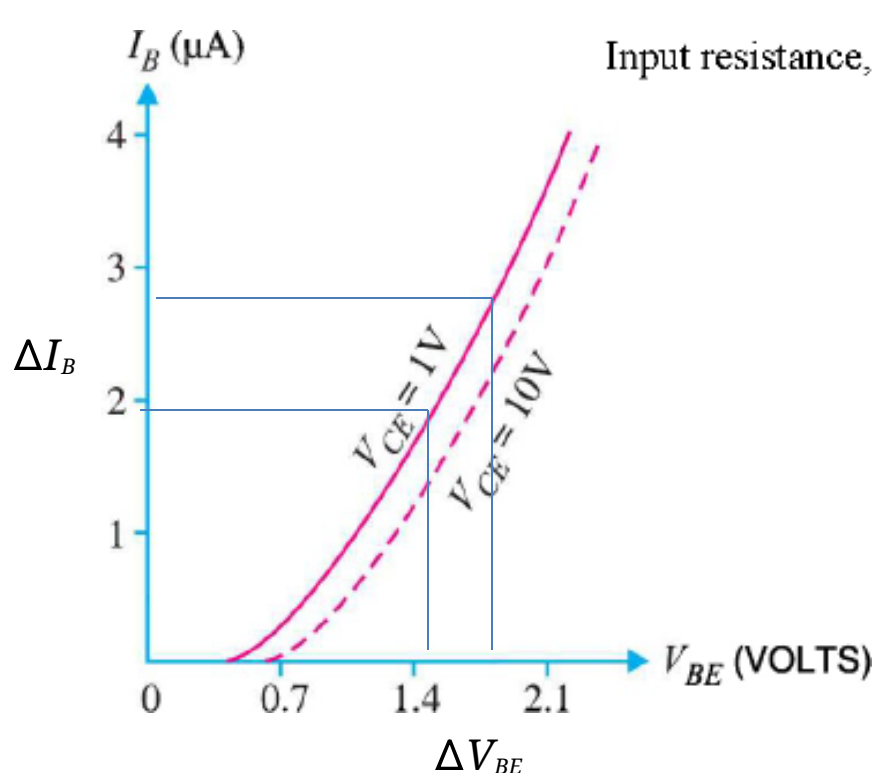
Characteristics of CE configuration

1. **Input characteristic.** *It is the curve between base current I_B and base-emitter voltage V_{BE} at constant collector-emitter voltage V_{CE}*



Characteristics of CE configuration

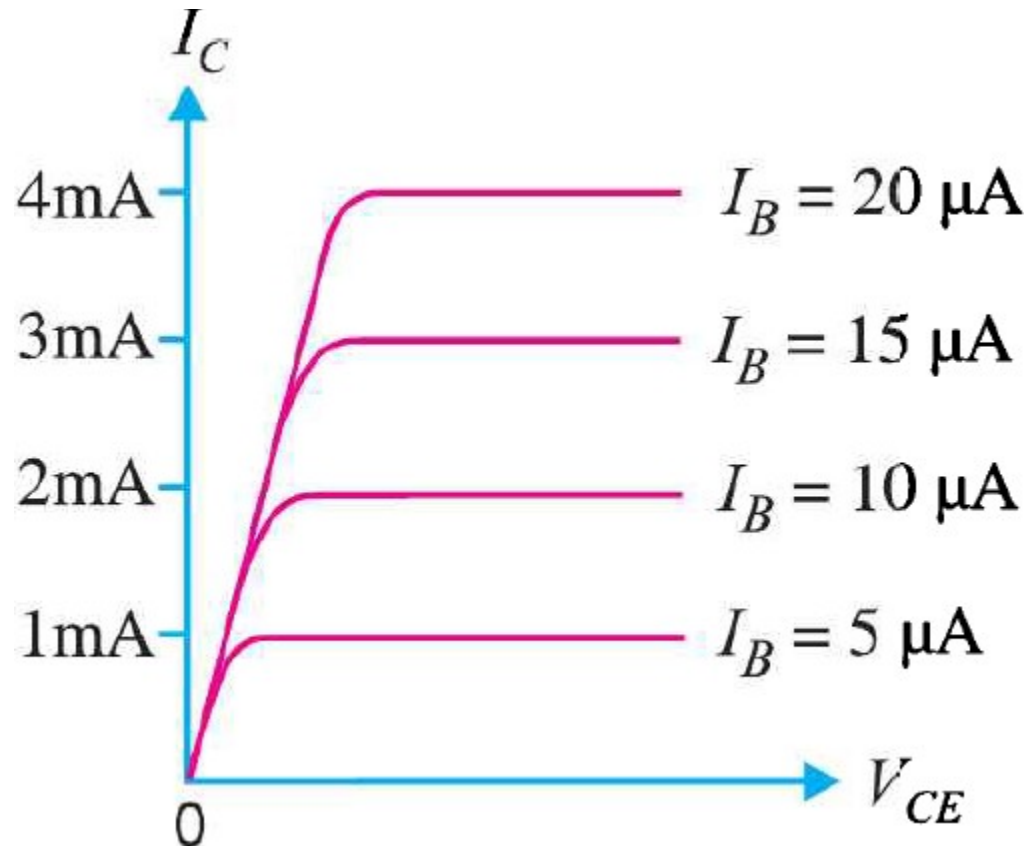
Input resistance. It is the ratio of change in base-emitter voltage (ΔV_{BE}) to the change in base current (ΔI_B) at constant V_{CE} i.e.



Input resistance, $r_i = \frac{\Delta V_{BE}}{\Delta I_B}$ at constant V_{CE}

Characteristics of CE configuration

2. Output characteristic. *It is the curve between collector current I_C and collector-emitter voltage V_{CE} at constant base current I_B .*



Characteristics of CE configuration

Output resistance. It is the ratio of change in collector-emitter voltage (ΔV_{CE}) to the change in collector current (ΔI_C) at constant I_B i.e.

$$\text{Output resistance, } r_o = \frac{\Delta V_{CE}}{\Delta I_C} \text{ at constant } I_B$$

